

Sub-grid Combustion Modeling for Compressible Two-Phase Reacting Flows

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dedicated to
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NOMENCLATURE

Roman Symbols

a	speed of sound
c_p	specific heat at constant pressure
c_v	specific heat at constant volume
Da	Damköhler number
D_{ij}	binary diffusion coefficient
D_t	turbulent diffusivity
E	total energy per unit mass
e	internal energy per unit mass
F	Flame property
$\mathbf{F}$	flux vector
h_1, h_2	principle curvatures
h_{ij}	curvature tensor
h_m	enthalpy of chemical species m
k^{sgs}	subgrid kinetic energy
L	Integral Length scale
$\mathcal{L}$	characteristic wave amplitude
Le	Lewis number
P	pressure
Pr	Prandtl number
q_i	heat flux vector
Re	Reynolds number
Ref	Reference quantity

R_u	universal gas constant
S_L	Laminar Flame Speed
S	source terms
T	temperature
t	time
u	velocity vector
u	x component of velocity vector
u'	r.m.s. velocity, turbulence intensity
u_i	Cartesian velocity component
v	y component of velocity vector
V	volume of control volume
$V_{i,m}$	diffusion velocity vector for constituent m
W_m	molecular weight
w	reaction rate, or z component of velocity vector
Y_m	chemical species mass fraction
x, y, z	Cartesian coordinate directions

Greek Symbols

α	thermal diffusivity
δ	Thickness
γ	ratio of specific heats or surface tension
ε	turbulent kinetic energy dissipation rate
η	Kolmogorov Scale
ρ	density
$\Delta h_{f,m}^o$	species heat of formation
δ_{ij}	Kronecker delta

κ	thermal conductivity
η	Kolmogorov length scale
ν	kinematic viscosity
ν_t	eddy viscosity
μ_t	turbulent viscosity
μ	absolute viscosity
ω, Ω	vorticity vector
Φ	source term
τ_{ij}	viscous stress tensor
τ_{ij}^{sgs}	subgrid stress tensor

Subscripts

0	reference quantity
i, j, k	Cartesian tensor indices
m	chemical constituent
sgs	subgrid scale
t	turbulent quantity
tot	total

Superscripts

n	time step index
sgs	subgrid scale

Other Symbols

∂	partial derivative
Σ	summation
$\sim$	Favre spatial filter or Favre time average
$-$	mixture averaged or simple spatial average

Abbreviations

CFD	Computational Fluid Dynamics
CFL	Coourant-Friedrichs-Lewy
DNS	Direct Numerical Simulation
LES	Large-Eddy Simulation
MPI	Message Passing Interface
PDF	Probability density function
RANS	Reynolds Averaged Navier-Stokes
RNG	Renormalization Group Theory
SGS	Subgrid Scale

SUMMARY

A generic formulation for modeling the sub-grid combustion in compressible, high Reynolds number, two-phase, reacting flows has been developed and validated. A sub-grid mixing/combustion model called Linear Eddy Mixing (LEM) model has been extended to compressible flows and used inside the framework of Large Eddy Simulation (LES) in this LES-LEM approach. The LES-LEM approach is based on the proposition that the basic mechanistic distinction between the convective and the molecular effects should be preserved for accurate prediction of the complex flow-fields such as those encountered in many combustion systems. In LES-LEM, all the physical processes such as molecular diffusion, small and large scale turbulent convection and chemical reaction are modeled separately but concurrently at their respective time scales. This multi-scale phenomena is solved using a two-scale numerical approach, wherein molecular diffusion, small scale turbulent convection and chemical reaction are grouped as small scale processes and the convection at the (LES grid) resolved scales are deemed as the large scale processes. Small-scale processes are solved using a hybrid finite-difference Monte-carlo type approach in a one-dimensional domain. Large-scale advection on the three-dimensional LES grid is modeled in a Lagrangian manner that conserves mass.

Liquid droplets (represented by computational parcels) are tracked using the Lagrangian approach wherein the Newton's equation of motion for the discrete particles are integrated explicitly in the Eulerian gas field. The gas phase LES velocity fields are used to estimate the instantaneous gas velocity at the droplet location. Drag effects due to the droplets on the gas phase and the heat transfer between the gas and the liquid phase are explicitly included. Thus, full coupling is achieved between

the two phases in the simulation.

Validation of the compressible LES-LEM approach is conducted by simulating the flow-field in an operational General Electric Power Systems' combustor (LM6000). The results predicted using the proposed approach compares well with the experiments and a conventional (G-equation) thin-flame model.

Particle tracking algorithms used in the present study are validated by simulating droplet laden temporal mixing layers. Comparison of the energy growth in the fundamental and sub-harmonic mode in the presence and absence of the droplets shows excellent agreement with spectral DNS. Quantitative comparison of the Stokes number effect on particle dispersion also exhibits good agreement with the spectral DNS studies.

Finally, to test the ability of the present two-phase LES-LEM in simulating partially premixed combustion, a LES of freely propagating partially premixed flame in a droplet-laden isotropic turbulent field is conducted. LES-LEM along with the spray models correctly captures the flame structure in the partially premixed flames. It was found that most of the fuel droplets completely vaporize before reaching the flame, and hence provides a continuous supply of reactants, which results in an intense reaction zone similar to a premixed flame. Some of the droplets that did not evaporate completely, traverse through the flame and vaporize suddenly in the post flame zone. Due to the strong spatial variation of equivalence ratio a broad flame similar to a premixed flame is realized. Triple flame structure are also observed in the flow-field due to the equivalence ratio fluctuations.

CHAPTER I

INTRODUCTION

Nearly all flows in nature and engineering applications are turbulent and many involve complex chemical reactions. This complexity is aggravated if the flow-field includes more than one phase (gas + liquid/solid phase) such as those encountered in gas turbines, liquid fueled rocket engines, diesel engines and industrial furnaces. The kind of combustion in such two-phase flows is predominantly non-premixed, in which the the heat release is controlled by rate of fuel-air mixing. Since, turbulence plays an important role in increasing the scalar gradients, and thus to increase the mixing rates, turbulence/chemistry interaction is much more stronger in non-premixed flames compared to premixed flames. This sensitivity to turbulence makes the non-premixed flames more susceptible to quenching (stretch induced extinction) and incomplete combustion. Therefore, the products of non-premixed combustion often contains unburnt hydrocarbons, nitrogen oxides (NO_x) and particulate matter in the form of soot. This just illustrates the effect of turbulence/chemistry interaction in non-premixed combustion. Inclusion of droplets, exasperates the situation, by introducing several other processes with numerous time and length scales. To quote Williams [1988] “*A typical sequence of events would be the injection and atomization of liquid fuel, mixing of droplets with oxidizing gas, heat transfer to droplets producing evaporation of liquid, mixing of fuel vapor with gas possibly followed by gas-phase ignition processes and certainly by either non-premixed or premixed gas-phase combustion all accompanied by various additional aspects of finite-rate chemistry, such as fuel-pyrolysis, production of oxides of nitrogen, and extinction phenomena*”.

1.1 Motivation

It is therefore, of great importance to understand the properties of these turbulent multi-phase reacting flows, to achieve clean, quiet and efficient combustion. In fact the research on spray combustion is driven by the need for fuel efficiency and reducing the pollutants emission. Understanding of the spray combustion phenomena requires a knowledge of

- (1) turbulent non-premixed combustion
- (2) statistical methods for characterizing the spray and
- (3) the mutual interaction between the droplets, turbulence and combustion, and the influences of each of these interactions.

In spite of its importance, our understanding of many aspects of spray combustion phenomena is relatively limited because of the complexity involved in turbulent spray combustion. The complexity is so pronounced for a comprehensive analytical treatment and hence, the analytical studies are often restricted to simple flows. Experiments on the other hand complement analytical studies to a great extent. Especially, with the development of non-intrusive optical measurement techniques such as Laser-Doppler Velocimetry (LDV), Particle Image Velocimetry (PIV), and spectrometry the measurements in complex flow fields have become more feasible. However, simultaneous measurement and processing of multitude of flow parameters with myriads of spatial and temporal scales is still a daunting task in experiments. Hence, combustor designers usually resort to empirical methods and experimental optimization of numerous configurations (Mongia [1994]).

Mathematical models and numerical simulation techniques have become useful tools for analyzing and conducting “experiments” that are not feasible hitherto using analytical or experimental techniques. Numerical approaches for combustion modeling have become more tractable of-late that, solution to an acceptable accuracy can be achieved for a wide variety of problems of technological importance. The three

most important numerical methods are briefly discussed below.

Direct Numerical Simulations The most straight-forward and accurate numerical approach is to solve the governing Navier-Stokes equations without averaging or approximation. This is called direct numerical simulation (DNS). But this approach is computationally too expensive, even for solving canonical flows like homogeneous isotropic turbulence. For instance, in homogeneous turbulence, the number of grid points required in each direction must be of the order of $Re^{3/4}$ and hence, the cost scales as $Re^{9/4}$ (Pope [2000]). Unfortunately, most combustion processes of practical interest involve high Reynolds number ($Re > 10^6$), compressible, swirling flows. As a result, the flow variables are strongly fluctuating in space and time. The spectrum of length scales varies from the dimension of the combustor to the flame thickness or the Kolmogorov scale (smallest fluid dynamic length scale) depending on the flow conditions. Resolving every single length and time scales involved in such complex flows would require computational resources beyond those that are available currently. This prevents the numerical description of these processes in “full detail”, as provided by DNS. As a result, it is not feasible to use DNS for predicting such complex flows. Moreover, in case of multi-phase flows, it is still not computationally tractable to simultaneously resolve the spatial scales interior and adjacent to the individual droplets when the spatial flow scales are much larger than the droplets. In this sense, two-phase DNS also involve averaging of the time and spatial scales that are of the order of droplets. Sirignano [1999], quotes that “ *The DNS methods as applied to two-phase flows have not been truly without any modeling in a fashion analogous to single phase flows. Applications have been confined to situations in which the smallest scale of turbulence is considerably larger than the droplet or the particle size. Since the velocity gradients on the scales of the droplet diameter and boundary-layer thickness are not resolved, the vorticity generated by means of the droplet-gas interaction*

are not determined. So the current two-phase DNS are useful only for special classes of turbulent two-phase flows.”

Reynolds’ Averaged Navier-Stokes Equation At the other extreme of “engineering” modeling is the Reynolds Averaged Navier Stokes (RANS) approach, which is primarily used to predict the mean flow. In the past, most of the numerical studies (Gosman and Ioannides [1983], Faeth [1987]) on spray combustion have been based on steady-state RANS models. In this method, time averaged governing equations for the flow variables are solved numerically, using models for unclosed terms. The most popular model is the $k - \epsilon$, which solves a modeled equation for turbulent kinetic energy (k) and its dissipation rate (ϵ) along with the time averaged equations for mass, motion and energy. Boussinesq gradient diffusion (Wilcox [1993]) approximation is typically employed to close the Reynolds stress tensor in terms of a turbulent eddy viscosity. Although, the $k - \epsilon$ model performs reasonably well in predicting attached flows, the model’s predictions in the presence of an adverse pressure gradient and separation are not so reliable (Driver and Seegmiller [1985]). In addition, the model’s limitation in predicting flows with swirl and strong streamline curvature is also well known (Bradshaw [1973]). In the presence of separation and streamline curvature, the Boussinesq approximation is no longer valid. This is because, for flows with rapid changes in mean strain rate (as in a separation bubble), the Reynolds stresses adjust to changes in the flow on a time scale unrelated to mean rate of strain (Bradshaw [1973]), contrary to the RANS assumptions. Another major drawback of RANS, is that, by using a global model for all turbulent length and time scales, it is very difficult to predict highly unsteady phenomena such as the vortex shedding and formation of large scale coherent structures (Hussain [1983]). Even though second-order closure modeling accommodates the complex effects such as streamline curvature, rigid body rotation, etc., for a three-dimensional flow, the second-order closure model

introduces seven additional equations, one for the turbulent length scale (or an equivalent quantity) and six equations for the components of the Reynolds' stress tensor (Wilcox [1993]). Hence, the computational cost can increase drastically with the use of second-order closure models. These are just the issues related to the modeling of momentum transport.

The presence of multiple, reactive scalars introduces more unclosed terms such as the species-velocity correlations, temperature-species correlations, and terms associated with the averaging of molecular diffusive fluxes (Libby et al. [1980]). In addition, it is well known that the gradient diffusion approximation (used in RANS methods) introduces larger errors in the presence of variable density effects and multiple scalars since counter-gradient diffusion is significant in reacting flows (Libby et al. [1980]). Further more, the Reynolds averaging process in RANS methods, de-couples the fluid mechanics from the chemistry (Libby et al. [1980]) and completely eliminates the effect of scalar mixing at small scale and molecular diffusion while it is well known that the chemical reaction can proceed only if the species mix at the molecular level.

Large Eddy Simulations An alternative to DNS and RANS is Large Eddy Simulation (LES). From a theoretical viewpoint, LES can be considered as a spatially filtered solution to the Navier-Stokes equation. In LES, the flow variables are decomposed into a sum of resolved (spatially filtered) component $\tilde{u}_i(x_i, t)$ and a sub-grid-scale (unresolved) component $u_i''(x_i, t)$. This filtering operation is performed using convolution integrals. The filtered equations represent the evolution of the time dependent resolved scale motion. Due to the truncation of scales by filtering, models are needed to incorporate the effect of the unresolved scales on the resolved scale motion. Since, the small scales motions are relatively universal in character, and carry only a small portion of the total turbulent kinetic energy, they can be modeled more reliably than in situations where the full range of length scales must be modeled (as

in RANS approach). Also, since, the flow/geometry dependent large scale motion are resolved explicitly in LES, results obtained from the LES are expected to be more accurate when compared to RANS results. Moreover, the accuracy of the combustion models and the models for spray depends on the accuracy of the fluid-dynamic fields. Since, the LES approach offers an accurate solution at an affordable cost compared to DNS and more accurate than RANS approaches the present study uses the LES methodology for modeling the reacting turbulent flows.

In the past, LES has been used successfully to predict a variety of flows ranging from homogeneous, isotropic turbulent flows, channel flows, to spatial and temporal mixing layers. A brief list of the various studies and the key developments in LES sub-grid modeling are listed below.

Smagorinsky [1963]

Earliest application of the LES sub-grid scale modeling was performed by Smagorinsky [1963] using an eddy viscosity model for the sub-grid stress. Since then, this model has been used quite extensively for a variety of incompressible turbulent flows (Rogallo and Moin [1984]).

Germano et al. [1990]

A self-consistent dynamic approach to evaluate the coefficients that appear in the sub-grid model is developed by Germano et al. [1990] for incompressible flows.

Erlebacher et al. [1992]

The first known extension of LES sub-grid models for compressible non-reacting flows was developed by Erlebacher et al. [1992] based on the mixed models of Bardina et al. [1980].

Moin et al. [1991]

Dynamic approach developed by Germano et al. [1990] for incompressible flows was successfully extended to compressible flows by Moin et al. [1991]

Menon [1992]

A one-equation model for the sub-grid turbulent kinetic energy, was introduced by Menon [1992] and successfully used in compressible flows. An alternative dynamic modeling formulation for compressible flows that removes the mathematical inconsistency in the Germano et al. [1990] model was introduced by Kim and Menon [1995]

LES of Reactive Flows

In spite of the large success of LES in predicting non-reacting flow fields, only a relatively few studies have been attempted on LES of reactive flows. This is due to the following reason. Even though the major part of the energy containing scales are resolved using LES, a sub-grid model is still needed to represent the larger unresolved scales. But it is well known that the chemical reactions in non-premixed combustion

occur at the smallest dissipative scales (by molecular mixing) and not at the larger unresolved scales. Hence, the combustion process occurs entirely at the sub-grid level, and has to be modeled entirely in LES. This explains why sub-grid modeling techniques, which have successfully been used in predictions of sub-grid momentum transport are not as successful in modeling the combustion. Nevertheless, LES is a promising approach for combustion studies. A few of the interesting studies that used LES for reacting flows are listed below.

Menon [1992]

Active control of combustion in a ramjet configuration was studied by Menon [1992] using LES and observed the presence of two types of instabilities. The data obtained using the LES was used to develop active control strategy to suppress the instability.

Smith and Menon [1996c]

LES was applied to predict the reacting flow-field in a three-dimensional dump combustor using a thin-flame model by Smith and Menon [1996c]. Analysis of various statistics about the vorticity-strain rate alignment, probability density of curvature shape factor etc., obtained using 3-D LES showed good agreement with the DNS of premixed flame propagation in isotropic turbulence and in temporal mixing layers.

Fureby et al. [1993]

The Eddy-Dissipation Model originally developed for RANS was extended for LES by Fureby et al. [1993]. This model is based on the premise that, the combustion process is either kinetics controlled or mixing controlled depending the relative time-scales between them. As a result, at any instant the reaction rate is computed as the minimum of the kinetic reaction rate and the mixing rate estimated in terms of sub-grid kinetic energy. Fureby et al. [1993] applied this model to bluff body stabilized flames using reduced kinetics for propane flames and obtained overall good agreement for temperature with the experiments. However, the agreement for reactive scalars was less encouraging.

Frankel et al. [1993]

Frankel et al. [1993] carried out a Large Eddy Simulation for computing the incompressible reactive mixing layers using assumed PDF methods. The chemistry was modeled using a global one-step reaction. Even though Frankel et al. [1993] obtained a good qualitative agreement with DNS, significant difference was observed in the species co-variance predictions.

Gao and O'Brien [1993]

A general methodology that combines LES and transported PDF approach was developed by Gao and O'Brien [1993] for simulating reacting flow problems. Even though this formulation provides a closed form representation for the sub-grid turbulent transport and small scale molecular mixing it is yet to be implemented in an actual LES to appraise this model.

Menon et al. [1993]

The stand-alone Linear Eddy Model developed by Kerstein [1988a,b, 1989] was extended to be a sub-grid combustion model for LES by Menon et al. [1993]. LEM was successfully used to simulate turbulent mixing (McMurtry et al. [1992]), non-premixed combustion (Menon and Calhoon [1996]) and premixed combustion (Chakravarthy and Menon [2001]).

Cook and Riley [1998]

A sub-grid combustion model for multi-step, finite rate Arrhenius reactions was proposed by Cook and Riley [1998]. It accounts for the finite-rate chemistry by invoking a laminar flamelet approximation and employs an assumed β function for the sub-grid mixture fraction PDF. Even though the model was developed for compressible turbulent flows, it was evaluated only for an incompressible decaying turbulence, with a single-step chemistry. Results indicate that that, as the activation energy is increased, the accuracy of the model degrades except in the case of equilibrium chemistry.

It should be noted here that the list of studies cited above is restricted to LES of

reacting flows and are far from complete. The purpose is to provide an idea of the various approaches used for combustion modeling in LES and the challenges to be faced during the numerical modeling of combustion. At the risk of over stating, it is repeated here that it is the presence of multiple time scales that makes the simulation of turbulent combustion very challenging. Combustion makes the system stiff, by adding faster time scales (compared to the turbulent time scales). In situations where the fast time scales are important, one must either resolve them (i.e., with a small time step) or model their effects. Often, however, the details of the fast time scales are not important. These are situations in which there exists a near balance among the various processes, and the dynamical time scale of the overall evolution is slow. For such problems, it is desirable (from the point of view of computational efficiency) to resolve only the dynamical (slow) time scales, without overstepping the fast time scales.

Consequently, the basic requirement for any turbulent combustion model is that it should truly represent multi-scale phenomena, while preserving the dynamical balance between the various processes to avoid large errors in the solutions. It will be shown later that, using LEM all the physical processes in turbulent combustion (such as large-scale advection, sub-grid scale mixing, molecular diffusion and chemical reaction) can be elegantly incorporated while still being affordable. It can therefore, capture the turbulence-chemistry interaction at all the scales including those below the grid resolution. Furthermore, additional physical processes such as the two fluid models for spray combustion (Pannala and Menon [1998]), radiative heat loss and soot modeling (Zimberg et al. [1998]) can be included in a straightforward manner. Even though LEM has been used as a LES sub-grid combustion model in some of studies conducted in the past, many refinements are still needed in order to use LEM for fundamental studies and for design level simulation with equal ease. Some of the short-comings of the previous LEM studies are listed below before identifying the

goals of the current research.

- **McMurtry et al. [1992]**

LES-LEM was applied to H_2 -Air combustion. Assumptions made include unity Lewis number and no heat release effects.

- **Calhoon [1996], Menon and Calhoon [1996]**

LES-LEM was applied to two-dimensional mixing layer with negligible heat release. Assumptions include equal molecular weight, infinite-rate chemistry was used in this study.

- **Smith [1998]**

LES-LEM was applied to two-dimensional stagnation point premixed flames with low heat release. A thin-flame model employing G-equation was used to study flame structure in the flamelet regime.

- **Chakravarthy and Menon [2000]**

A three-dimensional, low Mach Number incompressible LES-LEM was applied to stagnation point and bluff body stabilized premixed flames. A thin-flame model employing G-equation was used to study flames in the flamelet regime. This was the only study that validated the LES-LEM approach by comparing with the experiments.

- **Pannala and Menon [1998]**

A three dimensional, low Mach Number incompressible LES-LEM was applied to droplet-laden temporal mixing layers and isotropic turbulent flows. Even though droplet vaporization effects were included, effect of heat release due to combustion was not included.

1.2 Research Objectives

The scope of the present thesis is defined based on the refinements needed for the LES-LEM approach. The goal for the present work is to develop and validate an accurate unsteady simulation capability for studying spray combustion processes. The proposed **Two-Phase-LES-LEM** model will include

1. Compressibility effects for subsonic reacting flows
2. Realistic heat release
3. Multi-species formulation
4. Non-unity Lewis number effects
5. Finite-rate kinetics
6. Three dimensional LES formulation
7. Application to complex geometry e.g., gas-turbines
8. Application to complex flow-fields (swirling high Re flows)
9. Application for spray-combustion

1.3 Thesis-Outline

This thesis is organized as follows

- **Chapter 1** serves to introduce the need to develop tools to understand the spray combustion processes. This chapter is also used to illustrate the motivation for using LES, and the motivation for developing a sub-grid combustion model named LEM for two-phase flows.
- **Chapter 2** introduces the exact and the filtered governing equations for the gas phase, closure models for the sub-grid momentum terms, and a small review on the various combustion models that are in use and their strength and limitations.
- **Chapter 3** is used to provide information on the numerical method employed for solving the governing partial differential equation, and the application of initial and boundary conditions.
- **Chapter 4** explains in detail the rationale behind new sub-grid LEM model, and the various implementation issues.
- **Chapter 5** shows the results of the LES-LEM validation studies, for a three dimensional complex flow in an operational General Electric gas turbine combustor, LM6000.
- **Chapter 6** demonstrates the accuracy of the particle tracking algorithms used in this study for the two-phase flows. Validation is carried out using DNS and LES of droplet laden temporal mixing layers and the results are compared with spectral DNS of Ling et al. [1998]
- **Chapter 7** discusses a test-case spray combustion under realistic swirling conditions in a gas-turbine configuration. A single-step global mechanism with infinite rate chemistry is used to model the combustion. Results are reported

on the effect of swirl, particle size and heat release on the fine-scale structure of the turbulent flow field.

- **Chapter 8** reports on the development and simulation of a three-dimensional, spray combustion using a LES-LEM approach. LES of a freely propagating turbulent partially premixed flame in a decaying isotropic turbulence with droplets is used to demonstrate the accuracy and the ability of the new methodology developed here.
- **Chapter 9 & 10** summarizes the conclusions of this research and the future recommendations.

CHAPTER II

MATHEMATICAL FORMULATION AND MODELING

"That is the total and absolute generalization ... well, almost."

-Anonymous

This chapter summarizes all the relevant equations and the models used to supplement the governing equations used for reacting flows.

2.1 The Gas (Eulerian) Phase

The continuum equations of motion for a compressible, multi-phase, multi-species, reacting fluid are the Navier-Stokes equations describing the conservation of mass, momentum, total energy, and chemical species. These equations are given in the following section:

2.1.1 Exact governing equations

$$\begin{aligned}\frac{\partial \rho}{\partial t} + \frac{\partial \rho u_i}{\partial x_i} &= \dot{\rho}_s \\ \frac{\partial \rho u_i}{\partial t} + \frac{\partial}{\partial x_j} [\rho u_i u_j + p \delta_{ij} - \tau_{ij}] &= \dot{F}_{s,i} \\ \frac{\partial \rho E}{\partial t} + \frac{\partial}{\partial x_i} [(\rho E + p) u_i + q_i - u_j \tau_{ji}] &= \dot{Q}_s \\ \frac{\partial \rho Y_m}{\partial t} + \frac{\partial}{\partial x_i} [\rho Y_m (u_i + V_{i,m})] &= \dot{w}_m + \dot{S}_{s,k} \quad \text{where } m = 1, N-1\end{aligned}\tag{2.1}$$

In the above equations, all terms on the right hand side with subscript “s” are the source terms from the dispersed phase. Attention is drawn to the fact that the above equation is exact only in the absence of the dispersed phase source terms. This is because, the presence of the dispersed phase (particles/droplets) introduces fine scale

vortical motions in the boundary layer and/or in the wake of the particles/droplets. The characteristic scales of these vortical motions are usually smaller than the Kolmogorov scale ¹ that would be present in the flow, in the absence of the dispersed phase, for the same Reynolds' number. However, for the sake of numerical tractability, it is advantageous to filter such fine scale motions induced by the droplets and represent their average effect on the continuum phase. Note that the size of this filter must be of the order of the largest droplet radius and hence, smaller than the Kolmogorov scale, so as to avoid averaging the small scale turbulent motions. Thus, even though the fluid dynamic scales are not averaged (i.e., the gas phase equations are exact), the flow-field around the droplets have been averaged (and hence the governing equations with the source terms for the dispersed phase are not exact in a rigorous sense.) In this sense, the same limitation applies to DNS of two-phase flows as well.

Furthermore, in writing the equations in the form shown in Eqn. 2.1, there is an implicit assumption that the spray field is dilute. This is because of the fact that the volume occupied by the liquid phase compared to the gas phase has been neglected in writing Eqn. 2.1. Therefore, the above equations can be used only in a situation where the droplets do not interact (collision/break-up/coalescence) with each other but interact only with the gas phase.

In the above equations, ρ is the mass density, p is the pressure, E is the total energy per unit mass, T is the temperature, u_i ($i = 1, 2, 3$) are the velocity components, δ_{ij} is the Kronecker delta, $\kappa = \bar{c}_p \mu / Pr$ is the mixture averaged thermal conductivity, $\bar{c}_p = \sum_{m=1}^N Y_m c_{p,m}$ is the mixture averaged specific heat at constant pressure, and Pr is the Prandtl number. The individual species mass fraction, diffusion velocities, specific heat at constant pressure, enthalpy and mass reaction rate per unit volume

¹smallest turbulent fluctuation for a given Reynold's number

are, respectively, Y_m , $V_{i,m}$, $c_{p,m}$, h_m , and $\dot{w}_m$. The viscous stress tensor is given by:

$$\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} \mu \frac{\partial u_k}{\partial x_k} \delta_{ij} \quad (2.2)$$

where μ is the molecular viscosity coefficient. The mass reaction rate is

$$\dot{w}_m = W_m \sum_{k=1}^M (\nu''_{m,k} - \nu'_{m,k}) A_k T^{\alpha_k} e^{-(E_k/R_u T)} \prod_{n=1}^N \left(\frac{X_n P}{R_u T} \right)^{\nu'_{n,k}} \quad m = 1, \dots, N \quad (2.3)$$

where M is the total number of chemical reactions and N is the total number of species, A_k , α_k and E_k are the Arrhenius rates, temperature exponents, and activation energies of the “ k th” reaction respectively. Also, the universal gas constant is R_u , and W_m and X_n are the species molecular weight and molar fraction, respectively. The heat flux vector contains contributions from thermal conduction, diffusive enthalpy flux, Dufour heat flux and radiation heat flux. Neglecting the Dufour and radiation effects the heat flux vector is given by

$$q_i = -\kappa \frac{\partial T}{\partial x_i} + \rho \sum_{m=1}^N h_m Y_m V_{i,m} \quad (2.4)$$

The pressure is determined from the equation of state for a perfect gas mixture

$$p = \frac{\rho T}{W_m} \sum_{m=1}^N Y_m R_u \quad (2.5)$$

and the total energy per unit volume is determined from

$$\rho E = \rho \left(e + \frac{1}{2} u_k u_k \right) \quad (2.6)$$

and e is internal energy per unit mass given by

$$e = \sum_{m=1}^N Y_m h_m - \frac{p}{\rho} \quad (2.7)$$

The diffusion velocities are approximated by Fickian diffusion as,

$$V_{i,m} = -\frac{D_m}{Y_m} \frac{\partial Y_m}{\partial x_i} \quad (2.8)$$

where D_m is the mixture averaged molecular diffusion coefficient. In this work, pressure diffusion (Dufour) and temperature diffusion (Soret) effects are neglected and no body forces are included. Viscosity is approximated using Sutherland's Law,

$$\mu = S_0 \frac{T^{3/2}}{T + S_1} \quad (2.9)$$

where $S_0 = 1.458 \text{ kg}/(\text{m s } \sqrt{\text{K}^o})$ and $S_1 = 110.4 \text{ K}$, where μ_0 is a reference viscosity at T_0 . The caloric equation of state is given by,

$$h_m = \Delta h_{f,m}^0 + \int_{T_0}^T c_{p,m}(T') dT' \quad (2.10)$$

where $\Delta h_{f,m}^0$ is the standard heat of formation at temperature T_0 . Finally, the species mass fractions, diffusion velocities and mass fractions are constrained by the following identities:

$$\sum_{m=1}^N Y_m = 1 \quad (2.11)$$

$$\sum_{m=1}^N V_{i,m} = 0 \quad (2.12)$$

2.1.2 Filtered equations for LES

In this section, governing equations and the modeling techniques used in LES are presented. Following Erlebacher et al. [1992], the flow variables are decomposed into the resolved (super-grid scale) and unresolved (sub-grid scale) components by a spatial filtering operation, such that $f = \tilde{f} + f''$, where the tilde ($\sim$) denotes resolved scale and double prime ($''$) denotes unresolved sub-grid scale quantities. The Favre filtered variable is defined as $\tilde{f} = \overline{\rho f} / \bar{\rho}$ where the over bar represents a spatial filtering, which is defined as:

$$\overline{\rho f(x_i, t)} = \int_D \rho f(z_i, t) G(x_i - z_i, \Delta) dz_i. \quad (2.13)$$

Here, G is the filter kernel, D is the domain of the flow and Δ is filter size, defined as $\Delta = (\Delta x \Delta y \Delta z)^{1/3}$, where Δx , Δy and Δz are the grid size in axial, transverse and

span wise directions, respectively. The filtering operation is normalized by requiring that

$$\int_D G(x_i - z_i, \Delta) dz_i = 1. \quad (2.14)$$

It follows from Eqns [2.13] and [2.14] that for equally spaced and weakly stretched grids, $\overline{\partial f}/\partial t = \partial \bar{f}/\partial t$ and $\overline{\partial f}/\partial x_i = \partial \bar{f}/\partial x_i$. Filtering reduces the high wave number Fourier components of the flow variables and separates the resolved-scale components from the sub-grid scales. In this study, a box filter is employed which is appropriate for finite volume schemes (Schumann [1975]). The filter function G (in any direction, x) is defined as:

$$G = \begin{cases} 1/\Delta & -\frac{\Delta}{2} \leq (x - z) \leq \frac{\Delta}{2} \\ 0 & \text{otherwise} \end{cases} \quad (2.15)$$

Contrary to the more traditional Favre temporal averaging, $\tilde{\tilde{f}} \neq \tilde{f}$ and in general, $\tilde{f}'' \neq 0$. Applying the filtering operator to the conservation equations of mass momentum, energy and species equations (Eqn. [2.1]) results in the following LES equations for two-phase flows:

$$\begin{aligned} \frac{\partial \bar{\rho}}{\partial t} + \frac{\partial \bar{\rho} \tilde{u}_i}{\partial x_i} &= \tilde{\dot{\rho}}_s \\ \frac{\partial \bar{\rho} \tilde{u}_i}{\partial t} + \frac{\partial}{\partial x_j} [\bar{\rho} \tilde{u}_i \tilde{u}_j + \bar{p} \delta_{ij} - \bar{\tau}_{ij} + \tau_{ij}^{sgs}] &= \tilde{\dot{F}}_{s,i} \\ \frac{\partial \bar{\rho} \tilde{E}}{\partial t} + \frac{\partial}{\partial x_i} [(\bar{\rho} \tilde{E} + \bar{p}) \tilde{u}_i + \bar{q}_i - \tilde{u}_j \bar{\tau}_{ji} + H_i^{sgs} + \sigma_i^{sgs}] &= \tilde{\dot{Q}}_s \\ \frac{\partial \bar{\rho} \tilde{Y}_k}{\partial t} + \frac{\partial}{\partial x_i} [\bar{\rho} \tilde{Y}_k \tilde{u}_i - \bar{\rho} \tilde{Y}_k \tilde{V}_{i,k} + Y_{i,k}^{sgs} + \theta_{i,k}^{sgs}] &= \tilde{\dot{w}}_k + \tilde{\dot{S}}_{s,k} \quad k = 1, N_s \end{aligned} \quad (2.16)$$

where N_s is the total number of species present in the system and $\bar{q}_i$ is the heat flux vector given by

$$\bar{q}_i = -\bar{\kappa} \frac{\partial \tilde{T}}{\partial x_i} + \bar{\rho} \sum_{k=1}^{N_s} \tilde{h}_k \tilde{Y}_k \tilde{V}_{i,k} + \sum_{k=1}^{N_s} q_{i,k}^{sgs}.$$

The diffusion velocities are approximated using Fickian diffusion as $\tilde{\tilde{V}}_{i,k} = (-\bar{D}_k/\tilde{Y}_k)(\partial \tilde{Y}_k/\partial x_i)$.

The sub-grid terms in Eqns. [2.16] that require closure are:

$$\begin{aligned}
\tau_{ij}^{sgs} &= \bar{\rho} (\widetilde{u_i u_j} - \tilde{u}_i \tilde{u}_j) \\
H_i^{sgs} &= \bar{\rho} (\widetilde{E u_i} - \tilde{E} \tilde{u}_i) + (\overline{p u_i} - \bar{p} \tilde{u}_i) \\
\sigma_i^{sgs} &= \widetilde{u_j \tau_{ij}} - \tilde{u}_j \bar{\tau}_{ij} \\
Y_{i,k}^{sgs} &= \bar{\rho} [\widetilde{u_i Y_k} - \tilde{u}_i \tilde{Y}_k] \\
q_{i,k}^{sgs} &= [\overline{h_k D_k \partial Y_k / \partial x_i} - \tilde{h}_k \tilde{D}_k \partial \tilde{Y}_k / \partial x_i] \\
\theta_{i,k}^{sgs} &= \bar{\rho} [\widetilde{V_{i,k} Y_k} - \tilde{V}_{i,k} \tilde{Y}_k]
\end{aligned} \tag{2.17}$$

The pressure is determined from the filtered equation of state,

$$\bar{p} = \bar{\rho} R_u \sum_{k=1}^n \frac{\tilde{Y}_k \tilde{T}}{W_k} + \frac{\widetilde{Y_k T} - \tilde{Y}_k \tilde{T}}{W_k}$$

which reduces to $\bar{p} = \bar{\rho} R \tilde{T}$ after neglecting the temperature-species correlation term, $T^{sgs} = [\widetilde{Y_k T} - \tilde{Y}_k \tilde{T}]$. For low heat-release, T^{sgs} can be expected to be negligible (Fureby [1995]) but this may not be true for high heat release. However, due to the difficulty in modeling these terms they are generally neglected (Fureby [1995], Veynante and Poinso [1996]).

The filtered total energy per unit volume is given by $\bar{\rho} \tilde{E} = \bar{\rho} \tilde{e} + \frac{1}{2} \bar{\rho} \tilde{u}_i \tilde{u}_i + \bar{\rho} k^{sgs}$ where, the sub-grid kinetic energy (to be discussed later) is defined as, $k^{sgs} = (1/2) [\widetilde{u_k u_k} - \tilde{u}_k \tilde{u}_k]$. The filtered internal energy is given by

$$\tilde{e} = \sum_{k=1}^{N_s} [\tilde{Y}_k \tilde{h}_k + (\widetilde{Y_k h_k} - \tilde{Y}_k \tilde{h}_k)] - \bar{p} / \bar{\rho} \tag{2.18}$$

For calorically perfect gases, this reduces to

$$\tilde{e} = \sum_{k=1}^{N_s} [c_{v,k} \tilde{Y}_k \tilde{T} + c_{v,k} T^{sgs} + \tilde{Y}_k \Delta h'_{f,k}] \tag{2.19}$$

where, $\Delta h'_{f,k} = \Delta h_{f,k}^0 - c_{p,k} T^0$ and $\Delta h_{f,k}^0$ is the standard heat of formation at a reference temperature T^0 . Upon neglecting T^{sgs} , this further reduces to

$$\tilde{e} = \sum_{k=1}^{N_s} [c_{v,k} \tilde{Y}_k \tilde{T} + \tilde{Y}_k \Delta h'_{f,k}]$$

2.1.3 Momentum closure

Model for sub-grid stress tensor τ_{ij}^{sgs} is derived usually by drawing an analogy between the viscous stresses, (τ_{ij}) in the unfiltered Navier-Stokes equation and the sub-grid stresses, (τ_{ij}^{sgs}) in the filtered equations. Hence, the deviatoric part of the sub-grid stresses is assumed to be proportional to the deviatoric part of the resolved rate of strain, $\tilde{S}_{ij} = (1/2)(\partial\tilde{u}_i/\partial x_j + \partial\tilde{u}_j/\partial x_i)$. If the sub-grid stress tensor can be split into deviatoric and isotropic parts $\tau_{ij}^{sgs} = \tau_{ij,d}^{sgs} + \tau_{kk}^{sgs}$, then the deviatoric part $\tau_{ij,d}^{sgs}$ is modeled as follows;

$$\tau_{ij,d}^{sgs} = -2\nu_t[\tilde{S}_{ij} - \frac{1}{3}\tilde{S}_{kk}\delta_{ij}].$$

Noting that $\tau_{kk}^{sgs} = (2/3)k^{sgs}$, the total sub-grid stress tensor can be expressed as

$$\tau_{ij}^{sgs} = -2\nu_t[\tilde{S}_{ij} - \frac{1}{3}\tilde{S}_{kk}\delta_{ij}] + \frac{2}{3}\bar{\rho}k^{sgs}\delta_{ij}. \quad (2.20)$$

Therefore, to complete the closure for the sub-grid stresses, the sub-grid eddy viscosity ν_t and the sub-grid kinetic energy, k^{sgs} need to be modeled.

If the production and dissipation of sub-grid kinetic energy are in equilibrium (implying that the sub-grid field responds instantaneously to the changes in the resolved scales), then an algebraic eddy viscosity model as proposed by Smagorinsky [1963] can be used. However, in many engineering simulations, this may not be true. This is because, if the fraction of energy carried by sub-grid scales is significant compared to the resolved scales then there will be a finite relaxation time associated with the sub-grid scales. Since, most of the computations for engineering applications resolve the flow-field using coarse grids, the sub-grid scales can have significant portion of the total turbulent kinetic energy. Hence, Smagorinsky model for sub-grid stresses can induce large errors. Realizing this, Schumann [1975] derived a non-equilibrium model using a transport equation for the sub-grid kinetic energy, to represent the characteristic velocity scale. The evolution equation for sub-grid kinetic energy (following Menon et al. [1996]) is given as :

$$\frac{\partial \bar{\rho} k^{sgs}}{\partial t} + \frac{\partial}{\partial x_i} (\bar{\rho} \tilde{u}_i k^{sgs}) = P^{sgs} - \epsilon^{sgs} + \frac{\partial}{\partial x_i} \left(\bar{\rho} \frac{\nu_t}{Pr_t} \frac{\partial k^{sgs}}{\partial x_i} \right) + F_k \quad (2.21)$$

In the above formulation, F_k represents the work done due to the two-phase coupling force term $\dot{F}_{s,i}$ and provides a coupling between the turbulent motion of the droplets and the evolution of the sub-grid kinetic energy. Thus, it can be seen that the dispersed phase interacts with the sub-grid scales in two ways. By way of the source terms in Eqn. 2.21, a direct coupling is provided, while the source term in filtered momentum equation (in Eqns. 2.16), alters the velocity field due to particle drag and vaporization, which in turn modifies the sub-grid kinetic energy. This term is modeled as $F_k = \langle u_i \dot{F}_{s,i} \rangle - \tilde{u}_i \dot{F}_{s,i}$ (Pannala and Menon [1998]). The angled brackets represent the volume averaging over all droplets crossing a LES cell.

The terms, P^{sgs} and ϵ^{sgs} in Eqn. [2.21] are respectively, production and dissipation of sub-grid kinetic energy. The coefficient Pr_t is the turbulent Prandtl number and is taken to be 0.9. Spectral closure theories of Kraichnan [1976] and Lesieur [1998] predict the following expression for eddy viscosity at large scales

$$\nu_t = 0.261 \sqrt{E(k_c)/k_c} \quad (2.22)$$

where $k_c (\approx 1/\Delta)$ is the cut-off wave number corresponding to the smallest resolved wave-number in LES. By substituting the following form of the energy spectrum given by Pao [1965]

$$E(k) = \alpha \epsilon^{2/3} k^{-5/3} \exp[-(3/2)\alpha(k\eta)^{4/3}] \quad (2.23)$$

which is valid in both the inertial and dissipation range in Equation (2.22), Chakravarthy and Menon [2001] obtained the following expression for the eddy viscosity:

$$\nu_t = 0.067 \sqrt{k^{sgs}} \Delta \exp[-(3/2)\alpha(k\eta)^{4/3}]$$

where $\alpha (= 1.4)$ is the Kolmogorov constant, $\eta = \epsilon^{-1/4} \nu^{3/4}$ is the Kolmogorov length scale and ϵ is the total dissipation rate. The contribution of the exponential term

is negligible if k_c lies in the inertial range and hence, an approximate form of the sub-grid eddy viscosity (Chakravarthy and Menon [2001]) is given as

$$\nu_t = 0.067\sqrt{k^{sgs}}\Delta \quad (2.24)$$

The sub-grid dissipation, ϵ^{sgs} is obtained by integrating the dissipation spectrum ($D(k) = -2\nu k^2 E(k)$) over the unresolved wavenumbers (Chakravarthy and Menon [2001]), to get

$$\epsilon^{sgs} = C_\epsilon (k^{sgs})^{3/2} / \overline{\Delta} \quad (2.25)$$

where, $C_\epsilon = 0.916$. Finally the sub-grid production term is modeled as $P^{sgs} = -\tau_{ij}^{sgs}(\partial \tilde{u}_i / \partial x_j)$. The coefficients 0.067 (in the expression for ν_t) and 0.916 (in the expression for ϵ^{sgs}) can also be obtained as a part of the solution by using the dynamical procedure, as shown earlier by Kim et al. [1999]. This is explained in the following section.

2.1.4 Localized dynamic K-equation model - LDKM

The dynamic modeling provides a method to compute the model coefficients C_ν and C_ϵ as a function of the flow conditions. Dynamic models are based on the assumption that there exists a *scale similarity* in the inertial sub-range turbulence. The basic premise of scale similarity is that the largest unresolved sub-grid scales are statistically similar to the smallest resolved scales. To relate the scales near the LES cut-off limit (k_Δ), a test-filter, k_Δ is defined such that ($k_\Delta < k_\Delta$). Note, the grid-filter at the LES cut-off limit operates on the exact equations, whereas a test-filter operates on the grid-filtered (LES) variables. Stresses due to filtering at the grid-filter and the test-filter are defined as follows.

$$\tau_{ij}^{sgs, test} = \overline{\rho \tilde{u}_i \tilde{u}_j} - \frac{\widehat{\rho \tilde{u}_i \tilde{u}_j}}{\widehat{\rho}} \quad (2.26)$$

$$\tau_{ij}^{sgs} = \overline{\rho} (\widetilde{u_i u_j} - \tilde{u}_i \tilde{u}_j) \quad (2.27)$$

Now, it is assumed that the same model (Eqn. [2.20]) for the stresses holds at the original LES filtering level and the test filter level. Also, it is worthwhile to note that the stresses due to test-filtering (Eqn. [2.26]) can be exactly computed in terms of LES filtered variables. Hence, using Eqn. [2.26] and Eqn. [2.20] an expression for the *rms* of the model error is obtained which is then minimized with respect to the model coefficient C_ν to obtain,

$$C_\nu = -\frac{L_{ij}D_{ij}}{2D_{ij}D_{ij}}. \quad (2.28)$$

where,

$$\begin{aligned} L_{ij} &= \widehat{\bar{\rho} \tilde{u}_i \tilde{u}_j} - \frac{\widehat{\bar{\rho} \tilde{u}_i} \widehat{\bar{\rho} \tilde{u}_j}}{\widehat{\bar{\rho}}} - \frac{1}{3} \left(\widehat{\bar{\rho} \tilde{u}_k \tilde{u}_k} - \frac{\widehat{\bar{\rho} \tilde{u}_k} \widehat{\bar{\rho} \tilde{u}_k}}{\widehat{\bar{\rho}}} \right) \delta_{ij} \\ D_{ij} &= 2C_\nu \widehat{\Delta} \widehat{\bar{\rho}} \left(\frac{\widehat{\bar{\rho} \tilde{u}_k \tilde{u}_k}}{2\widehat{\bar{\rho}}} - \frac{\widehat{\bar{\rho} \tilde{u}_k} \widehat{\bar{\rho} \tilde{u}_k}}{2\widehat{\bar{\rho}}^2} \right)^{\frac{1}{2}} \left(\widehat{\tilde{S}_{ij}} - \frac{1}{3} \widehat{\tilde{S}_{kk}} \delta_{ij} \right) \end{aligned} \quad (2.29)$$

Using similar arguments, an expression can be obtained at the test-filter level for the error between the modeled sub-grid dissipation and the exact sub-grid dissipation. While the former is computed using the modeled expression Eqn. [2.25], the latter is computed from the LES filtered quantities. Using this expression for the model error, C_ϵ can be obtained as follows.

$$C_\epsilon = \frac{\widehat{\mu} \widehat{\Delta}}{\widehat{\bar{\rho}}} \frac{\left[\frac{\partial \widehat{\bar{\rho} \tilde{u}_j}}{\partial x_i} \frac{\partial \tilde{u}_j}{\partial x_i} - \frac{\partial}{\partial x_i} \left(\frac{\widehat{\bar{\rho} \tilde{u}_j}}{\widehat{\bar{\rho}}} \right) \frac{\partial}{\partial x_i} \left(\frac{\widehat{\bar{\rho} \tilde{u}_j}}{\widehat{\bar{\rho}}} \right) \right]}{\left(\frac{\widehat{\bar{\rho} \tilde{u}_k \tilde{u}_k}}{2\widehat{\bar{\rho}}} - \frac{\widehat{\bar{\rho} \tilde{u}_k} \widehat{\bar{\rho} \tilde{u}_k}}{2\widehat{\bar{\rho}}^2} \right)^{\frac{3}{2}}} \quad (2.30)$$

where $\widehat{\mu}$ is the test filtered molecular viscosity. More details about LDKM can be found elsewhere (Kim and Menon [1999]).

2.1.5 Scalar transport closure

In addition to τ_{ij}^{sgs} , several unclosed terms appear in the LES filtered energy and species equations given in Eqn. (2.17), such as:

$$\begin{aligned}
H_i^{sgs} &- \text{sub-grid enthalpy flux} \\
\sigma_i^{sgs} &- \text{sub-grid viscous work} \\
Y_{i,k}^{sgs} &- \text{convective species flux} \\
q_{i,k}^{sgs} &- \text{sub-grid heat flux} \\
\theta_{i,k}^{sgs} &- \text{sub-grid species diffusive flux}
\end{aligned} \tag{2.31}$$

The sub-grid total enthalpy flux H_i^{sgs} is also modeled using the eddy viscosity model as follows:

$$H_i^{sgs} = \frac{-\bar{\rho}\nu_t}{Pr_t} \frac{\partial \tilde{H}_i}{\partial x_i}$$

Note that, since large-scale motion is resolved in LES, the associated counter-gradient processes in the resolved scales are also resolved (even though a gradient closure is employed for H_i^{sgs}). The other unclosed terms like σ_i^{sgs} , $q_{i,k}^{sgs}$ and $\theta_{i,k}^{sgs}$, are often neglected and there exists no model for these terms (Poinsot and Veynante [2001]). These terms are the sub-grid contribution of the molecular diffusive flux and are often neglected assuming that their contributions are small in high Reynolds number flows (Fureby [1995], Veynante and Poinsot [1996]).

The sub-grid convective species flux $Y_{i,k}^{sgs}$, given in equation Eqn. [2.17] is modeled using the gradient diffusion assumption as follows.

$$Y_{i,k}^{sgs} = \frac{-\bar{\rho}\nu_t}{Sc_t} \frac{\partial \tilde{Y}_k}{\partial x_i}$$

The coefficient Sc_t is the turbulent Schmidt Number, and is taken to be unity. Nevertheless, theory and experiments (Dimotakis [1989]) have shown that this gradient diffusion assumption for species transport can lead to significant errors.

Most of these assumptions for neglecting the sub-grid terms such as $q_{i,k}^{sgs}$ and $\theta_{i,k}^{sgs}$, and the gradient diffusion modeling for $Y_{i,k}^{sgs}$ are often not justifiable. It will be shown in this thesis, that using the LES-LEM approach (to be described in Chapter 4), most of these assumptions can be relaxed and can be elegantly included in the model for sub-grid scalar transport.

2.1.6 Filtered reaction rate closure

The final term to be modeled in the LES reacting flow equations Eqn. (2.16) is the filtered reaction rate term, $\bar{\dot{w}}_k$. The closure for $\bar{\dot{w}}_k$ is complicated due to its highly non-linear nature. Consider a simple global mechanism $F + O \rightarrow P$, for which the production rate of the product can be expressed as

$$\dot{w}_P = AT^b \exp(-T_A/T) \rho^2 Y_F Y_O$$

where A , b and T_A are pre-exponential factor, temperature index and the activation energy, respectively. Also, ρ , Y_F , Y_O and Y_P are density, mass fraction of fuel, oxidizer and the products, respectively. Filtering the reaction rate term $\dot{w}_P$ gives rise to higher order correlations of the form, $\overline{\rho Y_F'' Y_O''}$, $\overline{\rho Y_F'' T''}$, $\overline{\rho T'' T''}$, etc., that cannot be ignored in a turbulent reacting flow. Hence, proper treatment of this term is a major research issue in LES of reacting flows. Excellent review on the various modeling approaches used to approximate the $\bar{\dot{w}}_k$ in the context of LES is given in Poinso and Veynante [2001]. In the following, few of the conventional approaches used to model $\bar{\dot{w}}_k$ are discussed briefly.

Reaction-rate based on filtered quantities

The simplest approach is to neglect the sub-grid scale fluctuations and assume perfect mixing at the sub-grid scales. The filtered reaction rate can then be expressed purely in terms of the filtered variables as follows.

$$\bar{\dot{w}}_P = A \bar{T}^b \exp(-T_A/\bar{T}) \bar{\rho}^2 \bar{Y}_F \bar{Y}_O$$

This expression implicitly assumes that the turbulent sub-grid time scales are shorter than all the chemical time scales. This may not be true in most combustion applications and has been shown to produce significant errors (Poinso and Veynante [2001]).

Eddy dissipation model

This model attempts to include the sub-grid mixing effects on the assumption that, the combustion process can be assumed to be either kinetics controlled or mixing controlled (depending the relative time-scales between them). As a result, at any instant the reaction rate is computed as the minimum of the kinetic reaction rate and the mixing rate. This is explained as follows

For a general reaction mechanism comprising of N_r reactions given by

$$\sum_{k=1}^{N_s} \nu'_{k,j} M_{k,j} \rightleftharpoons \sum_{k=1}^{N_s} \nu''_{k,j} M_{k,j}$$

the filtered reaction rates for species “k” is computed as follows

$$\bar{\dot{w}}_k = W_k \sum_{j=1}^{N_r} (\nu''_{k,j} - \nu'_{k,j}) \min[q_j^{mix}, q_j^{kin}] \quad (2.32)$$

where q_j^{mix} and q_j^{kin} are the molar reaction rates based on mixing and Arrhenius chemical kinetic rate, respectively. The Arrhenius reaction rates for step “j” is given by,

$$q_j^{kin} = k_{f,j} \prod_{k=1}^{N_s} [M_k]^{\nu''_{k,j}} - k_{b,j} \prod_{k=1}^{N_s} [M_k]^{\nu'_{k,j}}$$

and the mixing rates for the forward and the backward reactions are given by

$$q_{j,f}^{mix} = \frac{1}{\tau_M} \min \left(\frac{[M_k]}{\nu''_{k,j}} \right) \quad \text{and} \quad q_{j,b}^{mix} = \frac{1}{\tau_M} \min \left(\frac{[M_k]}{\nu'_{k,j}} \right)$$

Here, τ_M is the mixing time scale and is related to the sub-grid turbulence as follows

$$\tau_M = C_{EDC} \frac{\bar{\Delta}}{\sqrt{k^{sgs}}}$$

The most severe draw-back of this approach is the use of heuristic non-universal calibration constants. This description of the chemical reaction in turbulent flows is too simple to yield any reasonably good predictions, even though it is computationally very inexpensive.

PDF modeling

PDF modeling, introduced by Pope [1985] has been used for premixed and non-premixed combustion. PDF methods assume that the chemical state of a reacting mixture can be completely defined if the joint PDF of the species and temperature is known. Hence, these models attempt to predict the shape or form of the PDF, from which any quantity such as the reaction rate can be computed as follows.

$$\overline{\dot{\omega}_k} = \int_{Y_1=0}^1 \int_{Y_2=0}^1 \int_{Y_N=0}^1 \int_T \dot{\omega}_k(Y_1, Y_2 \dots Y_N, T) P(Y_1, Y_2 \dots Y_N, T) dY_1 \dots dY_N \quad (2.33)$$

Two most commonly used approaches to describe the PDF $P(Y_1, Y_2 \dots Y_N, T)$ are given below.

Presumed pdf models A given shape is assumed for the PDF based on the observation of mixing and reaction in experiments. The $P(Y_1, Y_2 \dots Y_N, T)$ is commonly assumed to follow an analytical distribution such as a β function distribution, parameterized by the first two moments of the species mass fractions, i.e. its mean and variance. Frankel et al. [1993] reported good qualitative agreement with DNS data for an incompressible reacting mixing layer. However, they also noted significant quantitative disagreement for the species co-variance and mixing layer thickness. These disagreements were primarily attributed to uncertainties associated with the shape of the presumed PDF. In addition it is even harder to presume the shape of a joint PDF parametrized by more than two variables. More limitations on using presumed PDF approach can be found in Goldin [1996], Goldin and Menon [1998].

Transported PDF models An exact transport equation for the PDF is derived for the joint PDF of velocity and composition Pope [1985]. In this equation, the reaction rate term appears in a closed form, whereas the molecular mixing term and the sub-grid turbulent transport terms are unclosed, and hence require modeling. This is the major draw-back of this approach since the accurate modeling of molecular mixing is extremely difficult, if not impossible. Since, molecular mixing and fine scale turbulent convection controls chemical reaction and heat release, inaccuracies in modeling the mixing process gets carried over to the evaluation of the reaction-rate term, and hence affects the over-all solution. Moreover, the numerical solution of PDF equation is generally based on the Monte-Carlo algorithms and is usually very time consuming (Poinsot and Veynante [2001], Peters [2000]). A general methodology that combines LES and transported PDF approach was developed by Gao and O'Brien [1993] for simulating reacting flow problems. Even though this formulation provides a closed form representation for the sub-grid turbulent transport and small scale molecular mixing, there have been no LES study yet, to appraise this model.

Linear-Eddy scalar closure

The Linear Eddy Model (LEM) is a stochastic approach that attempts to simulate turbulent mixing, molecular diffusion and chemical reaction in a one-dimensional domain that is embedded in a turbulent flow. Kerstein [1988a] originally formulated LEM for non-reacting flows and later extended it to reactive flows (Kerstein [1992]). In LEM, the evolution of the scalar field, by molecular diffusion and chemical reaction is solved numerically in a one-dimensional domain. This is followed by a stochastic rearrangement event (Kerstein [1988a]) which represents the action of an individual eddy on a scalar field. LEM has been used with a reasonable success to simulate scalar mixing (McMurtry et al. [1992]), non-premixed combustion (Calhoon [1996], Menon and Calhoon [1996]) and premixed combustion (Smith [1998], Chakravarthy

and Menon [2000]). This sub-grid combustion model forms the crux of this thesis and a complete description of this model for premixed and non-premixed combustion is given in chapter 4.

2.2 The Liquid (Lagrangian) Phase

The dispersed phase has been modeled using both Eulerian and Lagrangian approach in the past. In the Eulerian approach, the dispersed phase is treated as a second continuum, and two sets of equations, one for the gas phase and another for the liquid (pseudo-fluid) phase are solved together to obtain the complete solution. Two-fluid models of Crowe et al. [1997], Druzhinin and Elgobashi. [1998] are based on this approach. On the other hand, in the Lagrangian model (also referred to as Multi-continua formulation by Sirignano [1999]), Newton's equation of motion for the discrete parcels (moving droplet clouds) are solved explicitly to track an average particle in the neighborhood of the Eulerian gas field (Sirignano [1999]). In this approach, the Lagrangian or characteristic path is the trajectory of an average droplet that represents a group of droplets that (at least initially) is in the same neighborhood (Sirignano [1999]). Discrete-particle models of Gosman and Ioannides [1983], Sirignano [1983, 1999], Faeth [1987] are based on this approach. There are advantages and disadvantages to both approaches.

The primary advantage of the two-fluid model is that it is not limited by the number of particles that can be tracked in the system since discrete particles are not tracked. Therefore, for problems involving bubbly flows like cavitation flow modeling, where millions of bubbles need to be tracked, it is advantageous to use Eulerian two-fluid model instead of Lagrangian models (Crowe et al. [1997]). Another advantage of this approach is that the numerical algorithm for the gas phase may be utilized for the pseudo-fluid phase as well. But this model has a major problem with the closure of the particle Reynolds stress and heat flux. The impact of size distribution

on the spray interaction process is lost since the particle size effects are ignored in this approach. Appropriate (and correct) boundary conditions for the pseudo-fluid phase is also a problem. For instance, the application of no-slip condition at the wall for the pseudo-fluid phase is not justified, since the particle velocity need not be zero.

On the other hand Lagrangian models provide detailed information on the particle-fluid interaction (Sirignano [1983, 1999], Faeth [1987]) and do not require any model for the effective viscosity or thermal conductivity due to the inclusion of the particulate phase. Further more, since the Lagrangian schemes have lesser numerical error due to artificial diffusion (Sirignano [1999]), it is preferred over the Eulerian two-fluid models. In addition, application of boundary condition for the particles is relatively straight-forward in Lagrangian models (Crowe et al. [1997]). But this model is limited by the number of particles that can be tracked in the system, as the computational cost rises steeply with the number of particles. Also, the presence of smaller particles in the flow may impose a severe time-step restriction for the integration of Lagrangian equations of particle motion.

Most RANS models used for engineering studies in the past, have employed the Lagrangian approach since the ability to include droplet size effects (which directly impacts evaporation, fuel-air mixing and hence, combustion), is very important for accurate prediction. The present LES approach also employs this model for the very same reason. Studies using this approach within an LES formulation for two-phase mixing layer flows have been demonstrated earlier (Oefelein and Yang [1996], Pannala and Menon [1998]). Solving for the particle evolution in space and time within the gas field can be computationally expensive if a large number of particles have to be tracked. Therefore, a sampling technique is employed, whereby characteristic groups of droplets are represented as “computational parcels”. Each of these computational parcels represents a group of particles, all having identical size, location, velocity and temperature. For completeness, some of the assumptions used in the dispersed phase

modeling are presented here, although more details can be found elsewhere (Faeth [1987]).

2.2.1 Modeling assumptions for the liquid phase

- The spray field simulated here is assumed to be dilute. The dilute spray approximation implies that while particles interact with the gas phase, direct interaction between drops can be neglected. Dilute spray approximation also implies that the volume occupied by the liquid phase is significantly smaller than the volume of the gas phase. Ratio of liquid volume to gas volume is usually of the order of 0.001
- The dilute spray approximation also precludes the droplet breakup effects and coalescence processes (which might be significant in a dense spray situation). This implies that the present model is not suitable for simulating the near field of the liquid atomizers/injectors where the droplet collisions are high. However, the physical processes of interest, such as droplet dispersion, evaporation, fuel-air mixing and combustion, that occur farther away from the injector (where the dilute spray approximation is valid) can be simulated using this model.
- The droplet is assumed to be spherical in this formulation. It is known that moving droplets tend to deform; however, since the drag correlations (used in the present model) implicitly incorporate this effect, and treat the droplets as equivalent spheres, this assumption is reasonable.
- The pressure at the location of the droplet is assumed to be constant during its interaction with the gas phase and equal to the local ambient pressure. This assumption is satisfactory, except at very low pressure.
- Only single component liquid fuels are considered in this study.

- Liquid phase mass diffusivities are assumed to be negligible. This is true because mass diffusion in the liquid phase is very slow compared to with the heat diffusion in the liquid. Moreover for a single component liquid (at a pressure lower than the critical pressure) liquid-phase mass diffusivity can be neglected (Sirignano [1999]).
- The circulation velocity inside the droplet are assumed to be negligible.
- A droplet moving in a turbulent medium is subjected to various external forces such as
 - (1) gravity ($\vec{g}$) and centrifugal force (w_d^2/r_d),
 - (2) Coriolis force ($u_d w_d/r_d$) - Force that causes an apparent deflection of the path of an object that moves within a rotating coordinate system.
 - (4) Basset force - Force due to history effect, which is associated with the correction to the temporal variation of the relative velocity of the particle (Sirignano [1999])
 - (5) virtual mass effect - The force that accounts for the inertia of the gas in the boundary layer and the wake of the droplet (Sirignano [1999])

All such external forces on the droplet are neglected in this study. These approximations are valid, when the liquid-gas density ratio ($\rho_d/\rho \gg 1$) is very high (Faeth [1987], Sirignano [1999]) (which is valid here).

Note that, within the Lagrangian formulation, many of these assumptions can be relaxed using appropriate models. However, since these extensions require empirical correlations (that may not apply to every flow), these effects are ignored at present.

2.2.2 Droplet momentum transport equations

In modeling the spray field, it is assumed that the Kolmogorov scale is of the same order or larger than the largest droplets in the spray field. For such a situation, the

interaction between gas and liquid phases is dominated by laminar fluid dynamics. Given these assumptions, the governing equations for the transport of dispersed phase, (following Oefelein and Yang [1996]) are given by:

$$\frac{dx_{i,d}}{dt} = u_{i,d} \quad (2.34)$$

$$\frac{du_{i,d}}{dt} = \frac{3}{16} \frac{C_D \mu Re_d}{\rho_d r_d^2} (u_i - u_{i,d}) \quad (2.35)$$

Here, $u_{i,d}$ is the i -th component of the droplet velocity, ρ_d is the droplet density and r_d is the radius of the droplet. Re_d is the droplet Reynolds number defined as $Re_d = \sqrt{(u_i - u_{i,d})(u_i - u_{i,d})} d_d / \nu$, where C_D is the drag coefficient. The drag coefficient is obtained from empirical correlations for evaporating droplets (assuming that they retain the spherical shape). This results in the following expression for C_D given by Faeth [1987]:

$$C_D = \begin{cases} \frac{24}{Re_d} (1 + \frac{1}{6} Re_d^{2/3}) & Re_d \leq 1000 \\ 0.424 & Re_d > 1000 \end{cases} \quad (2.36)$$

2.2.3 Droplet mass transfer equations

The equation governing the mass conservation for the droplet is given by

$$\frac{dm_d}{dt} = -\dot{m}_d \quad (2.37)$$

where the mass of the particle is given by $m_d = (4/3)\pi r_d^3 \rho_d$ and $\dot{m}_d$ is the net mass transfer rate for a droplet in a convective flow field given by (Faeth [1987], Lefebvre [1989]):

$$\frac{\dot{m}_d}{\dot{m}_{Re_d=0}} = 1 + [0.278 \sqrt{Re_d} Sc^{(1/3)}] / [1 + \frac{1.232}{Re_d Sc^{(4/3)}}]^{(1/2)} \quad (2.38)$$

Here, $Re_{d=0}$ is the Reynolds number for the droplet at rest. The governing equation for mass transfer under quiescent condition reduces to $\dot{m}_{Re_d=0} = 2\pi \rho_s D_{sm} d_d \ln(1 + B_M)$

where ρ_s and D_{sm} respectively, are the gas mixture density and the mixture diffusion coefficient at the droplet surface. Also, B_M is the Spalding number given by (Chen and Shuen [1993], Lefebvre [1989])

$$B_M = \frac{(Y_{fs} - Y_{f,\infty})}{(1 - Y_{fs})}$$

where W_α is the molecular weight of the gas (excluding fuel vapor), W_f is the molecular weight of the fuel and X_{fs} is the mole-fraction of the fuel at the droplet surface. The mole-fraction X_{fs} is obtained from Raoult's law (Chen and Shuen [1993]), which assumes that the mole-fraction at the droplet surface is equal to the ratio of the partial pressure of the fuel vapor (p_{vp}) to the total pressure. For the present spray calculations, the partial pressure of the fuel vapor is computed based on the following empirical correlation (Reid et al. [1988])

$$\ln\left(\frac{p_{vp}}{p_c}\right) = \frac{1}{(1-x)}[P_{v1}x + P_{v2}x^{1.5} + P_{v3}x^3 + P_{v4}x^6]$$

where $x = 1 - [T/T_c]$, and T_c and p_c are the critical temperature and critical pressure of the fuel vapor, respectively. The coefficients P_{v1} , P_{v2} , P_{v3} , P_{v4} , for a given fuel vapor can be found elsewhere (Reid et al. [1988]).

2.2.4 Droplet temperature equations

As a single droplet enters a hot environment, a small region of the droplet near the surface will heat up quickly while the central core of the droplet remains “cold”. The heat conduction to its interior increases as the droplet penetrates further into the hot ambient gas. Eventually, the temperature within the droplet will become nearly uniform before the end of its lifetime. To solve this transient phenomena within a droplet is not trivial, and various models have been proposed (Faeth [1987]) of which the simplest is the uniform temperature model. In this model, the thermal conductivity is assumed infinite inside the liquid and therefore, the bulk liquid temperature is same as the surface temperature. However, the species mass fraction at the surface is allowed to be different from the bulk liquid and the ambient condition.

The equation governing the internal temperature distribution based on this *uniform temperature model* is

$$m_d C_L \frac{dT_d}{dt} = h_d \pi d_d^2 (\tilde{T} - T_d) - \dot{m}_d L_v \quad (2.39)$$

The heat transfer coefficient, h_d for a droplet in a convective flow field with mass transfer is modeled following the proposed correlation of Faeth and Lazar [1971] :

$$\frac{h_d}{h_{Re_d=0}} = 1 + 0.278 Re_d^{1/2} Pr^{1/3} / [1 + \frac{1.232}{Re_d Pr^{4/3}}]^{1/2}$$

Here, the heat transfer coefficient at quiescent condition, $h_{Re_d=0} = \frac{\kappa Nu_{Re_d=0}}{d_d}$

$Nu_{Re_d=0}$ is the Nusselt's number at quiescent conditions given by

$Nu_{Re_d=0} = 2 \ln(\frac{\alpha}{\alpha-1})$, where $\alpha = (1 + B_M)^{Le^{-1}}$ and Le is the Lewis number of the gas mixture evaluated using the Schmidt number and Prandtl number as $Le = Sc/Pr$.

In Eqn. 2.39, L_v is the latent heat of vaporization defined as (Miller and Bellan [1999])

$$L_v = h_{v,s} - h_{L,s}$$

$$h_{v,s} = h_v^o + \int_{T_o}^{T_d} C_{p,v}(T') dT' \quad \text{and} \quad h_{L,s} = h_L^o + \int_{T_o}^{T_d} C_L(T') dT'$$

where

$h_{v,s}$ = enthalpy of the evaporated vapor at the droplet surface temperature, T_d

$h_{L,s}$ = enthalpy of the liquid at the droplet surface temperature, T_d

h_v^o = reference enthalpy of the vapor at the reference temperature, T_o

h_L^o = reference enthalpy of the liquid at the reference temperature, T_o

upon simplification, this yields

$$L_v = (h_v^o - h_L^o) + \int_{T_o}^{T_d} (C_{p,v}(T') - C_L(T')) dT'$$

The reference enthalpy of the liquid is taken to be zero at $T_o = 0$ i.e., $h_L^o = 0$ (Miller and Bellan [1999]). In addition for a calorically perfect gas the above expression for L_v reduces to

$$L_v = h_v^o + (C_{p,v} - C_L) T_d \quad (2.40)$$

2.2.5 Continuum/Dispersed phase coupling

Coupling between the dispersed phase and the gas phase are provided by the inter-phase exchange terms or the source terms that appear on the right hand side of the Eqns. [2.16] These are computed as follows (Oefelein and Yang [1996], Faeth [1987], Chen and Shuen [1993]):

$$\begin{pmatrix} \dot{\rho}_s \\ \dot{F}_{s,i} \\ \dot{Q}_s \\ \dot{S}_{s,k} \end{pmatrix} = - \begin{pmatrix} \frac{dm_d}{dt} \\ \frac{dm_d u_i}{dt} \\ \frac{dm_d e_d}{dt} \\ \frac{dm_d Y_k}{dt} \end{pmatrix} = - \begin{pmatrix} \rho_d \frac{dV_d}{dt} + V_d \frac{d\rho_d}{dt} \\ m_d \frac{du_{i,d}}{dt} + u_{i,d} \frac{dm_d}{dt} \\ m_d \frac{de_d}{dt} + e_d \frac{dm_d}{dt} \\ m_d \frac{dY_{k,d}}{dt} + Y_{k,d} \frac{dm_d}{dt} \end{pmatrix}$$

In all the expressions that follow, subscript d refers to droplet quantities. Here, ρ_d , V_d , $u_{i,d}$, e_d , and $Y_{k,d}$ are the density, volume, “i-th” component of the velocity, total energy and species mass fraction of the fuel in the droplets, respectively.

The following relations are used to further simplify the above equations.

droplet mass, $m_d = \rho_d V_d$

droplet volume, $V_d = \frac{4\pi r_d^3}{3}$

droplet internal energy, $e_d = C_L T_d$, where C_L is the liquid heat capacity

droplet evaporation rate, $\frac{dm_d}{dt} = -\dot{m}_d$

where $\dot{m}_d$ is computed using Eqn. (2.38). Further, simplifications can be made by realizing that only single component fuels are considered in this study. This implies that the mass fraction of fuel species in the liquid droplet is always unity, i.e., $Y_{k,d} = 1$ and $\frac{dY_{k,d}}{dt} = 0$.

Source terms for the droplet energy equations are simplified as follows

$$\frac{dm_d e_d}{dt} = m_d \frac{de_d}{dt} + e_d \frac{dm_d}{dt} = \underbrace{m_d C_L \frac{dT_d}{dt}}_A + \dot{m}_d C_L T_d$$

Using Eqn. (2.39) and Eqn. (2.40) to replace the term A in the above equation yields

$$\frac{dm_d e_d}{dt} = h_d \pi d_d^2 (\tilde{T} - T_d) - \dot{m}_d (h_v^o + (C_{p,v} - C_L) T_d) - \dot{m}_d C_L T_d$$

Noting that the enthalpy of the evaporated vapor at the droplet surface is given by, $h_{v,s} = h_v^o + C_{p,v} T_d$ (Miller and Bellan [1999]), the final expressions for the source term of a single droplet is given by

$$\begin{pmatrix} \dot{\rho}_s \\ \dot{F}_{s,i} \\ \dot{Q}_s \\ \dot{S}_{s,k} \end{pmatrix} = \begin{pmatrix} \dot{m}_d \\ \dot{m}_d u_{i,d} - \frac{4\pi}{3} \rho_d r_d^2 \frac{du_{i,d}}{dt} \\ \dot{m}_d h_{v,s} - h_d \pi d_d^2 (\tilde{T} - T_d) \\ \dot{m}_d \end{pmatrix}$$

At this juncture,

- (1) Note that, all the equations and the droplet models stated so far corresponds to that of a single droplet moving in a flow-field.
- (2) It should be recalled from Section 2.2 that groups of particles (computational parcels), rather than individual particles are tracked in this approach. Therefore, the inter-phase exchange terms should be multiplied by the number of particles present per group for mass conservation. This approach, besides allowing for a statistically significant number of particles to be tracked, also helps to control the mass flow rate of the liquid.

Therefore, if n_d number of particles are present per group, then the **volume averaged** source terms for all the droplet group trajectories that cross a computational cell is computed by summing the contribution from every group as follows.

$$\begin{pmatrix} \widetilde{\dot{\rho}_s} \\ \widetilde{\dot{F}_{s,i}} \\ \widetilde{\dot{Q}_s} \\ \widetilde{\dot{S}_{s,k}} \end{pmatrix} = \begin{pmatrix} \sum_m n_d \dot{m}_d \\ \sum_m n_d [\dot{m}_d u_{i,d} - \frac{4\pi}{3} \rho_d r_d^2 \frac{du_{i,d}}{dt}] \\ \sum_m n_d [\dot{m}_d h_{v,s} - h_d \pi d_d^2 (\tilde{T} - T_d)] \\ \sum_m n_d \dot{m}_d \end{pmatrix}$$

where the summation index “ m ” is over all the droplet groups crossing a computational cell volume.

2.2.6 Time-step restriction for two-phase flow

Inclusion of spray introduces new time scales in the flow. To accurately calculate the particle trajectories, size and temperature, the Lagrangian time step used for the integration has to be the smallest of the various physical time scales. At any instant, the smallest of the following time scales is used for particle evolution (Faeth [1987], Chen and Shuen [1993], Oefelein and Yang [1996]).

- (1) LES gas phase CFL time
- (2) droplet velocity relaxation time
- (3) droplet life time
- (4) turbulent eddy droplet interaction time
- (5) droplet surface temperature constraint time

Droplet velocity relaxation time: The local linearized droplet equations of motion of a particle in a uniform flow have an exact solution in terms of the droplet Reynolds’s number, which gives the following expression for the *droplet relaxation time*,

$$\tau_{relax} = [16\rho_d r_d^2 (C_D Re_d)^{-1}] / [3\rho_g \nu] \quad (2.41)$$

Droplet life time: In flows involving vaporizing droplets, to ensure that the droplet size remains positive in a given time step, *droplet life time* is estimated based on the evaporation rate of the droplet as follows:

$$\tau_{life} = \frac{4\pi r_d^3 \rho_d}{3\dot{m}_d}$$

Eddy-droplet interaction time: A particle is assumed to interact with an eddy for a time which is the minimum of eddy life time and the transit time for the

particle to traverse the eddy. The *eddy life time* is determined by assuming that the characteristic size of a randomly sampled eddy is the dissipation length scale, proportional to the LES filter width, Δ and is given by

$$\tau_{eddy} = \Delta / \sqrt{(2k^{sgs}/3)}$$

and the *transit time* is determined from the following equation

$$\tau_{transit} = \tau_{relax} \ln\left(1 - \frac{\Delta}{(\tau_{relax}|u_i - u_{i,d}|)}\right).$$

Therefore, the *Eddy-droplet interaction time* is estimated as,

$$\tau_{int} = \begin{cases} \tau_{eddy} & , \quad \Delta > \tau_{relax}|u_i - u_{i,d}| \\ \min(\tau_{eddy}, \tau_{transit}), & \Delta \leq \tau_{relax}|u_i - u_{i,d}| \end{cases}$$

Droplet surface temperature constraint or heating time: If the boiling temperature of the fuel is very low or close to the room temperature, then the droplet surface temperature rise becomes another constraint. If the Lagrangian time-step is large, then the droplet temperature rises abruptly, leading to an increased evaporation and mass loading locally. This might affect the stability of the calculation. For the uniform temperature model, the *droplet heating time* is thus given by Chen and Shuen [1993]

$$\tau_{evap} = (\rho_d C_v d_d) / (6h_d)$$

Drop/Turbulence interaction: Ignoring the effect of (sub-grid) turbulence on the drop is not realistic since the droplet motion are limited to deterministic trajectories prescribed by their initial condition and mean gas properties (Faeth [1996]). As a result, predictions from such an approach (usually named as Deterministic Separated Flow model) exhibit unphysical, laminar like behavior where drops collect in regions of low velocity and which leads to low turbulent mixing (Faeth [1996]).

Motivated by the preceding discussion, the effect of turbulence on the droplet motion is simulated using the Stochastic Separated Flow model (Faeth [1987], Gosman and Ioannides [1983]). The stochastic dispersion of droplets due to turbulent motion is incorporated by representing the gas phase velocity at a particle location as $u_i = \tilde{u}_i + X\sqrt{2k^{sgs}/3}$, where X is a random number sampled from a uniform distribution (with zero mean). Note that, particles are not restricted to lie on the Eulerian grid points where the gas-phase properties are known. Therefore, an interpolation function S , of the following general form

$$\phi(x_d) = \sum_c S(x_d, x_c)\phi(x_c)$$

is used to estimate the gas phase velocities at the particle location. Here, x_c is the Eulerian grid cell center and ϕ is any gas phase property known at the position x_c . An eight-point, volume-weighted averaging of the adjacent cells is used to interpolate the gas phase properties to the droplet locations, and also for the redistribution of the spray source terms from the particle position to the Eulerian grid (gas phase).

CHAPTER III

NUMERICAL METHOD

*"In order to solve the differential equation
you look at it till a solution occurs to you"*

-Polyaccute George

In this chapter the numerical implementation of the governing equations is presented. The conservation laws of fluid motion may be expressed mathematically in either differential form or integral form. When a numerical scheme is applied to the differential equation, the domain of solution is divided into discrete points, upon which the finite difference equations are solved. On the other hand, when the integral form of the equations are utilized, the domain of solution is divided into small volumes. Subsequently, the conservation laws in the integral form are applied to these elementary volumes. In the integral form, the conservation is satisfied in every infinitesimal volume in the solution domain, whereas in the discretized integral form, the strict conservation is satisfied only up to the discretized elemental volumes.

The finite-volume approach has several advantages compared to finite-difference schemes. Finite-volume schemes can handle skewed grid/meshes as long as the cell face fluxes are adequately reconstructed. Second, finite volume schemes provide greater flexibility in terms of available choices for the selection of discrete volumes. Hence, the geometrical boundaries can be truly reconstructed.

3.1 Finite-Volume Scheme

Consider the governing equations written in Cartesian co-ordinates

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} + \frac{\partial \mathbf{G}}{\partial y} + \frac{\partial \mathbf{H}}{\partial z} = \Phi \quad (3.1)$$

where $\mathbf{U}$ is the state vector and $\mathbf{F}$, $\mathbf{G}$ and $\mathbf{H}$ are the flux vectors in the X , Y and Z directions respectively. Φ represents the vector of source terms. Integrating Eqn. 3.1 over a control volume such as the one shown in fig 3.1 gives,

$$\int_V \frac{\partial \mathbf{U}}{\partial t} dV + \int_V \left(\frac{\partial \mathbf{F}}{\partial x} + \frac{\partial \mathbf{G}}{\partial y} + \frac{\partial \mathbf{H}}{\partial z} \right) dV = \int_V \Phi dV \quad (3.2)$$

Volume integration and the time differentiation operators are commutative for stationary grids, hence the time derivative can be taken outside the volume integral. Applying Green's Theorem, to convert the volume integral to surface integral, Eqn. [3.2] can be written as

$$\frac{\partial}{\partial t} \int_V \mathbf{U} dV + \oint_S (\mathbf{F} dS_x + \mathbf{G} dS_y + \mathbf{H} dS_z) = \int_V \Phi dV \quad (3.3)$$

The vectors $\mathbf{U}$, $\mathbf{F}$, $\mathbf{G}$, $\mathbf{H}$ and Φ are given by,

$$\mathbf{U} = \begin{pmatrix} \bar{\rho} \\ \bar{\rho} \tilde{u} \\ \bar{\rho} \tilde{v} \\ \bar{\rho} \tilde{w} \\ \bar{\rho} \tilde{E} \\ \bar{\rho} k^{sgs} \\ \bar{\rho} \tilde{Y}_k \end{pmatrix} \quad \mathbf{\Phi} = \begin{pmatrix} \tilde{\hat{\rho}}_s \\ \widetilde{\dot{F}_{s,1}} \\ \widetilde{\dot{F}_{s,2}} \\ \widetilde{\dot{F}_{s,3}} \\ \widetilde{\dot{Q}_s} \\ P^{sgs} - D^{sgs} + \widetilde{\dot{F}_{s,1}} + \widetilde{\dot{F}_{s,2}} + \widetilde{\dot{F}_{s,3}} \\ \widetilde{\dot{\omega}_k} \end{pmatrix} \quad (3.4)$$

In the vector Φ , all terms with subscript “s” are the source terms from the spray.

$$\mathbf{F} = \begin{pmatrix} \bar{\rho} \tilde{u} \\ \bar{\rho} \tilde{u}\tilde{u} + \bar{p} \\ \bar{\rho} \tilde{u}\tilde{v} \\ \bar{\rho} \tilde{u}\tilde{w} \\ \bar{\rho} \tilde{u}(\tilde{E} + \bar{p}) \\ \bar{\rho} \tilde{u}k^{sgs} \\ \bar{\rho} \tilde{u}\tilde{Y}_k \end{pmatrix} - \begin{pmatrix} 0 \\ \tilde{\tau}_{xx} \\ \tilde{\tau}_{xy} \\ \tilde{\tau}_{xz} \\ \tilde{u}\tilde{\tau}_{xx} + \tilde{v}\tilde{\tau}_{xy} + \tilde{w}\tilde{\tau}_{xz} - \tilde{q}_x \\ 0 \\ \bar{\rho} D_k \frac{\partial \tilde{Y}_k}{\partial x} \end{pmatrix} + \begin{pmatrix} 0 \\ \tau_{xx}^{sgs} \\ \tau_{xy}^{sgs} \\ \tau_{xz}^{sgs} \\ H_x^{sgs} \\ K_x^{sgs} \\ Y_{1,k}^{sgs} + \theta_{1,k}^{sgs} \end{pmatrix}$$

$$\mathbf{G} = \begin{pmatrix} \bar{\rho} \tilde{v} \\ \bar{\rho} \tilde{v}\tilde{u} \\ \bar{\rho} \tilde{v}\tilde{v} + \bar{p} \\ \bar{\rho} \tilde{v}\tilde{w} \\ \bar{\rho} \tilde{v}(\tilde{E} + \bar{p}) \\ \bar{\rho} \tilde{v}k^{sgs} \\ \bar{\rho} \tilde{v}\tilde{Y}_k \end{pmatrix} - \begin{pmatrix} 0 \\ \tilde{\tau}_{yx} \\ \tilde{\tau}_{yy} \\ \tilde{\tau}_{yz} \\ \tilde{u}\tilde{\tau}_{yx} + \tilde{v}\tilde{\tau}_{yy} + \tilde{w}\tilde{\tau}_{yz} - \tilde{q}_y \\ 0 \\ \bar{\rho} D_k \frac{\partial \tilde{Y}_k}{\partial y} \end{pmatrix} + \begin{pmatrix} 0 \\ \tau_{yx}^{sgs} \\ \tau_{yy}^{sgs} \\ \tau_{yz}^{sgs} \\ H_y^{sgs} \\ K_y^{sgs} \\ Y_{2,k}^{sgs} + \theta_{2,k}^{sgs} \end{pmatrix}$$

$$\mathbf{H} = \underbrace{\begin{pmatrix} \bar{\rho} \tilde{w} \\ \bar{\rho} \tilde{w}\tilde{u} \\ \bar{\rho} \tilde{w}\tilde{v} \\ \bar{\rho} \tilde{w}\tilde{w} + \bar{p} \\ \bar{\rho} \tilde{w}(\tilde{E} + \bar{p}) \\ \bar{\rho} \tilde{w}k^{sgs} \\ \bar{\rho} \tilde{w}\tilde{Y}_k \end{pmatrix}}_{\text{Inviscid-Flux}} - \underbrace{\begin{pmatrix} 0 \\ \tilde{\tau}_{zx} \\ \tilde{\tau}_{zy} \\ \tilde{\tau}_{zz} \\ \tilde{u}\tilde{\tau}_{zx} + \tilde{v}\tilde{\tau}_{zy} + \tilde{w}\tilde{\tau}_{zz} - \tilde{q}_z \\ 0 \\ \bar{\rho} D_k \frac{\partial \tilde{Y}_k}{\partial z} \end{pmatrix}}_{\text{Viscous-Flux}} + \underbrace{\begin{pmatrix} 0 \\ \tau_{zx}^{sgs} \\ \tau_{zy}^{sgs} \\ \tau_{zz}^{sgs} \\ H_z^{sgs} \\ K_z^{sgs} \\ Y_{3,k}^{sgs} + \theta_{3,k}^{sgs} \end{pmatrix}}_{\text{Sub-grid-Flux}}$$

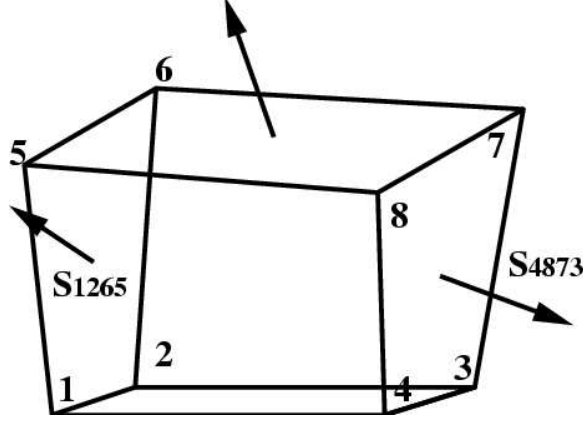


Figure 3.1: Finite-volume cell.

3.2 Spatial Discretization

A grid/mesh is defined by choosing a set of discrete points and connections among them. The grid divides the region bounded by the physical boundaries into a set of contiguous cells, where the grid points are now cell vertices and the connections form the edges of the faces common to the neighboring cells. For a structured grid like the one used in the present study, the cells are hexahedrons with eight vertices and six faces in three dimensions.

It is convenient and a common practice to transform the Navier-Stokes equation from the physical (x, y, z) space to the computational (ξ, η, ζ) space using a simple application of the chain rule such as,

$$\begin{aligned}\frac{\partial}{\partial x} &= \xi_x \frac{\partial}{\partial \xi} + \eta_x \frac{\partial}{\partial \eta} + \zeta_x \frac{\partial}{\partial \zeta} \\ \frac{\partial}{\partial y} &= \xi_y \frac{\partial}{\partial \xi} + \eta_y \frac{\partial}{\partial \eta} + \zeta_y \frac{\partial}{\partial \zeta} \\ \frac{\partial}{\partial z} &= \xi_z \frac{\partial}{\partial \xi} + \eta_z \frac{\partial}{\partial \eta} + \zeta_z \frac{\partial}{\partial \zeta}\end{aligned}$$

The coordinate transformation introduced here follows the development of Vinokur [1989]. The transformations are chosen so that the grid spacing in the transformed space is uniform and of unit length. This produces a computational space, $(\xi, \eta, \text{ and } \zeta)$ which is a rectangular domain with a regular uniform mesh so that

standard unweighted differencing schemes can be used in the numerical formulation. The original Cartesian space (x,y and z) will be referred to as the physical domain. Typically there will be a one-to-one correspondence between a physical point in space and a computational point, except for regions where there are singularities or cuts due to the topology. With this construction, a single computational code can be used for a wide variety of physical geometries and grid systems. Transforming the governing equations from the physical space to the computational space introduces the transformation derivatives known as metrics. Metrics represent the ratio of arc lengths in the computational space to that of the physical space. In most cases, the metric relations are not known analytically, rather they are generated numerically. Usually, only x, y and z coordinates of grid points are provided and the grid metrics are generated numerically using finite differences approximations.

Using grid metrics, Eqn. [3.1] can be written as

$$\frac{\partial \mathbf{U}'}{\partial t} + \frac{\partial \mathbf{F}'}{\partial \xi} + \frac{\partial \mathbf{G}'}{\partial \eta} + \frac{\partial \mathbf{H}'}{\partial \zeta} = \Phi \quad (3.5)$$

where

$$\mathbf{U}' = \frac{1}{J}[\mathbf{U}]$$

$$\mathbf{F}' = \frac{1}{J}[\xi_x \mathbf{F} + \xi_y \mathbf{G} + \xi_z \mathbf{H}]$$

$$\mathbf{G}' = \frac{1}{J}[\eta_x \mathbf{F} + \eta_y \mathbf{G} + \eta_z \mathbf{H}]$$

$$\mathbf{H}' = \frac{1}{J}[\zeta_x \mathbf{F} + \zeta_y \mathbf{G} + \zeta_z \mathbf{H}]$$

where, J is the Jacobian of the transformation and is given by

$$J = \frac{\partial(\xi, \eta, \zeta)}{\partial(x, y, z)} = [x_\xi y_\eta z_\zeta + x_\zeta y_\xi z_\eta + x_\eta y_\zeta z_\xi - x_\xi y_\zeta z_\eta - x_\eta y_\xi z_\zeta - x_\zeta y_\eta z_\xi]^{-1}$$

J can also be interpreted as the ratio between the volumes in the computational space and in the physical space.

A semi-discrete finite-volume formulation of the Eqn. [3.5] is written as

$$\begin{aligned}
\frac{\partial}{\partial t}(\frac{\mathbf{U}'}{J})_{ijk} + (\mathbf{F}'_{i+\frac{1}{2},j,k} - \mathbf{F}'_{i-\frac{1}{2},j,k}) + \\
(\mathbf{G}'_{i,j+\frac{1}{2},k} - \mathbf{G}'_{i,j-\frac{1}{2},k}) + \\
(\mathbf{H}'_{i,j,k+\frac{1}{2}} - \mathbf{H}'_{i,j,k-\frac{1}{2}}) = \\
\Phi_{ijk}
\end{aligned} \tag{3.6}$$

In the above equation, the spatial indices (i, j, k) is used to refer a cell or cell center. Fractional subscripts such as $(i + \frac{1}{2}, j, k)$ are used to indicate cell faces. The state vector variables, $\mathbf{U}'$ normally refer to the average value of $\mathbf{U}$ in that cell at some average point in the cell. Since the precise location of this average point is not needed in a finite volume scheme (Vinokur [1989]), it is preferable to define this point as the cell-center (average of the cell vertices), which is easier to calculate. Φ_{ijk} is the volume averaged source term in a cell ijk .

Eqn. [3.6] is a consistent approximation of the integral form of the conservation law, given by the Eqn.[3.3], since the surface integral in Eqn.[3.3] can be written as the sum of the contributions of the flux from six faces of the hexagonal cells. Figure 3.1 shows a typical hexahedron with surface area vector S_{1265} and S_{4873} pointing in (outward normal) positive directions. The directed area is calculated as one-half of the vector cross product of the two diagonal vectors connecting the opposite vertex points of a cell face.

3.3 Explicit MacCormack Scheme

Eqn. [3.6] is solved using a MacCormack explicit predictor-corrector scheme that is second-order accurate in time and fourth-order accurate in space. The MacCormack explicit predictor-corrector scheme is:

$$\mathbf{U}_i^* = \mathbf{U}_i^n -$$

$$\frac{\Delta t}{\Delta V} \left[(\mathbf{F}_{i+1/2,j,k}^{+(n)} - \mathbf{F}_{i-1/2,j,k}^{+(n)}) + (\mathbf{G}_{i,j+1/2,k}^{+(n)} - \mathbf{G}_{i,j-1/2,k}^{+(n)}) + (\mathbf{H}_{i,j,k+1/2}^{+(n)} - \mathbf{H}_{i,j,k-1/2}^{+(n)}) \right]$$

$$\mathbf{U}_i^{n+1} = \frac{1}{2}(\mathbf{U}_i^n + \mathbf{U}_i^*) -$$

$$\frac{\Delta t}{2\Delta V} \left[(\mathbf{F}_{i+1/2,j,k}^{-(*)} - \mathbf{F}_{i-1/2,j,k}^{-(*)}) + (\mathbf{G}_{i,j+1/2,k}^{-(*)} - \mathbf{G}_{i,j-1/2,k}^{-(*)}) + (\mathbf{H}_{i,j,k+1/2}^{-(*)} - \mathbf{H}_{i,j,k-1/2}^{-(*)}) \right]$$

where,

n denotes discrete time levels

$*$ denotes the predicted value

$n + 1$ denotes the corrected value

"+" and "-" signs denote forward and backward differencing, respectively.

In finite-volume schemes, the flux at the cell face is constructed from the known values at the cell center. Therefore, an extrapolation procedure is required. Simplest extrapolation that yields second order spatial accuracy is given by

Forward	Backward	
$F_{i+1/2,j,k}^+ = F_{i+1,j,k}$	$F_{i+1/2,j,k}^- = F_{i-1,j,k}$	(3.7)
$G_{i,j+1/2,k}^+ = G_{i,j+1,k}$	$G_{i,j+1/2,k}^- = G_{i,j-1,k}$	
$H_{i,j,k+1/2}^+ = H_{i,j,k+1}$	$H_{i,j,k+1/2}^- = H_{i,j,k-1}$	

This sequence of forward-backward differencing is reversed every other time step to prevent a bias in the solution. In 3-D this leads to 2^3 variants of the scheme, each with

the same order of the accuracy. Higher order accuracy can be obtained by including more terms (i.e., by increasing the stencil size) used in the extrapolation. A fourth order accurate numerical scheme proposed by Nelson [1997] that is used in the present study is written as:

$$\begin{aligned}\mathbf{F}_{i+\frac{1}{2}}^+ &= \frac{1}{6} (2 \mathbf{F}_i + 5 \mathbf{F}_{i+1} - \mathbf{F}_{i+2}) \\ \mathbf{F}_{i+\frac{1}{2}}^- &= \frac{1}{6} (2 \mathbf{F}_{i+1} + 5 \mathbf{F}_i - \mathbf{F}_{i-1}).\end{aligned}\tag{3.8}$$

For stretched grids, interpolation of inviscid fluxes at the face is improved by accounting directly for the grid stretching in the interpolation formula (Nelson [1997]). Thus, derivatives for the primitive variables such as u, v, w, T, P , etc. are computed as

$$\begin{aligned}\frac{\partial u^+}{\partial \xi} &= \frac{1}{6} (-u_{i+2} + 8u_{i+1} - 7u_i) \\ \frac{\partial u^-}{\partial \xi} &= \frac{1}{6} (u_{i-1} + 8u_i - 7u_{i+1})\end{aligned}$$

Transverse derivatives on the i_{face} (face normal to the ξ direction) are computed using the interpolated $i - face$ values and a fourth-order central difference formula as follows.

$$\frac{\partial u}{\partial \eta} = \frac{1}{12} (-u_{j+2} + 8(u_{j+1} - u_{j-1}) + u_{j-2}).$$

Grid metrics are computed using the following finite difference approximations, for example on an $i - face$,

$$\frac{\partial x}{\partial \xi} = \frac{1}{12} (-x_{i+2} + 15(x_{i+1} - x_{i-1}) + x_{i-2})\tag{3.9}$$

Time-Step Restriction

The MacCormack explicit scheme is subjected to stability requirements. For simple wave equations, stability analysis can be performed to yield conditions for time-steps that would provide consistent solutions. But for viscous, three dimensional equations, derivation of the stability condition leads to a quite complicated expression. No analytically derived stability condition is known for the mixed partial differential

equations (Navier-Stokes eqns.) (MacCormack [1969]). Hence, based on physical arguments and numerical experimentation, the following requirement has been reported in the literature (MacCormack [1969]).

$$\Delta t = \frac{1}{\left[\frac{|\mathbf{u}|}{\Delta x} + \frac{|\mathbf{v}|}{\Delta y} + \frac{|\mathbf{w}|}{\Delta z} + a \sqrt{\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2} + \frac{1}{\Delta z^2}} + 2 \left(\frac{4}{3} \frac{\gamma \nu}{Pr} \right) \left(\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2} + \frac{1}{\Delta z^2} \right) \right]}$$

where a is the speed of sound, γ is the ratio of specific heat capacities, Pr is the Prandtl number and $\Delta x, \Delta y, \Delta z$ are the local grid spacing in x, y, and z directions.

3.4 Boundary Conditions

A set of information with regard to the the dependent variable and/or its derivative must be specified along the boundaries of the domain. This set of information, known as *boundary conditions* (B.C) may be categorized as follows.

- Dirichlet B.C: The value of the dependent variables is specified. Dirichlet B.C. are typically associated with problems involving inflow phenomena and isothermal walls.
- Neumann B.C: The normal gradient of the dependent variable is specified. Neumann B.C. are often associated with symmetry boundaries, adiabatic walls or non-reactive walls.
- Mixed B.C: A combination of the Dirichlet and Neumann type B.C is specified.

In a finite volume formulation, flow variables are defined on the cell center. But physical boundaries such as wall usually coincide with the cell faces. Hence, for the purpose of specifying the B.C at the correct location, “ghost cells” are defined. These ghost cells have the same geometric properties as that of the interior cell, adjacent to the physical boundaries. Boundary conditions are set at these ghost cells to obtain physically correct conditions on the domain physical boundary. The boundary conditions at the wall, symmetry and at the periodic boundaries are specified as follows.

- At walls, no slip conditions are prescribed such that the normal and the tangential velocities at the wall are equal to zero.
- For slip or symmetry conditions, values are prescribed for ghost cells such that velocities normal to the face at the boundary are zero.
- Adiabatic wall implies zero gradient for temperature. Hence the ghost cell is set a value equal to the temperature of the cell adjacent to the adiabatic wall. For non-catalytic walls, equivalent conditions are specified for species mass fractions.
- Periodic boundary conditions are prescribed as follows. Suppose the domain is bounded by $[x(1, j), x(imax, j)]$ in one dimension. The first cell is $i = 1$ and its ghost cell is $i = 0$. Similarly the last cell is at $i = imax - 1$ and its ghost cell is $i = imax$. Then the periodic b.c implies $u(imax, j) = u(1, j)$ and $u(0, j) = u(imax - 1, j)$.

3.5 Inflow/Outflow Boundary Conditions

Inflow and outflow boundaries are much more complicated and require special attention in order to prescribe physically meaningful boundary conditions. In a subsonic flow, at any boundary, some waves are propagating into the computational domain and some are propagating out of it (Thompson [1987]). The outward propagating waves have their behavior defined entirely by the solution at and within the boundary. The inward propagating waves depend on solution/flow conditions exterior to the computational volume and therefore require boundary conditions to completely specify their behavior. Hence, in both inflow and outflow boundaries, a reduced set of governing equations, recast into characteristic form normal to the boundary are solved simultaneously with the interior domain solution procedure.

Characteristic Boundary Conditions

The wave like character, possessed by the compressible Navier-Stokes equations, allows for the identification of characteristic waves whose direction and magnitude of propagation can be determined from interior information.

Consider the model equation for one-dimensional Euler equation

$$\frac{\partial Q}{\partial t} + \frac{\partial(AQ)}{\partial x} = 0 \quad (3.10)$$

Assuming that A is a constant coefficient matrix, Eqn. [3.10] can be diagonalized using the eigenvector matrix X , so that

$$\frac{\partial(X^{-1}Q)}{\partial t} + \Lambda_A \frac{\partial(X^{-1}Q)}{\partial x} = 0 \quad \text{or} \quad \partial W / \partial t + \mathcal{L} = 0 \quad (3.11)$$

where $W = X^{-1}Q$, and $\mathcal{L}$ represents the time variation of the characteristic wave amplitude, W at a given location. The characteristic speed Λ_A is given by

$$\Lambda_A = \begin{pmatrix} u & 0 & 0 \\ 0 & u + a & 0 \\ 0 & 0 & u - a \end{pmatrix}$$

where a is the speed of sound. Thus, from Eqn. [3.11] it can be seen that for a subsonic in-flow at the left boundary of the flow domain, two characteristic speeds $u, u + a$ are positive, implying incoming characteristic waves. Hence, two numerical boundary conditions must be specified at the inflow. The third characteristic, for which $u - a$ is negative, is handled such that its value is not constrained, i.e., it is determined from the interior solution. Similarly, for an outflow situated at the right boundary of the flow domain, two characteristics $(u, u + a)$ leave the domain, while one characteristic $(u - a)$ enter the domain. Therefore, one numerical boundary condition is needed at the outflow for this problem. Therefore, if the $\mathcal{L}$ can be estimated at the boundaries, the solution for the following time-step at the boundaries can be computed by solving Eqn. [3.11].

In the present study, a similar approach is used for three dimensional compressive Navier-Stokes equation based on the technique prescribed by Poinso and Lele [1992]. Using the characteristic analysis (Poinso and Lele [1992]), to modify the hyperbolic terms, the governing Navier-Stokes equation can be recast into the system of following equations.

$$\begin{aligned}
\frac{\partial \bar{\rho}}{\partial t} + d_1 \frac{\partial(\bar{\rho}\tilde{u}_2)}{\partial x_1} + \frac{\partial(\bar{\rho}\tilde{u}_3)}{\partial x_2} &= 0 \\
\frac{\partial(\bar{\rho}\tilde{u}_1)}{\partial t} + \tilde{u}_1\tilde{d}_1 + \bar{\rho}\tilde{d}_3 + \frac{\partial\bar{\rho}\tilde{u}_1\tilde{u}_2}{\partial x_2} &= \frac{\partial\tau_{1j}}{\partial x_j} \\
\frac{\partial(\bar{\rho}\tilde{u}_2)}{\partial t} + \tilde{u}_2\tilde{d}_1 + \bar{\rho}\tilde{d}_4 + \frac{\partial\bar{\rho}\tilde{u}_2\tilde{u}_2}{\partial x_2} + \frac{\partial\bar{\rho}\tilde{u}_3\tilde{u}_2}{\partial x_3} + \frac{\partial\bar{\rho}}{\partial x_2} &= \frac{\partial\tau_{2j}}{\partial x_j} \\
\frac{\partial(\bar{\rho}\tilde{u}_3)}{\partial t} + \tilde{u}_3\tilde{d}_1 + \bar{\rho}\tilde{d}_5 + \frac{\partial\bar{\rho}\tilde{u}_2\tilde{u}_3}{\partial x_2} + \frac{\partial\bar{\rho}\tilde{u}_3\tilde{u}_3}{\partial x_3} + \frac{\partial\bar{\rho}}{\partial x_3} &= \frac{\partial\tau_{3j}}{\partial x_j} \\
\frac{\partial(\bar{\rho}\tilde{E})}{\partial t} + \bar{\rho}\tilde{u}_1\tilde{d}_3 + \bar{\rho}\tilde{u}_2\tilde{d}_4 + \bar{\rho}\tilde{u}_3\tilde{d}_5 + \frac{1}{2}(\tilde{u}_{11}^2 + \tilde{u}_{22}^2 + \tilde{u}_{33}^2)\tilde{d}_1 + \frac{\tilde{d}_2}{\gamma-1} \\
+ \frac{\partial}{\partial x_2}(\bar{\rho}\tilde{E} + \bar{p})\tilde{u}_2 + \frac{\partial}{\partial x_3}(\bar{\rho}\tilde{E} + \bar{p})\tilde{u}_3 + \left[\tilde{e} - \frac{a^2}{\gamma(\gamma-1)}\right]\tilde{d}_1 \\
+ \sum_{m=1}^N \bar{\rho}\tilde{d}_{6+m} \left[\tilde{e}_m - \frac{\tilde{R}_m\tilde{T}}{W_m(\gamma-1)}\right] &= -\frac{\partial\tilde{q}_j}{\partial x_j} + \frac{\partial\tilde{u}_j\tau_{ij}}{\partial x_j} \\
\frac{\partial\bar{\rho}k^{sgs}}{\partial t} + k^{sgs}\tilde{d}_1 + \bar{\rho}\tilde{d}_6 + \frac{\partial\bar{\rho}\tilde{u}_2k^{sgs}}{\partial x_2} + \frac{\partial\bar{\rho}\tilde{u}_3k^{sgs}}{\partial x_3} &= \frac{\partial}{\partial x_j} \left(\rho\nu_t \frac{\partial k^{sgs}}{\partial x_j}\right) + P^{sgs} - D^{sgs} \\
\frac{\partial\bar{\rho}\tilde{Y}_m}{\partial t} + \tilde{Y}_m\tilde{d}_1 + \bar{\rho}\tilde{d}_{6+m} + \frac{\partial\bar{\rho}\tilde{u}_2\tilde{Y}_m}{\partial x_2} + \frac{\partial\bar{\rho}\tilde{u}_3\tilde{Y}_m}{\partial x_3} &= \frac{\partial}{\partial x_j} \left(\bar{\rho}\tilde{D}_m \frac{\partial\tilde{Y}_m}{\partial x_j}\right) + \tilde{w}_m \quad m = 1, N
\end{aligned}$$

The $\tilde{d}_i$'s in the above equations are functions of the characteristic wave amplitudes,

$\mathcal{L}_i$:

$$\mathbf{d} = \begin{pmatrix} \frac{1}{c^2} [\mathcal{L}_2 + \frac{1}{2}(\mathcal{L}_5 + \mathcal{L}_1)] \\ \frac{1}{2}(\mathcal{L}_5 + \mathcal{L}_1) \\ \frac{1}{2\bar{\rho}c}(\mathcal{L}_5 - \mathcal{L}_1) \\ \mathcal{L}_3 \\ \mathcal{L}_4 \\ \mathcal{L}_6 \\ \mathcal{L}_{6+m} \end{pmatrix} ; \mathcal{L} = \begin{pmatrix} \lambda_1 \left(\frac{\partial\bar{\rho}}{\partial x_1} - \bar{\rho}c \frac{\partial\tilde{u}}{\partial x_1} \right) \\ \lambda_2 \left(\bar{c}^2 \frac{\partial\bar{\rho}}{\partial x_1} - \frac{\partial\bar{\rho}}{\partial x_1} \right) \\ \lambda_3 \left(\frac{\partial\tilde{v}}{\partial x_1} \right) \\ \lambda_4 \left(\frac{\partial\tilde{w}}{\partial x_1} \right) \\ \lambda_5 \left(\frac{\partial\bar{\rho}}{\partial x_1} + \bar{\rho}c \frac{\partial\tilde{u}}{\partial x_1} \right) \\ \lambda_6 \left(\frac{\partial k^{sgs}}{\partial x_1} \right) \\ \lambda_{6+m} \left(\frac{\partial\tilde{Y}_m}{\partial x_1} \right) \end{pmatrix} ; \lambda = \begin{pmatrix} \tilde{u} - \bar{c} \\ \tilde{u} \\ \tilde{u} \\ \tilde{u} \\ \tilde{u} + \bar{c} \\ \tilde{u} \\ \tilde{u} \end{pmatrix}$$

3.5.1 Subsonic inflow

For a subsonic, reacting, three dimensional flow, $5 + m$ characteristics are entering the domain $(\mathcal{L}_2, \mathcal{L}_3, \mathcal{L}_4, \mathcal{L}_5, \mathcal{L}_6, \mathcal{L}_{6+m})$ while only one $\mathcal{L}_1$ is leaving the domain at the speed of $\lambda_1 = \tilde{u} - \bar{c}$. Therefore, density is determined by the flow itself. Hence $5 + m$ physical boundary conditions (for $u_1, u_2, u_3, T, k^{sgs}, Y_m$) and one numerical boundary condition (for ρ) are required. To advance the solution, only the equation for the density in the above system of equations has to be computed. The characteristic amplitude $\mathcal{L}_1$ needed to compute density at the inflow can be computed from the interior points. The amplitudes $(\mathcal{L}_2$ and $\mathcal{L}_5)$ can be recast in terms of $\mathcal{L}_1$ (Poinsot and Lele [1992]), as follows.

$$\begin{aligned}\mathcal{L}_5 &= \mathcal{L}_1 - 2\rho c \frac{\partial \tilde{u}}{\partial t} \\ \mathcal{L}_2 &= \frac{1}{2}(\gamma - 1)(\mathcal{L}_1 + \mathcal{L}_5) + \frac{\bar{\rho}c^2}{\bar{T}} \frac{\partial \tilde{T}}{\partial t}\end{aligned}$$

This completes the inflow treatment for a subsonic flow.

3.5.2 Subsonic partially reflecting outflow

For subsonic outflow, using perfectly non-reflecting (no incoming waves into the domain) boundary conditions might lead to an “ill-posedness” of the problem (Rudy and Strikwerda [1981]). For the problem to remain well-posed, the pressure at infinity, p_{infy} must be added to the boundary condition. This is done by linking the incoming wave amplitude to the pressure difference $p - p_\infty$. This correction to the non-reflecting boundary treatment makes it partially reflecting (Poinsot and Lele [1992] and thus controls the mean pressure in the flow domain. The amplitude of the incoming acoustic wave is approximated by the asymptotic expression which sets the pressure at infinity using, $\mathcal{L}_1 = \sigma(1 - M^2)c(p - p_\infty)/L$ where M is the maximum Mach number in the domain and L is the characteristic length of the domain, c is the speed of sound and σ is a constant to control the amount of wave reflection. While this condition just sets one physical boundary condition all the other dependent variables

at the boundary is estimated by solving the system of conservation equations. This completes the outflow boundary condition treatment for subsonic flows.

3.6 Numerical Method for the Dispersed Phase

The Lagrangian tracking of the dispersed phase given by Equations 2.34, 2.35, 2.38 and 2.39 are integrated in time using a fourth-order Runge-Kutta (R-K) scheme. Runge-Kutta algorithm computes the solution of the initial value problem $y = f(t, y)$, with the initial condition given by $y(t_0) = y_0$. RK method is similar in principle to backward Euler's method, in that the $n + 1^{th}$ term of the approximating sequence is only a function of the n^{th} term. However, the difference is that whereas Euler's method only uses the value of the function f at the left point in the interval, RK explores its behavior throughout the interval. This extra information makes it more accurate. The solution at the new time step $n + 1$ is given by

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) \quad (3.12)$$

where y_{n+1} is the approximation to the solution at t_{n+1} , given by $t_{n+1} = t_n + \Delta t$. k_1 , k_2 , k_3 , and k_4 are defined as follows

$$\begin{aligned} k_1 &= hf(t_n, y_n) \\ k_2 &= hf(t_n + \frac{1}{2}h, y_n + \frac{1}{2}k_1) \\ k_3 &= hf(t_n + \frac{1}{2}h, y_n + \frac{1}{2}k_2) \\ k_4 &= hf(t_n + h, y_n + k_3) \end{aligned} \quad (3.13)$$

Thus, the solution for the Lagrangian equations for the dispersed phase is advanced until the LES time-step is attained. The solution of the Lagrangian equations for the dispersed phase are advanced (integrated in time) by choosing the smallest of the various time-steps discussed in Section 2.2.6. Thus for the combined evolution of the various phases, each explicit advancement of the LES equations at the fluid dynamic/acoustic global time step is followed by several explicit fractional steps for the dispersed phase as governed by the stability of the numerical schemes.

CHAPTER IV

LINEAR EDDY MODEL FOR TURBULENT COMBUSTION

*We all agree that your theory is crazy,
but is it crazy enough ?*

-Niels Bohr

The modeling of turbulent fluid flow in realistic engineering geometries, even in the non-reacting case, remains one of the great scientific challenges. It is well known that large scale advective transport breaks down the in-homogeneity in the scalar field into smaller scales, and molecular diffusion dissipate the scalar fluctuations at small scales by acting on the local gradients. Chemical reaction and the associated heat release introduce fine-scale density and velocity fluctuations which in turn, couple the small scales events back to the larger fluid-dynamical scales. This close coupling between the various processes and the presence of wide separation between the relevant scales makes turbulent combustion modeling more complex compared to turbulent modeling for non-reacting flows. Consequently, the basic requirement for any turbulent combustion model is that it should truly represent the interplay between the small scale and large-scale turbulence, molecular diffusion and finite-rate kinetics. Inaccuracies in the models for small scale scalar transport and/or the closure for the chemical source terms would lead to large errors in the solutions.

In Chapter 2, a brief description of the various conventional sub-grid models and their shortcomings was discussed. In this chapter, a more sophisticated model that attempts to simulate the various physical processes at their relevant length and time

scales, is described. In the following section, rationale behind the model, equations employed and the implementation issues are addressed.

4.1 LEM: Sub-grid Scalar Closure for LES

The development of the Linear Eddy Mixing (LEM) model (Kerstein [1988a, 1989, 1991a,b]) was motivated by the understanding that, for an accurate representation of the turbulent combustion, a model should distinguish and resolve the various physical processes, such as large scale advection, small scale mixing, molecular diffusion and chemical reaction. In addition, each of the above mentioned processes must be resolved at their relevant length and time scales, to get the correct coupled solution.

Three-dimensional resolution of all possible length and time scales that occur in reacting flows require tremendous computational resources. Therefore, a “one-dimensional” description of the scalar field, in which all length scales can be resolved is used in the LEM formulation. The unique feature of the LEM model is that although the scalar evolution is simulated in 1D, the effect of turbulence on the scalar fields is modeled using 3D scaling laws. As a result, the combustion occurs at spatial and temporal scales, that mimic the effects of realistic 3D turbulent eddies on the flame. It has been shown that, the scalar power spectrum obtained from the LEM simulations of homogeneous mixing in a sustained mean gradient by Kerstein [1991a] closely resembles the 3D-power spectrum and shows a strong Sc dependence. This, and the numerous other LEM simulations Kerstein [1989, 1990, 1991a,b], Kerstein and Ashurst [1992], Kerstein [1992] provides good credibility to LEM as an accurate and economic mixing model. LEM approach applied to LES, also known as LES-LEM was first applied to premixed and non-premixed combustion by Menon et al. [1993] and since then has gone through several modifications (McMurtry et al. [1992], Calhoon [1996], Menon and Calhoon [1996], Smith [1998], Chakravarthy and Menon [2000], Pannala and Menon [1998]).

This kind of reduced dimensional representation of the molecular mixing and chemical reactions is not entirely new and has been attempted quite successfully in the past. Marble [1985], Karagozian and Marble [1986] first studied analytically, the growth of a diffusion flame interacting with a vortex using a fast chemistry assumption and identified the key parameters that enhance the combustion rate. In this study, using an appropriate transformation for the transverse co-ordinate and for time, along with the assumption that the normal strain has a greater influence on the molecular diffusion process than the shear strain, Marble [1985] obtained a one-dimensional unsteady species diffusion equation. Cetegen and Sirignano [1988, 1990] extended this approach to investigate molecular mixing and chemical reaction in a two-dimensional viscous vortex representing a mixing layer. In this study, two species initially separated across a plane (material surface) are allowed to diffuse and react in the presence of a line vortex (Cetegen and Sirignano [1990]) or an infinite row of two-dimensional vortices (Cetegen and Sirignano [1988]). The initially planar material surface is stretched and twisted around the viscous vortex. Molecular diffusion and chemical reaction that takes place across the material elements along the interface is computed in a Lagrangian manner. The computation of the concentration field across this interface rely on the knowledge of the flow field. For this purpose, an analytical solution of the flow-field (viscous line vortex or the infinite row of point vortices) is utilized and the successive positions of the material element are computed by integrating the flow-field with time. Using the same transformation as that of Marble [1985], Cetegen and Sirignano [1988, 1990] obtained unsteady diffusion equation for which exact solution is known. In the case of finite-rate chemical reactions, Cetegen and Sirignano [1988, 1990] obtained an analytical solution for the reacting species equation using an approximate method based on the Green's function solution. Unfortunately its extension to multi-species, variable lewis number formulation

is not straight-forward, since no known analytical solution exists for the variable coefficient, non-linear diffusion equation. Nevertheless, it is worthwhile to consider this approach for future extension of the LEM. The interesting feature of this approach is that, similar to LEM, this approach also distinguishes the various physical processes such as molecular diffusion, chemical reaction and turbulent transport and elegantly incorporates their effect using a deterministic approach.

In LES-LEM the governing filtered conservation equations for mass, momentum and energy are numerically integrated using a time and space accurate numerical scheme on an appropriate three-dimensional grid. It should be noted that the equations for mass, momentum and energy solved on the resolved grid are in the Eulerian form (i.e. Eulerian frame of reference). Attention is drawn to the fact that, at this stage, no conservation equation for species are solved on the resolved LES grid.

The species/scalar field evolution is tracked using a **two-scale** numerical approach. In this technique, turbulent convection of the scalars is split into two parts: **large scale advection** and **turbulent convection** at the small (sub-grid) scales. Large scale advection is the convection that happens above the unresolved scales. This is modeled by using a Lagrangian scheme which explicitly transports mass across the finite-volume cell surfaces in a manner that is consistent with the mass transport in the Eulerian continuity equation solved on the resolved grid. Small-scale advection is the turbulent stirring by scales smaller than the resolved grid. Note that, turbulent convection at the sub-grid scales only transports fluid mass over sub-grids length scales.

To describe this two-scale numerical method, consider the exact (i.e., unfiltered) evolution equation for a reactive scalar Φ , written in the Eulerian form:

$$\rho \frac{\partial \Phi}{\partial t} + \underbrace{\rho u_i \frac{\partial \Phi}{\partial x_i}}_{\text{(Large + Small) scale convection}} + \underbrace{\rho \frac{\partial}{\partial x_i} [-D_\Phi \frac{\partial \Phi}{\partial x_i}]}_{\text{molecular diffusion}} = \underbrace{\dot{w}_\Phi}_{\text{chemical source term}} + \underbrace{\dot{S}_\Phi}_{\text{spray source term}} \quad (4.1)$$

Note that, $\dot{S}_\Phi$ is a non-zero source term for the fuel species due to phase change from the liquid phase. Now, split $u_i = (\tilde{u}_i + u'_i)$ where, $\tilde{u}_i$ and u'_i are the contribution to the total convection from the resolved (large) and sub-grid (small) scales. By grouping the molecular diffusion and the chemical reaction as small scale processes one can split Eqn. [4.1] into two equations, one corresponding to the large scale processes and another corresponding to the small-scale processes as follows.

$$\rho \frac{\partial \Phi}{\partial t} + \underbrace{\rho \tilde{u}_i \frac{\partial \Phi}{\partial x_i}}_{\text{Large scale convection}} = 0 \quad (4.2)$$

$$\rho \frac{\partial \Phi^*}{\partial t} + \underbrace{\rho u'_i \frac{\partial \Phi^*}{\partial x_i}}_{\text{Small scale convection}} + \rho \frac{\partial}{\partial x_i} [-D_\Phi \frac{\partial \Phi^*}{\partial x_i}] = \dot{w}_\Phi + \dot{S}_\Phi \quad (4.3)$$

The large-scale advection processes (governed by Eqn. [4.2]) and the small scale advection (governed by Eqn. [4.3]) account for the complete evolution of the scalar fields. Here, large scale implies the size of scales that range from the dimension of the combustor to the resolution of the grid. Small scales are those whose size range from the grid resolution down to the Kolmogorov (smallest fluid dynamic scale) or Batchelor micro-scale (approximate size of the smallest scalar fluctuation) or the flame thickness. The chemical reaction and heat release occurs at these small scales and hence, to capture the correct physics of combustion, these small scales must be resolved along with the small scales. As stated earlier, resolving all the length scales in 3-D will in general be computationally intractable.

In LES-LEM, this conflicting requirement is met by resolving the small scale processes (sub-grid scale turbulent mixing, molecular diffusion and chemical reaction)

in one-dimension. This idea relies on the following.

- (1) small scale turbulence is locally homogeneous and isotropic
- (2) heat-release and the associated thermal expansion is isotropic
- (3) the model in its basic form, has been shown (Kerstein [1988a]) to capture the correct physics of scalar mixing at small-scales in homogeneous, isotropic turbulence.

It should be noted that, isotropic behavior required by (1) and (2) are questionable in high Mach number flows, since information does not propagate at equal speeds in all directions. However, for subsonic flows (that are of interest here) it is reasonable.

Therefore, within the context of LES, the LEM approach can be used to model the small scale processes (in reduced dimension) while the large scales of the flow are calculated directly from the LES equations of the motion with an appropriate coupling procedure.

Organization of LES-LEM

Rest of the section on LES-LEM is organized as follows.

- **Section 4.2.1** presents the sub-grid LEM simulation approach. Deterministic, one-dimensional reaction-diffusion equations for species and temperature are introduced in this section along with the assumptions.
- **Section 4.2.2** is used to introduce the stochastic model for turbulent stirring at the sub-grid scales. Physical reasons behind the model and the implementation issues are also discussed.
- **Section 4.2.3** explains the algorithm for the implementation of volumetric expansion due to heat release from chemical reaction. This process is explicitly modeled in the sub-grid LEM simulation.
- **Section 4.2.4** provides details on the numerical method employed to solve the governing reaction-diffusion Eqn. [4.3]. This section also describes the boundary and initial conditions along with the resolution requirements.
- **Section 4.3** describes the modeling procedure for the large-scale advection governed by Eqn. [4.2]. Algorithm and the implementation techniques are explained in detail in this section.
- **Section 4.4** explains the method for coupling the 1-D LEM simulation (small-scale processes) with the 3-D LES solution (large scale processes) to obtain a complete time-accurate solution.

4.2 Sub-grid LEM Simulation

In the LES-LEM approach, the scalar evolution in the sub-grid is simulated by solving the equation Eqn. [4.3] in an one-dimensional space rather than in three-dimensions. This reduced-dimensional representation of the sub-grid processes in LEM, offers its greatest advantage, viz., the ability to perform a localized, affordable DNS and makes it an ideal candidate for high Re and Sc reacting flows.

For the purpose of solving Eqn. 4.3, within each LES cell, a 1-D line segment known as the LEM domain is defined and discretized. On this line segment (LEM domain) 1-D reaction-diffusion equation of the form shown in Eqns. 4.4 and 4.5 are solved numerically.

4.2.1 Molecular diffusion and chemical reaction

The 1-D reaction-diffusion equation for the species and the temperature are written in non-conservative form as follows:

$$\rho \frac{\partial Y_k}{\partial t} + F_{k\,stir} + \frac{\partial}{\partial s}(-\rho D_k \frac{\partial Y_k}{\partial s}) = \dot{\omega}_k W_k \quad (4.4)$$

$$\rho c_{p\,mix} \frac{\partial T}{\partial t} + F_{T\,stir} - \sum_{k=1}^N \rho c_{p,k} D_k (\frac{\partial Y_k}{\partial s})(\frac{\partial T}{\partial s}) - \frac{\partial}{\partial s}(\bar{\kappa} \frac{\partial T}{\partial s}) = - \sum_{k=1}^N h_k \dot{\omega}_k W_k \quad (4.5)$$

Note, Eqns. [4.4] and [4.5] are the equivalent to Eqn.[4.3] written in terms of sub-grid species and temperature field. $F_{k\,stir}$ and $F_{T\,stir}$ are the turbulent convection of the species and the temperature, at the sub-grid scales, respectively.

Here, s denotes the co-ordinate direction on the 1-D domain. The orientation of the 1D domain is usually aligned in the direction of the maximum scalar gradient, but the physical interpretation of this 1-D domain is dependent on the application considered (Kerstein [1989]). Note, the actual orientation and physical interpretation of the LEM domain is not a model input, and hence the arbitrary nature of its interpretation does not affect the performance of the model.

Assumptions & Definitions

- Sub-grid pressure (inside the LEM domain) is assumed to be constant and same as the resolved grid pressure. In the absence of strong pressure gradient such as those in shocks, and in highly compressible flows, it is reasonable to assume that the pressure in the sub-grid remains constant.
- The contribution from the sub-grid viscous work is neglected.
- Calorically perfect gas model assumed. But, it is straight-forward to extend LEM for thermally perfect model.
- Radiation effects are neglected in this study. But it can be included easily as shown in Zimberg et al. [1998].
- Density for the linear eddy cells are computed using the equation of state for the scalar mixture $\bar{p} = \rho T \sum_{k=1}^N Y_k R_u / W_k$ and the caloric relation is given by $h_k = \Delta h_{f,k}^o + c_{p,k}(T - T^o)$
- T , $\bar{p}$ and ρ are the sub-grid temperature, resolved pressure, and the sub-grid mass density, respectively.
- Y_k , W_k , $c_{p,k}$ and R_u , are the mass fraction, molecular weight, specific heat at constant pressure, and universal gas constant respectively.
- $\dot{\omega}_k$, h_k , V_k and $\Delta h_{f,k}^o$ are mass reaction rate, enthalpy, diffusion velocity and standard heat of formation (at standard temperature, T^o) of the k -th species.
- $c_{p_{mix}}$, κ and D_k are the mixture averaged specific heat at constant pressure, thermal conductivity and diffusivity of the k -th species respectively.
- The length of the LEM domain is taken to be equal to that of the local LES filter width, Δ . The one-dimensional linear eddy domain is further discretized

(uniformly) to resolve the characteristic mixing scales and/or the Kolmogorov or the Batchelor scales based on the nominal sub-grid Reynolds number and the Schmidt number in the computations.

4.2.2 Sub-grid scale turbulent convection

Turbulent convection at the sub-grid scales of the form $u\partial Y_k/\partial s$ and $u\partial T/\partial s$ are symbolically represented as F_{kstir} and F_{Tstir} , respectively in Eqns. [4.4] and [4.5] and these are implemented explicitly, as discussed in the next section.

Since the sub-grid velocity field is assumed isotropic, the sub-grid field does not convect in any particular direction (inside the LES control volumes). The effects of the sub-grid velocity field on the sub-grid scalar fields are modeled (numerically) using stochastic re-arrangement events called *triplet maps* ((Kerstein [1989])). These triplet maps represents the action of an isotropic turbulent eddy on the sub-grid scalar field. The intuitive rationale for adopting the triplet map is illustrated in Fig. 4.1. Consider a mixing layer configuration shown in Fig. 4.1(a) with a plane material surface separating species A from species B. The bold straight line running from the top to the bottom shown in the box (dashed line) represents the initial concentration profile (uniform gradient). Horizontal lines (red and blue) represents the concentration isopleths at the initial time. Fig. 4.1(b) shows the distortion (stretching + compression) of the concentration isopleths due to the action of the eddies/vortex roll-up. The initial linear concentration profile evolves to a form, qualitatively resembling the profile obtained by applying the triplet map to the linear profile. This scalar field produced by this is continuous and measure preserving (Note: scalar gradient field is not continuous). This analogy provides a physical basis for specifying the random processes governing the sequence of rearrangement events.

It has been shown by Kerstein [1989] that this mapping can capture correctly the physical increase in scalar gradient (without affecting the mean scalar concentration)

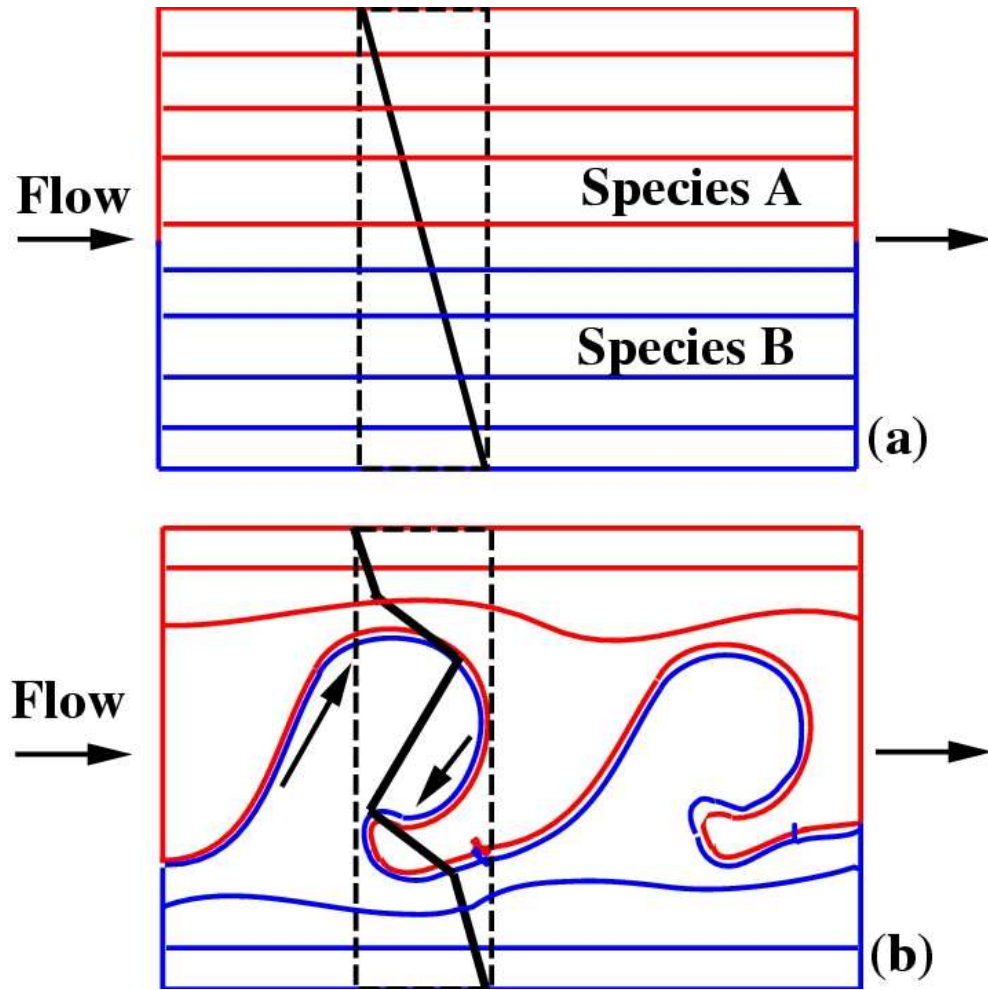


Figure 4.1: Schematic illustration of the effect of an eddy/vortex roll-up on a two-dimensional scalar field.

due to eddy motion. In addition, the scalar field produced by the map is continuous and measure preserving (even though the scalar gradient field is not continuous). Measure preserving property of the map ensures that the scalars are neither created nor destroyed, but merely re-arranged along the 1-D line by the rearrangement event. This is consistent with the notion that the eddies just transport fluid elements without altering their concentration. Although the mappings occur instantaneously and interrupt the continuous time integration of the species equation, the time between the mapping events may be interpreted as an eddy turn-over time (Kerstein [1991a]). Schematically, this mapping first creates three copies of the selected segment and then increases the spatial gradients of the copies by compressing each segment by a factor of three and reversing the middle segment. Finally, the original segment is replaced by the newly mapped segments. Mathematically, the *triplet map* transforms the scalar field $\Phi(x, t_o)$ to $\hat{\Phi}(x, t_o)$ according to,

$$\hat{\Phi}(x, t_o) = \begin{cases} \Phi(3x - 2x_o, t_o) & x_o \leq x \leq x_o + l/3, \\ \Phi(-3x + 4x_o + 2l, t_o) & x_o + l/3 \leq x \leq x_o + 2l/3, \\ \Phi(3x - 2x_o - 2l, t_o) & x_o + 2l/3 \leq x \leq x_o + l, \\ \Phi(x, t_o) & \text{otherwise} \end{cases} \quad (4.6)$$

where l is the size of the mapping event, x_o is the location of the mapping event and t_o is the time of the event.

An illustration of the triplet mapping and the competing actions of stirring and diffusion on a mapped segment of a scalar are shown in fig 4.2. The initial scalar field with a constant gradient is shown in fig 4.2(a). If the magnitude of the scalar can be represented schematically using numbers from 1 to 9, the effect of the triplet map, is to rearrange the scalar field as shown in fig 4.2(b) and (c). Molecular diffusion acts on the scalar field and smooth out the gradient as shown in Figure 4.2.

Three parameters are needed to implement these turbulent stirring events: eddy size l , the eddy location within the 1D domain and the stirring frequency (mean

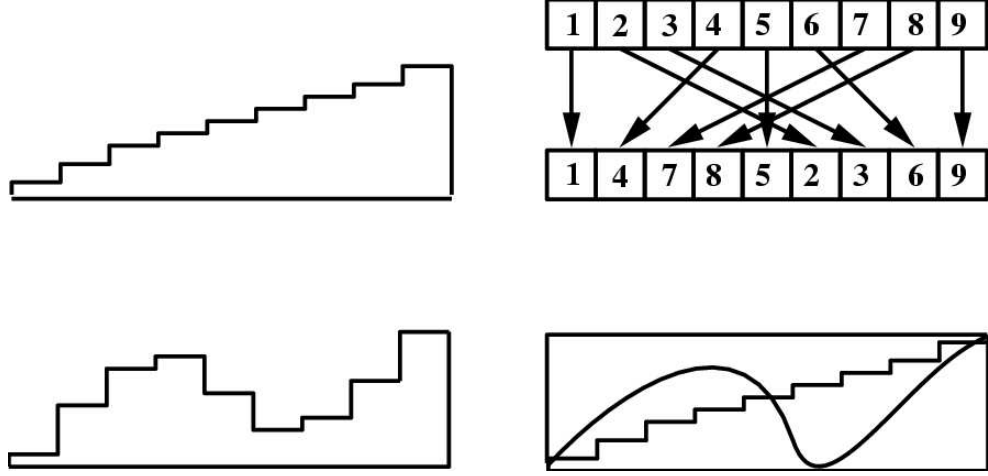


Figure 4.2: Schematic diagram illustrating an LEM triplet mapping event followed by molecular diffusion.

event rate per unit length of the mapping domain) λ . The eddy size in the range Δ to η (Kolmogorov scale) is determined randomly from an eddy size distribution $f(l)$, obtained using inertial range scaling in three-dimensional turbulence Kerstein [1989]:

$$f(l) = \frac{5}{3} \frac{l^{-8/3}}{(\eta^{-5/3} - \Delta^{-5/3})} \quad (4.7)$$

Here, η is determined from inertial range scaling law

$$\eta = N_\eta \frac{\Delta}{Re_\Delta^{3/4}}$$

where N_η is an empirical constant and Re_Δ is the sub-grid Reynolds' number based on the sub-grid turbulence intensity, kinematic viscosity and the local LES filter width, Δ . This constant reduces the effective range of scales between Δ and η but does not change the turbulent diffusivity, as described by Smith and Menon [1996a]. The event location is randomly chosen from a uniform distribution and the event (mapping) rate (mean frequency per unit length) are implemented as a Poisson process in time (Kerstein [1989])

$$\lambda = \frac{54}{5} \frac{\nu Re_\Delta}{C_\lambda \Delta^3} \frac{[(\Delta/\eta)^{5/3} - 1]}{[1 - (\eta/\Delta)^{4/3}]} \quad (4.8)$$

The time interval between events is then given as

$$\Delta t_{stir} = \frac{1}{\lambda \Delta} \quad (4.9)$$

where Δ is the length of the 1-D domain, which is also same as the local LES filter width. Note, that the mean rate parameter of the mapping event, λ is not a function of length scale, l which implies that the interval between the stirring events are the same for all the length scales. Strictly speaking this is not true, but the following considerations justify the proposition.

- (1) Assumption of the local isotropy in the sub-grid, implies that the range of the stirring length scales are closely spaced in the wave-number space. This implies that their time-scale (or the turn-over time) is nearly same.
- (2) The sub-grid Re and the filter width, Δ are varying spatially over the LES (resolved) grid. This implies that the the range of the stirring length scales and hence the frequency of the stirring events also vary spatially over the LES (resolved) grid, even though it is same within the sub-grid.

The above formulation has two constants: C_λ and N_η , both of which arise from the use of scaling laws. In the earlier studies (Smith and Menon [1996a,b]), these parameters were defined by comparing LEM predictions to experimental data (Abdel-Gayed et al. [1984]) in the flamelet regime. Chakravarthy [2000] argued that the scalar diffusivity due to triplet maps should equal the momentum diffusivity (by triplet maps), since inertial range eddies are responsible for both the processes. Therefore, the same value of the coefficient C_ν used in the expression for eddy viscosity expression (Eqn. [2.24]), should be used for scalar diffusivity C_λ . This value (0.067) turns out to be close to the values used by Smith and Menon [1996a,b] The present study, uses the same constant for all the simulations reported, even though the LDKM approach used here, yields dynamically varying coefficients.

4.2.3 Volumetric expansion

The final sub-grid process in the LEM, which needs to be described is the volumetric expansion due to heat release. As mentioned earlier, constant pressure is assumed in the sub-grid so that the heat release will cause volume expansion. Since the sub-grid convection terms are modeled explicitly using triplet maps, volumetric expansion must be included explicitly as well. Physically, volume expansion manifests itself as a radiation of pressure waves which causes local flow acceleration. Mathematically, the presence of the pressure gradient term in the momentum equation couples the velocity field with the pressure field. However, there is no implicit coupling between sub-grid velocity and the pressure field, in the the governing equation for sub-grid LEM simulation, Eqn. [4.3]. This is because, the effect of the sub-grid velocity field is modeled explicitly using a stochastic triplet-map, independent of the pressure. As a result, volumetric gas expansion caused by heat release is modeled explicitly by expanding each linear-eddy cell in the sub-grid domain by an amount equal to

$$\Delta V_{LEM,i}^* = \frac{\rho_i^n}{\rho_i^*}$$

where $\Delta V_{LEM,i}$ is the change in volume of LEM cell i . ρ_i^n and ρ_i^* are, respectively, the density of the “ $i - th$ ” cell at the previous and the current time integration levels in the sub-grid simulation (not at the fluid-dynamic time-step, (Δt_{LES}) at LES level,).

In the previous studies by Smith [1998] and Chakravarthy [2000], the domain containing the expanded cells is re-gridded so that each cell is returned to its initial volume. In the present study, re-gridding the cells is done after the large-scale advection process, and not at this stage. This is explained in the Section 4.3. Decoupling the volumetric expansion from the reaction-diffusion equations, is a limitation of this model and the quantification of its effect on the accuracy has not been evaluated yet.

4.2.4 Numerical method for sub-grid LEM simulation

Typical reaction-diffusion equation such as Eqn. [4.3] are characterized by large ranges of spatial and temporal scales. Explicit time integration of such equations has typically two bottlenecks: the stiffness¹ of the chemical source terms, and the diffusive stability restriction. Both lead to the need for small time steps, rendering detailed multidimensional flame computations very expensive. Implicit schemes, on the other hand has severe memory requirements (Knio et al. [1999]). Moreover, accuracy considerations suggest that the time step size must be kept smaller than the smallest physically relevant time scale in the flow, for which the acoustic/convective CFL number should be maintained less than unity. This renders a fully implicit implementation counter-productive.

The goal is then to integrate diffusion and chemical source terms accurately and efficiently, with explicit methods. In this study, an operator splitting method (Smith and Menon [1996a], Calhoon et al. [1995]) is used to integrate the stiff reaction-diffusion equation (Eqn. 4.3). The idea behind operator-splitting technique is the sequential application of individual operators at the respective time-scale of the operator in such a way as to ensure accuracy and numerical stability of the computation.

In LES-LEM, each of these operators, namely molecular diffusion, chemical reaction, thermal expansion and turbulent convection are implemented numerically as discrete events in time. The epoch or the instant at which each of these processes need to be implemented depends on the respective time-scale of each of these processes. These time-scales are identified below.

Diffusion time scale - Δt_{diff} : Time scale associated with the transport of species and temperature from a region of higher concentration to a lower concentration due to random motion of the molecules. This is the slowest physical process among the

¹simultaneous presence of small and large time scale processes

other sub-grid processes and is computed as

$$\Delta t_{diff} = \kappa \frac{\Delta s^2}{\max[D_k]}$$

where, Δs is the LEM grid size and D_k is the mass diffusivity of the species “k”. κ is set to 0.25 for numerical stability.

Chemical time scale - Δt_{chem} : Time scale associated with the rate of conversion of reactant species to intermediate and/or product species. Most of the time, this is the fastest time scale in the flow, barring few situations, when turbulent time scales can be competing with the chemical time scales. In the present study, an integration package called DVODE (Double-precision Variable-coefficient ODE solver) based on the CHEMKIN package (Kee et al. [1989]) is used to integrate the stiff non-linear ODE (ordinary differential equations). DVODE, adaptively chooses the largest time-step that would provide a solution within the specified error tolerance.

Thermal expansion time scale - Δt_{vol} : Time scale associated with the thermal expansion due to heat release. Ideally, $\Delta t_{vol} \approx \Delta t_{chem}$, since the thermal expansion happens soon after the heat release. But it is set equal to Δt_{diff} in the present study for the following reasons.

- (1) In case of fast chemistry ($\Delta t_{chem} < \Delta t_{diff}$), heat release (and hence the volumetric expansion) is limited by rate of molecular mixing occurring at Δt_{diff} .
- (2) In case of slow chemistry ($\Delta t_{chem} > \Delta t_{diff}$), Δt_{diff} is the smallest time scale and hence for the sake of numerical stability, it is better to implement volumetric expansion at the diffusion time scale.

Stirring time scale - Δt_{stir} : Time scale associated with the sub-grid turbulent convection, is estimated as shown in Eqn. 4.9. This is the time-scale at which the triplet maps are implemented. It is a strong function of sub-grid Reynolds number Re_Δ , i.e., higher Re_Δ implies a smaller stirring time scale.

Operator-Splitting

To illustrate this, consider the scalar equation

$$\rho \frac{\partial \Phi}{\partial t} = F_{\Phi, stir} + D_{\Phi} + R_{\Phi} \quad (4.10)$$

where, $F_{\Phi, stir}$, D_{Φ} and R_{Φ} , denote the stirring, diffusion and reaction operator, respectively.

1. Let Δt_{LES} be the fluid-dynamic (LES) time-step
2. Let $N_{diff-max}$ be the maximum number of diffusive sub-steps in Δt_{LES} , i.e

$$\Delta t_{diff} = \Delta t_{LES} / N_{diff-max}$$
3. Note, this implies $N_{chem} = N_{diff}$
4. Let N_{stir} be the number of stirring sub-steps in Δt_{LES} , i.e $\Delta t_{stir} = \Delta t_{LES} / N_{stir}$

5. **If** ($t_{diff} < t_{stir}$) **then**

$$\hat{\Phi}^{k+1} - \Phi^k = \Delta t_{diff} D_{\Phi}^k + \int_{\Delta t_{diff}} R(\Phi) dt$$

$$t_{diff} = t_{diff} + \Delta t_{diff}$$

$$N_{diff} = N_{diff} + 1$$

if ($N_{diff} > N_{diff-max}$) exit loop & Goto **End Sub-grid Simulation**

Else

$$\Phi^{k+1} - \hat{\Phi}^k = F_{\Phi, stir}$$

$$t_{stir} = t_{stir} + \Delta t_{stir}$$

Endif

Repeat Step 5

End Sub-grid Simulation

This splitting combines an explicit treatment of the LES resolved mass and momentum equations at the global time step with several explicit fractional steps for diffusion, reaction and turbulent stirring at the sub-grid scales. To solve the Eqns. [4.4]

and [4.5] numerically, all spatial derivatives are discretized using a second-order accurate central difference schemes.

Initial and boundary conditions To numerically integrate Eqn. [4.3] initial and boundary conditions are needed. At the start of the simulation, sub-grid LEM fields are set with the resolved grid values of the scalar field. As the sub-grid field evolves under the action of molecular diffusion, chemical reaction, turbulent stirring and large scale advection, the subsequent steps automatically derive their initial condition from the previous time-step.

In order to chose the boundary conditions, two assumptions are made

1. The 1-D sub-grid LEM domain has no preferred orientation with respect to the LES domain co-ordinates. This is strictly true only if the LES-filter lies in the inertial range.
2. The species and temperature diffusion at the LES-cell interface are negligible compared to the advection at the large scales. This assumption is justified in high Reynolds number flows.

Therefore, zero gradient boundary conditions are imposed for the species and the temperature equations at sub-grid simulation.

Sub-grid domain resolution

LEM-domain (1-D line segment) is initialized in every LES cell with a fixed number of linear-eddy cells. The number of linear-eddy cells is estimated as follows. To represent an eddy using a triplet map, a minimum of 6 points are needed (Menon and Calhoon [1996]). If the sub-grid Re_Δ is known, then using the expression $\eta \approx \Delta Re_\Delta^{-3/4}$, an estimate of the smallest length scale can be obtained. Then the maximum number of LEM cells needed to completely resolve all the sub-grid scales can be computed using

the expression,

$$N_{max} = \min(\frac{\Delta_{ijk}}{\eta_{ijk}})$$

where Δ_{ijk} and η_{ijk} are the LES filter width and the Kolmogorov scale at the cell “ijk”. It is possible to initialize every LES with varying number of LEM cells based on the local sub-grid Re_Δ . But for the purpose of reducing the computational complexity, every LES cell is initialized with the same number of LEM cells (N_{max}). The length of the linear eddy domain is set equal to the LES filter width Δ . At the beginning of each fluid-dynamic time step (Δt_{LES}), volume of each linear eddy cell is set to have an equal fraction of the LES cell volume i.e., $(\Delta V_{LEM})_{ijk} = (\Delta V_{LES})_{ijk} / N_{max}$

4.3 Large Scale Advection: Splicing

This section describes the procedure involved in modeling the large-scale advection process govern by Eqn. [4.2]. The approach used for this purpose is called **splicing**. It is important to realize that splicing models the large-scale advection process and does not solve Eqn. [4.2]. Nevertheless, splicing satisfies the basic condition for the conservation of mass.

After completing the sub-grid LEM simulations in each LES cell, the sub-grid scalar fields are advected by the LES-resolved velocity field, in a Lagrangian sense. This method involves the transfer of LEM cells between the LES control volumes to account for the mass flux across the LES cell faces.

Three quantities are needed for this procedure:

- magnitude or the amount of mass that has to be transported across each of the LES cell faces
- direction of the mass flux (influx or outflux) on each cell face
- ordering/priority of the advection operation for the 3-co-ordinate directions

In a finite-volume scheme, mass flux is known on each of the LES cell faces. In addition, the direction of the velocity field defines the direction of the mass flux on the cell-faces. Hence, the first two of the above mentioned three quantities are taken care of by the resolved grid continuity equation and the momentum equations. The third issue is resolved using the following arguments.

In the numerical implementation of the scalar advection, the three-dimensional advection operator is approximated by a sequence of three, one-dimensional advection operators. On a general three-dimensional grid, the mass transfer from any control volume to any neighboring control volume is also predominantly three-dimensional. As a result, different number of linear eddy cells are transported in different spatial directions. So the order in which these operators act on the scalar field can have a significant effect on the scalar field evolution. In the new algorithm, the linear eddy cells going out in the direction of largest outflux are fluxed out first from the right end of the sub-grid domain. Similarly the largest influx is added first to the left end of linear eddy domain. This upwind type flux update has been used successfully in the past by Chakravarthy and Menon [2001].

Next step in the large-scale advection process is the identification and actual transport of the linear-eddy cells. This is implemented as follows. Since the mass flux on each of the six control surfaces of a LES cell is known, it can be sorted in an ascending order. Note before sorting, the following sign-convention is imposed for the face mass-fluxes. Influx is given a positive sign and the outflux is given a negative sign along each co-ordinate directions. Thus, upon sorting, the mass fluxes would be arranged, from the maximum outflux (least negative number) to maximum influx (highest positive number). Now, for each outflux, every LES cell computes the number of LEM cells that contains the mass that has to be fluxed out. Since the density and the volume is known for each LEM cell, the LEM cell mass is computed simply as the product of the cell density times the cell volume. It should be noted

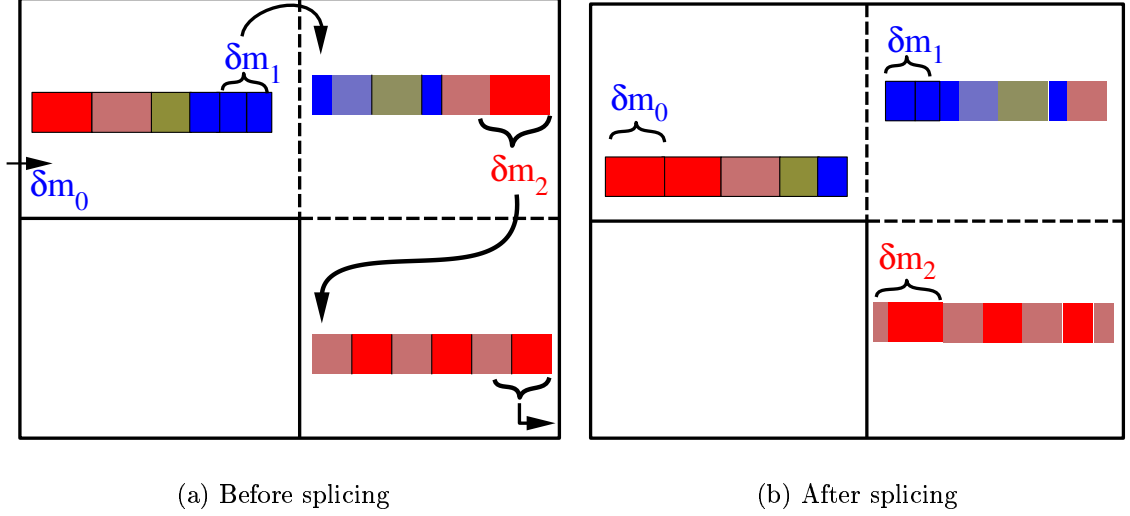


Figure 4.3: Schematic diagram illustrating the splicing process.

that, the amount of mass to be transported across a LES cell surface can be a fraction of the linear eddy cell mass. In the event of the mass being lesser than the mass of the linear-eddy cell that has to be transported, it is split into two cells, as follows.

- Let the temperature and mass fraction of the linear-eddy cell D be T_D and Y_{k_D}
- Let the density, volume and the mass of the cell D be ρ_D , V_D and Δm_D
- Let the amount of mass to be transported across an LES cell surface be δm
- If $\delta m < \Delta m_D$, then split D into two cells, D_L and D_R such that,

$$\rho_{D_L} = \rho_{D_R} = \rho_D$$

$$\Delta m_{D_R} = \delta m$$

$$\Delta m_{D_L} = \Delta m_D - \delta m$$

$$V_{D_R} = \delta m / \rho_D$$

$$V_{D_L} = V_D - V_{D_R}$$

$$T_{D_L} = T_{D_R} = T_D$$

$$Y_{k_{D_L}} = Y_{k_{D_R}} = Y_{k_D}$$

Figure 4.3 shows a schematic diagram of the scalar field before and after splicing.

Features of the New Splicing Algorithm

1. The present splicing algorithm advects the sub-grid scalar field based on the mass, as opposed to the volume of the LEM cell, that was used in the past by Menon and Calhoon [1996], Smith [1998], Chakravarthy [2000], Calhoon [1996].
2. Since the convection of the sub-grid field is based on mass, the volume of the linear eddy cells is also carried as an additional scalar. This is another modification to the previous splicing algorithms by Menon and Calhoon [1996], Smith [1998], Chakravarthy [2000], Calhoon [1996]
3. Also in the earlier approach (Smith [1998]), fractional LEM cells, that have to be convected are accumulated, until they reach an integer value. Since this would violate the continuity equation, the present approach convects the fractional cells as and when, it has to be convected. This ensures strict mass conservation at the resolved and sub-grid scales at each instant of the calculation.
4. The previous studies on compressible LES-LEM (Smith [1998], Calhoon [1996]), employed the convective time step ($\Delta t_{conv} > \Delta t_{LES}$) for splicing. The present study uses the acoustic time-step, Δt_{LES} . This provides a much closer coupling between the momentum and the scalar fields and avoids unphysical pressure oscillations that were observed in the earlier study (Smith [1998]).
5. Molecular diffusion across the adjacent LES cell-face is neglected in this study. In the earlier work by Smith [1998], the inter-LES cell molecular diffusion was proposed as a special case, when a fully burnt product LES cell is adjacent to a fully unburnt LES cell. In this situation, to initiate burning across the LES cell, inter-LES diffusion was used. According to the current author, this is entirely heuristic. In the situation when a burnt LES cell happens to be adjacent to an unburnt cell, the pressure difference between the LES cells, would convect

the flame to the unburnt LES cell by large scale advection. Following this, the sub-grid diffusion and reaction would initiate burning. Therefore, this should not be explicitly added to the model.

6. In general, discontinuities in the scalar field may be generated between the incoming sub-grid elements and the original field, due to splicing. This is an artifact of the splicing process, which may cause spurious diffusion. In the previous study by Calhoon [1996] partitions were introduced at such discontinuities and no diffusion was allowed across such partitions. In another study by Smith [1998] and Chakravarthy [2000], these discontinuities were avoided by using a set of heuristic rules that avoid the placement of burnt cell next to an unburnt cell, taking advantage of the binary nature of the G-field in their studies.

If the flow-field is smooth and continuous (well resolved, like the sub-grid LEM field), the scalar gradient can be expected to be smoothly varying between the adjacent LES cells. In such a situation, the discontinuities in the scalar field, generated by splicing can be expected to be minimal. Therefore, in the present study no such heuristic rules are employed and the incoming cells are placed in their order of their arrival.

7. In the previous study by Calhoon [1996], in order to maintain a constant number of linear eddy cells, additional linear-eddy cells generated by splicing were truncated. This process can induce severe mass-conservation error. But the author (Calhoon [1996]) argued that its effect is minimal if the number of the incoming cells are of the same order as that of the outgoing cells. No such artificial truncation methods are used in the present algorithm. The only criterion imposed on splicing is mass conservation.

Re-gridding the Linear-Eddy domain

Even though splicing conserves mass, it does not conserve the number of linear-eddy cells, due to the convection of fractional linear-eddy cells. So, the number of linear-eddy cells in an LES cell before and after splicing are not the same. Further more, volumetric expansion due to heat release and splicing by itself, lead to a non-uniform distribution of the volume of the linear-eddy cells. Since the numerical scheme employed for solving the sub-grid reaction-diffusion equations assumes a uniform grid, it is desirable to re-grid the non-uniform 1-D line segment to a uniform grid. Also, to avoid programming complexities in a parallel environment, it is desirable to have same number of linear eddy cells everywhere in the computational domain. So, to equalize the volume of the linear-eddy cells and to retain the same number of linear eddy cells everywhere in the computational domain, the linear eddy domain is re-gridded to have cells of equal volume after each splicing. Errors introduced due to the spurious diffusion associated with re-gridding has not been evaluated yet. A schematic diagram of the procedure is shown in figure 4.4. The letters “u”, “r” and “b” refers to the unburnt, reacting and completely burnt linear eddy cells, respectively. Fig. 4.4(a) shows the initially uniform linear eddy domain. Chemical reaction and the subsequent thermal expansion causes a non-uniform distribution of the cell volumes, as shown in Fig. 4.4(b). Re-gridding process, generates a uniform distribution of the volume in the linear eddy cells as shown in Fig. 4.4(c).

4.4 Large scale - Small scale Coupling

To complete the LES-LEM formulation, the simulated sub-grid field must be coupled to the resolved field variables. The resolved field provides the pressure and the sub-grid kinetic energy to the sub-grid LEM simulation. After the sub-grid simulation, the sub-grid LEM field provides the filtered species, temperature and the specific heats to the resolved field.

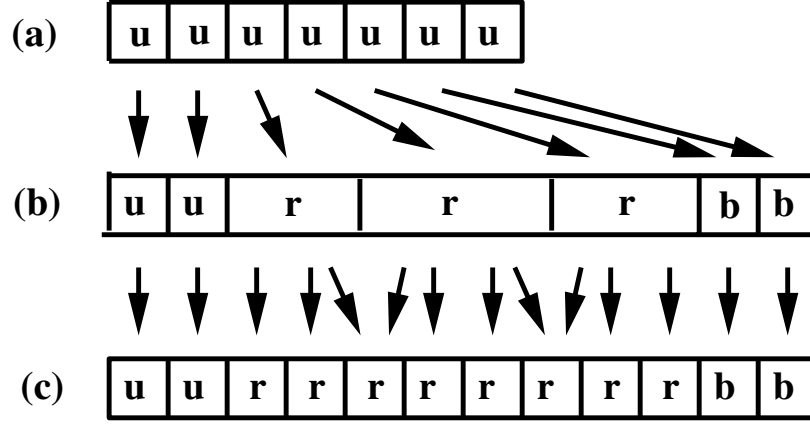


Figure 4.4: Schematic diagram of the re-gridding procedure used to model volumetric expansion of LEM cells.

Filtered quantities from the sub-grid simulation are computed as follows.

- Let Y_{k_i} be the sub-grid species mass fraction for species 'k' in a cell 'i'
- Let T_i and ρ_i be the sub-grid temperature and density in a cell 'i'
- Sub-grid averaged (filtered) species mass fraction is computed as

$$\widetilde{Y}_{k_i} = \frac{\sum_{i=1}^{N_{LEM}} \rho_i Y_{k_i}}{\sum_{i=1}^{N_{LEM}} \rho_i}$$

- Sub-grid averaged (filtered) temperature is computed as

$$\widetilde{T}_i = \frac{\sum_{i=1}^{N_{LEM}} \rho_i T_i}{\sum_{i=1}^{N_{LEM}} \rho_i}$$

This sub-grid averaged (filtered) temperature is redundant, since the solution of the resolved grid energy equation along with the filtered species (from LEM) also gives another temperature. The comparison of the two temperatures showed that they are in very good agreement with each other. In the present study, the temperature computed from the LES energy equation is used as the resolved grid temperature. However, the sub-grid temperature field is not altered and it is evolved continuously. The filtered density computed by the resolved continuity equation is retained instead of filtering the sub-grid density field.

4.5 Summary of LES-LEM Sub-grid Closure

One cycle of the LES-LEM simulation involves, the following computations.

- Solve and update the filtered conservation equation for mass, momentum and energy using an appropriate numerical scheme and a grid
- Store the mass flux across each LES control surface that is computed as a part of the LES finite volume scheme.
- Perform the sub-grid LEM simulation, i.e, in every LES cell, solve the equations 4.4 and 4.5
- Perform the large scale advection of the scalar field using the splicing algorithm
- Couple the sub-grid simulation to the resolved grid quantities, by filtering (Favre average) the sub-grid species and temperature fields.
- Compute the thermo-chemical properties, like mixture specific heats and characteristic gas constants using the sub-grid averaged species fields.
- Compute the filtered internal energy, filtered temperature and the filtered pressure, using the filtered species field (provided by LEM) and the density from the filtered continuity equation.

CHAPTER V

LES-LEM OF LM6000 COMBUSTOR

"I'll explain this by waving my hand in an appropriate manner"

-Anonymous

This chapter serves to validate the LES-LEM approach discussed in the previous chapter. LES-LEM is subjected to the most rigorous and the useful test, viz., to predict the reacting flow field in an actual gas turbine LM6000 combustor. Note, this chapter addresses the issues regarding the model performance, in the context of premixed combustion only and no spray combustion studies are considered here.

LM6000 is an operational hardware used by GE Power Systems (GEPS) company. LES-LEM is used to predict the highly swirling, turbulent reacting flow-field found in the the LM6000 combustor and the results are compared against the experimental/numerical studies of Hura et al. [1998]. If the combustion model (LEM in this case) fails to predict the correct mixing and burning rates, the flame cannot be anchored/stabilized at the operating conditions of the combustor. Earlier, Kim et al. [1999], Kim and Menon [2000] employed a G-equation combustion model within the (dynamic) LES frame work and obtained good agreement with the experimental predictions.

LM6000 is a rectangular can type combustor with a circular inlet from a pre-mixer/swirler configuration. A schematic diagram of the LM6000 combustor is shown in Fig. 5.1 The inlet consists of a highly swirling jet (Swirl number of 0.56) injected at high pressure (6 atm.) and with a pre-heated temperature of 644 K. Here, the Swirl number is defined as the ratio of axial flux of angular momentum to that of axial flux

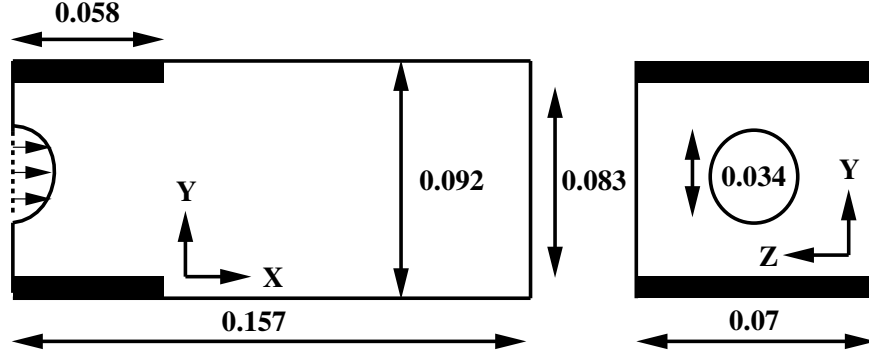


Figure 5.1: Schematic diagram of the GE Power Systems - LM6000 combustor. All dimensions are in meters.

of the axial momentum itself.

$$S = \frac{1}{R_c} \frac{\int_0^{R_c} \rho u w r^2 dr}{\int_0^{R_c} \rho u^2 r dr}$$

where u and w are the axial and azimuthal velocities respectively, and R_c is the radius of the combustor.

The conditions specified at the inlet are the axial, radial and tangential velocities and the turbulent kinetic energy. The inlet velocity profile determines the Swirl number, which in turn determines the key flow-field characteristics such as the formation of the *Vortex Break-down* phenomenon, location of the frontal stagnation point, the extent of the reverse mass flow and the strength of the external recirculation zone. These features in turn determine the flame location and the heat release. Thus, the accurate specification of the inlet conditions is a key-requirement for the correct prediction of the flow-fields. In the present study, the mean velocity profile provided by the GEPS company are used at the inlet and this is same as the profiles used in the earlier studies by Kim et al. [1999] and Kim and Menon [2000]. These inlet velocity profiles are developed (by GEPS) to match the LDV measurements and the observed flame impingement location at the nominal settings of the inlet turbulence parameters (Hura et al. [1998]). The dark region at the top and bottom part of the combustor shown in fig. 5.1 are the blowing sections. These sections are provided to

cool the optical window used for measurements. In the present study, similar to the previous studies (Kim et al. [1999] and Kim and Menon [2000]) hot products were injected through the blowing sections.

Only information provided about the inflow turbulence was the intensity of the incoming turbulence, and that is set equal to 7%. The Reynolds number based on the inlet diameter is 320,000. Based on the characteristic flow and chemical time scales, Kim et al. [1999], argued that flame conditions are that of those in the thin-reaction zone regime of the premixed combustion diagram.

No-slip, adiabatic wall boundary conditions are used for the combustor's walls. Characteristic boundary conditions were used for the subsonic inflow and outflow (Poinsot and Lele [1992]).

A structured rectangular computational grid that uses $151 \times 118 \times 92$ grid points is employed in the present study. The present finite-volume solver uses an embedded boundary method for handling the circular inflow in a Cartesian rectangular grid Peskin [2002]. The embedded boundary method is based on setting the values of the state vector at particular locations (in the finite volume cells) inside the solid body in such a way that the solid wall boundary conditions are satisfied at the solid wall. The method presents several advantages. The main one is its ability to manage complex geometries. In the case of a gas turbine combustor, since the geometry is circular in nature, a traditional approach will require a cylindrical grid, where the volume of the finite volume cells will be too small near the center line. This in turn would impose severe time-step restrictions for a numerically stable integration. An alternate method, the unstructured grid approach, will be able to handle the centerline problem, but will have overwhelming computational requirements, both in terms of CPU time, and of memory. Since the final goal is the LES of a reactive flow, which is very computationally demanding by itself, the computational expenses of an unstructured grid solver can be prohibitive.

Another advantage of the method is the fact that it allows the numerical scheme to employ a Cartesian grid irrespective of the complexity of the geometry involved. This allows for orthogonal cells, which diminishes the numerical error and improves the stability of the numerical solution. In addition, the aspect ratio of the cells is much easier to control. Since it is not in the scope of the present thesis, more details about the embedded grid approach are not given here and can be found elsewhere Peskin [2002]

5.1 Results and Discussion

Two simulations were performed to compare and contrast the features of the LES-LEM simulations. First simulation employs the dynamic-LES with a G - equation flamelet approach (G-LES) and the second simulation employs the LES-LEM approach. Before proceeding with the comparison, the rationale of the G-LES model is introduced here. Note, G-LES modeling is not in the scope of the present thesis and hence the details are only briefly summarized here. More information can be found elsewhere (Kim and Menon [2000]).

5.1.1 G-LES

In the G-equation approach, the flame-front is tracked as an infinitely thin surface that is convected by the flow and also propagates normal to itself at a characteristic flame speed S_F . The governing equation for this progress variable is based on the G -equation model:

$$\frac{\partial \rho G}{\partial t} + \frac{\partial}{\partial x_i} (\rho u_i G) = -\rho S_F |\nabla G| \quad (5.1)$$

Here, G is a progress variable whose value varies between $G = 1$ representing premixed fuel and $G = 0$ representing the burnt products. In laminar flow, $S_F = S_L$ which is the laminar burning speed. This speed contains the effect of the entire

thermo-chemical state of the fuel-air mixture. Since the equivalence ratio along with the thermo-dynamic properties, uniquely defines S_L in a laminar flow, the flame propagation can be tracked without any need for expensive finite-rate kinetic computations.

The propagation of the turbulent premixed flames in the thin-flame (flamelet) limit can also be modeled using Eqn. [5.1]. In the flamelet burning regime of turbulent premixed combustion, flame thickness (δ_L) is smaller than the smallest length scale (Kolmogorov scale η) and the burning time scale (τ_c) is smaller than the characteristic flow time (τ_f). In this case, the thin laminar flame is wrinkled by the turbulent eddies, thereby increasing the instantaneous flame surface area and hence, the effective burning or consumption rate. However, since even the smallest eddy is larger than the flame thickness they cannot penetrate into the flame and thus, the local burning speed is still given by the laminar flame speed S_L . It is worth noting that although flamelet burning is prevalent in gas turbine combustors, there are also local regimes where the turbulence levels are so high that the smallest eddy is smaller than the flame thickness. This regime of burning is called distributed reaction zone or thin reaction zone (Peters [2000]) and in this regime, the smaller eddies penetrate into the flame zone thereby, thickening the flame. However, Peters [2000] have shown that even in this regime, the reaction zone is very thin and hence, flamelet burning approximation can still be employed. Models that explicitly account for flame broadening effect in the thin-reaction zone regime have also been developed (Pocheau [1994]).

To model combustion in this regime, the above equation is filtered using the LES filter to obtain the filtered G -LES equation as follows:

$$\frac{\partial \bar{\rho} \tilde{G}}{\partial t} + \frac{\partial}{\partial x_i} (\bar{\rho} \tilde{u}_i \tilde{G}) = - \frac{\partial}{\partial x_i} \bar{\rho} (\widetilde{u_i G} - \tilde{u}_i \tilde{G}) - \overline{\rho S_F |\nabla G|} \quad (5.2)$$

Note that in the present form, the LES-resolved variable $\tilde{G}$ can range from 0 to 1 and represents a measure of the flame brush thickness. This is in contrast to the

original “laminar” interpretation of the G field where $G = 0$ denotes burned product and $G = 1$ denotes reactants and the flame is considered an infinitely thin interface between $G = 0$ and $G = 1$. In the LES implementation, the filtered $\tilde{G}$ represents the resolved flame brush which is an average location of the instantaneous thin flames.

The right hand side of Eqn. [5.2] needs to be modeled. The sub-grid convection term is modeled using a gradient assumption that incorporates the effect of the curvature of the flame as (Peters [2000]):

$$-\frac{\partial}{\partial x_i} \left(\widetilde{u_i G} - \tilde{u}_i \tilde{G} \right) = -\bar{\rho} D_T \tilde{\kappa} |\nabla \tilde{G}| \quad (5.3)$$

where $\tilde{\kappa}$ is the flame curvature defined as:

$$\tilde{\kappa} = \nabla \mathbf{n} \quad (5.4)$$

and

$$\mathbf{n} = -\frac{\nabla \tilde{G}}{|\nabla \tilde{G}|} \quad (5.5)$$

Here, $\mathbf{n}$ is the unit normal oriented in the direction of flame propagation. The flame front propagation term $\overline{\rho S_F |\nabla G|}$ is closed using a turbulent flame speed model.

Thermodynamic coupling appears through the filtered internal energy $\tilde{e} = c_v \tilde{T} + \Delta h_f \tilde{G}$, where $\Delta h_f = c_p(T_p - T_f)$ is the heat of formation, c_v and c_p are the specific heats at constant volume and pressure, respectively. T_p and T_f are the product and fuel temperatures, respectively. According to flamelet theory, $\tilde{T} = (\tilde{e} - \Delta h_f \tilde{G} H(G - G_0))/c_v$ where $H(G - G_0)$ is a Heavy-side function of G . However, this approach will result in numerical instability due to the sudden jump conditions across the flame. For this reason, the temperature is allowed to vary linearly with $\tilde{G}$ which broadens the heat-release zone.

5.1.2 Turbulent flame speed model

Basically, turbulent flame speed models are used to provide a functional relationship between the normalized turbulent flame speed S_T/S_L and normalized turbulence intensity u'/S_L . Unfortunately, the turbulent flame speed is not a well defined quantity unlike the laminar flame speed, which can be determined using detailed chemical kinetic schemes. The experimental data for turbulent flame speed exhibit a large scatter and depend on various parameters (Poinsot and Veynante [2001]) thus rendering it difficult to obtain a universal model, that is valid in all regimes of premixed combustion. Therefore, most of the available models use empirical correlations (Andrews et al. [1975]) and heuristic arguments obtained from calibrating the experimental data. Flame speed models based on RNG theory such as those used in the present G-LES study are mostly based on constant density flows (Yakhot [1988]).

Therefore, assuming that a turbulent flame speed S_T exists and that, $S_T = S_T(u', S_L)$, where u' is sub-grid turbulence intensity, the following model based on the Pocheau's model (Pocheau [1994]) can be written as

$$\frac{S_T}{S_L} = \left(1 + \beta \frac{u'^\alpha}{S_L^\alpha}\right)^{\frac{1}{\alpha}} \quad (5.6)$$

where $\alpha=2$ is chosen to conserve the energy and β is the minimum of 0.8165 u'/S_L and 16.56 (Pocheau [1994]). This minima is imposed such that S_T tends to zero when S_L tends to zero.

A localized dynamic model that extends S_T model to the thin-reaction zone is used here (Kim and Menon [2000]). Studies have shown that this closure allows the study of gas turbine combustion over a wide range of operating conditions (i.e., flamelet and thin-reaction zone regimes) using the same flame propagation model. This is important since in real gas turbine combustors, a wide range of flame characteristics can co-exist. However, when turbulence starts to penetrate the thin reaction zone, flame stretch by small eddies changes the chemical reaction rate and often lead to

quenching. The above flamelet model cannot be used in this regime and hence a domain of validity has to be enforced for the flame speed models.

In the following section the results obtained from G-LES and LES-LEM are compared.

5.1.3 Model performance comparison: G-LES Vs LES-LEM

Before proceeding with the comparison a note on the perspective of the comparison is due here. The G-LES approach along with the various sub-models such as the turbulent flame speed models have been fine tuned to capture the flame-turbulence interaction in the wrinkled and thin-reaction zone regimes. Several parameters such as β, γ for the broadened flame speed model, turbulent Schmidt number, Sc^G , parameter A in the relation between the broadened flame thickness and the broadened flame speed (Kim and Menon [2000]) appear in the flame-speed model. Even though some of the model constants such as the Sc^G can be estimated dynamically, earlier studies (Kim and Menon [2000]) have employed empirical constants. For instance, $\gamma = 2$ and $\beta = 20$ gave better results than the dynamically determined β . In fact, $\beta = 20$ is the only case for which the earlier study did not over predict the extent of the V-B zone. Hence, it is no surprise that these models perform well in predicting the flame-turbulence interactions in the flamelet and thin-reaction zone regimes. On the other hand LES-LEM does not use such “zone-specific ad-hoc” flame speed models and tune-up coefficients that depends on the regimes of the combustion. Therefore, a similar level of agreement (with experiments) using LES-LEM should imply that the LES-LEM approach is performing correctly, capturing the proper physics of flow. In addition, the fact that no empirical inputs are needed for LES-LEM should imply that this the level of agreement is remarkable.

Statistics such as the mean and the RMS quantities were computed over 5 flow through times. Before sampling the data for statistics, the flow was evolved for

approximately 2 flow through times. All the results presented in this section are ensemble averaged over 5 flow through times based on the mean center-line axial velocity at the inlet. All the velocity profiles shown are non-dimensionalized by the center-line velocity at the inlet U_O and the distance are non-dimensionalized by the inlet diameter D . The current simulations were carried out on a distributed memory parallel processing computers (Cray-T3E, 1200 MHz) using the Message Passing Interface (MPI). Typically 160 processors were employed for present simulations. For the present simulation using ≈ 1.6 million LES grid points and 12 LEM cells per LES cell, it required 3840 single processor hours on the Cray T3E per flow through time. LES-LEM without chemistry (scalar mixing alone) is very efficient and it is only 10 % of the total cost. It is the stiff chemistry integration that makes the simulation expensive. The memory requirements is 2.4 Gigabytes.

5.1.4 Mean velocity profiles

Figure 5.2 shows the variation of the axial mean velocity along the center-line of the combustor. The flow evolution is very similar to that of a swirling decaying jet. The swirl imparted to the flow at the inlet causes the familiar *Vortex Break-down* (V-B) phenomenon. V-B is a remarkable feature found in many swirling flows, such as those in gas turbine combustors. V-B is typically characterized by the presence of an appreciable axial velocity super-imposed upon the rotational motion. When the ratio of the azimuthal to axial velocity exceeds certain level (of order unity), V-B occurs, i.e. vortex core, abruptly widens into a new stable coherent structure. A predominant feature of the V-B is the abrupt deceleration of the flow near the axis leading to the formation of a stagnation point, as if the flows sees a solid obstacle in its path. Such a formation of stagnation point in the flow-field is of great significance for stabilizing the flame/reaction front in a combustor. This is the primary reason why most of the modern high speed combustors impart swirling motion to the inflow air. As an added

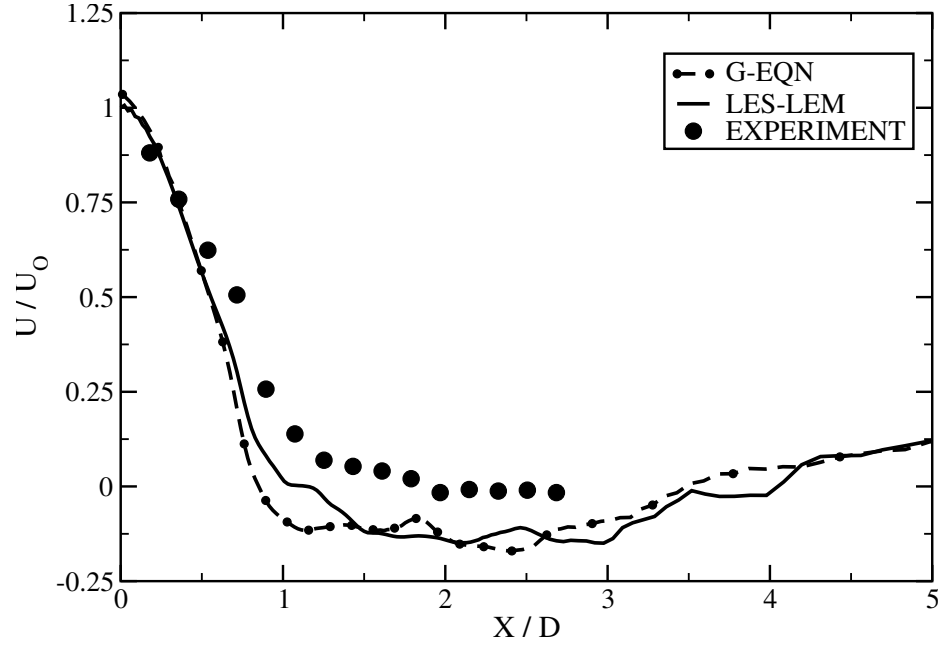


Figure 5.2: Center-line variation of mean axial velocity.

benefit swirl imparted to the air, also helps to minimize the length of the combustion zone, thus reducing the size of the combustors.

Comparison of the center-line axial velocity variation shows that even though both the combustion models (LES-LEM and G-LES) show reasonable agreement with the experimental data, G-LES slightly over-predicts the center-line decay compared to the LES-LEM. To obtain a similar level of agreement, Hura et al. [1998] obtained a similar match with LDV measurements using RANS models, but only after a significant number of iterative procedures that involved changing the inlet conditions and the model parameters. Considering the fact that these simulations were done without any need for ad-hoc adjustments, this level of agreement is considered excellent.

Figure 5.3 shows the contours of the axial velocity in the LM6000 combustor predicted using LES-LEM. The location of the V-B is marked by the dark-blue contours around the center-line. As it can be observed the region of break-down is asymmetric and extends far downstream. The extent of V-B measured in the experiment was ≈ 1 diameter. The axial extent of the VB predicted by LES-LEM is 2.2 diameters,

whereas the G-LES predicts it to be 2.7 diameters. The reason for the over-prediction of the extent of the V-B zone, is attributed to the incorrect inlet radial velocity profiles (Hura et al. [1998]). The premixer for the actual LM6000 hardware is designed in such a way that the radial component of velocity at the premixer exit plane is much less than the axial and the tangential components of velocity (Hura et al. [1998]). Hence, the radial velocity LDV measurements of the radial velocity at the inlet plane were not reliable. So, Hura et al. [1998] conducted a series of iterations with different radial velocities to match the LDV measurements. Hura et al. [1998] analyzed the sensitivity of the flow-field to the inlet radial velocity profiles by using 3 different radial profiles at the inlet and showed that the in-accurate radial velocity profiles can over-predict the size of the V-B region.

Even though the same profiles are used in the present study, it is not necessary that the boundary conditions (inlet velocity profiles) that give a certain solution in RANS should provide a similar solution in LES. RANS solutions are inherently dissipative and as mentioned earlier in the chapter 1, its prediction are not reliable in the presence of swirl and stream-line curvature. Hence, the inlet radial profiles used in RANS by Hura et al. [1998], that gave good agreement with the experiments need not be the correct inlet velocity profile that was observed in the experiments. In addition, the fact that 2 completely different models (G-equation and LES-LEM) using the same boundary conditions are predicting similar results (different from experiments), indicates that it is the boundary condition that is strongly influencing the results.

Figure 5.4 shows the radial profiles of axial velocity at $X/D = 0.18$ predicted using G-LES and LES-LEM. It can be observed that the LES-LEM predicts a better shear layer spread compared to the G-LES. Figure 5.5 shows the mean radial velocity profiles at $X/D = 0.18$ and $X/D = 0.72$. LES-LEM prediction is reasonably better (but not completely) than the G-LES approach, even though the over-all agreement

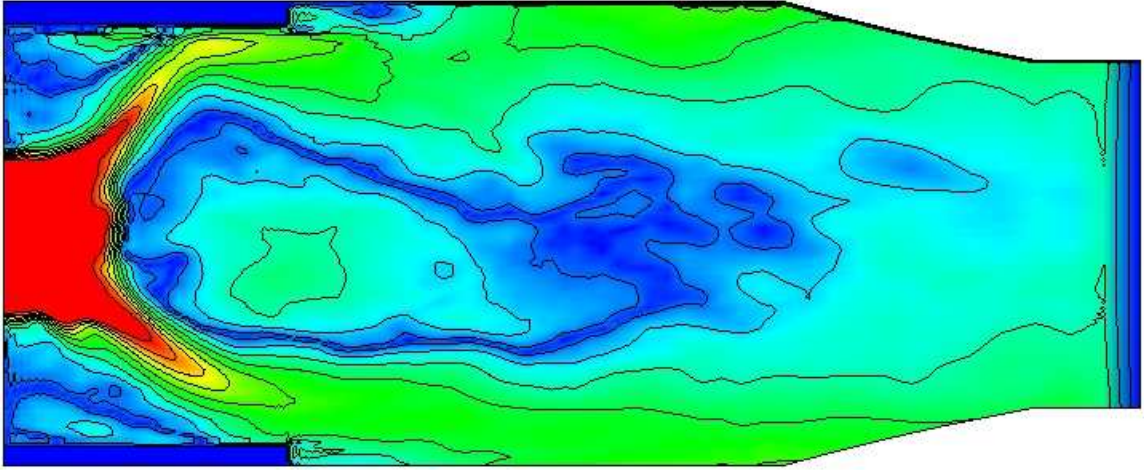


Figure 5.3: Mean axial velocity contours predicted using LES-LEM.

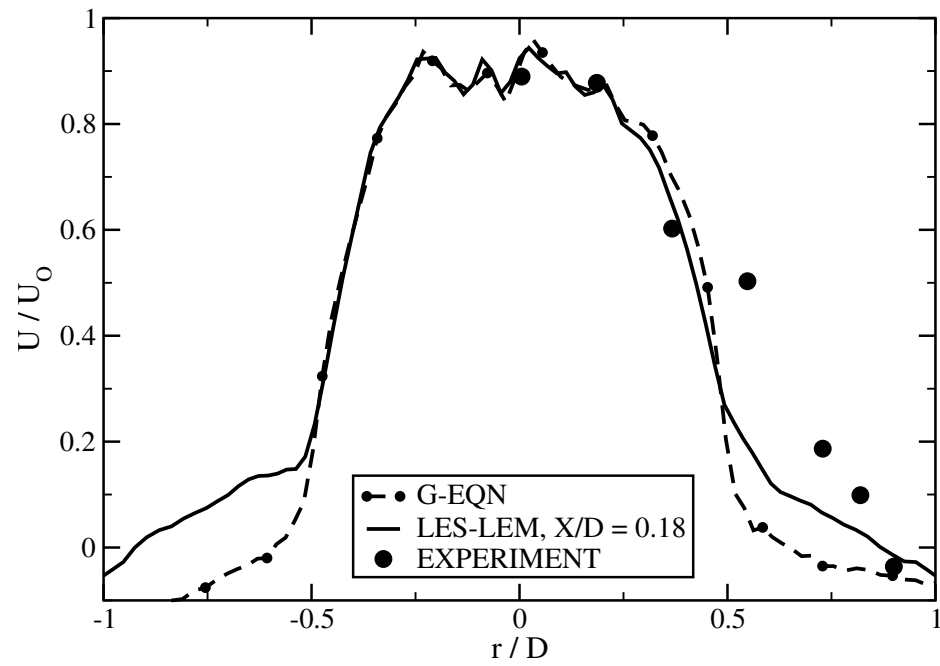


Figure 5.4: Radial variation of mean axial velocity profile.

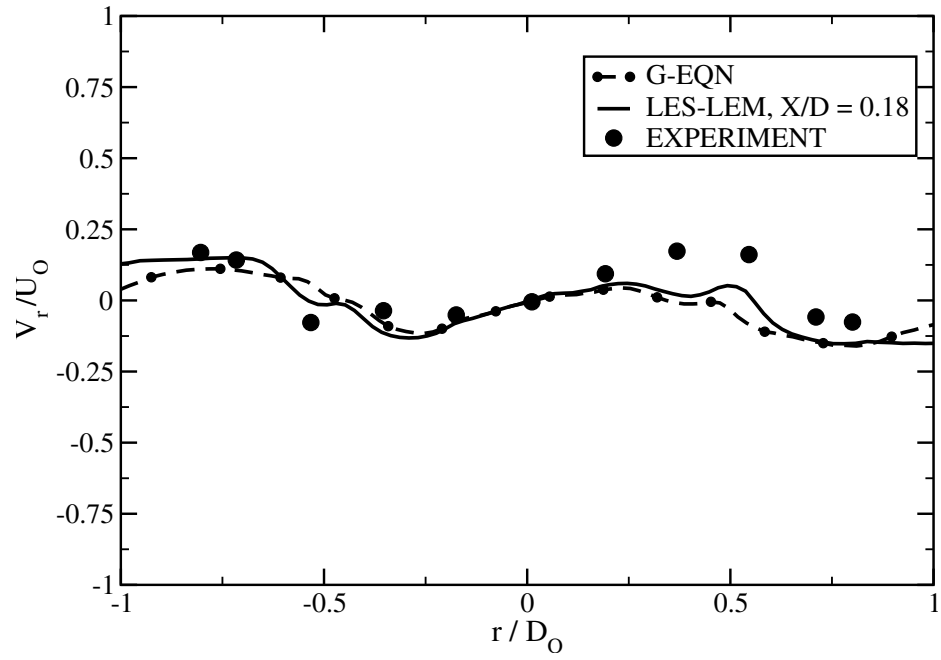
of both the models are good.

Figure 5.6 shows the radial profiles of the non-dimensionalized mean tangential velocity. Both G-LES and LES-LEM predict peaks at $r/D \approx 0.38$ similar to those in the experiments. Earlier study by Kim and Menon [2000] also showed similar peaks and it was attributed to the in-adequate resolution of the inlet velocity profile measurements by GEPS. The overall prediction of the mean velocity fields, (U, V , and W components) by LES-LEM is in good agreement with the LDV measurements and the G-LES approach. This good agreement in the velocity field also implies that the thermal field has been captured very well in the present simulations.

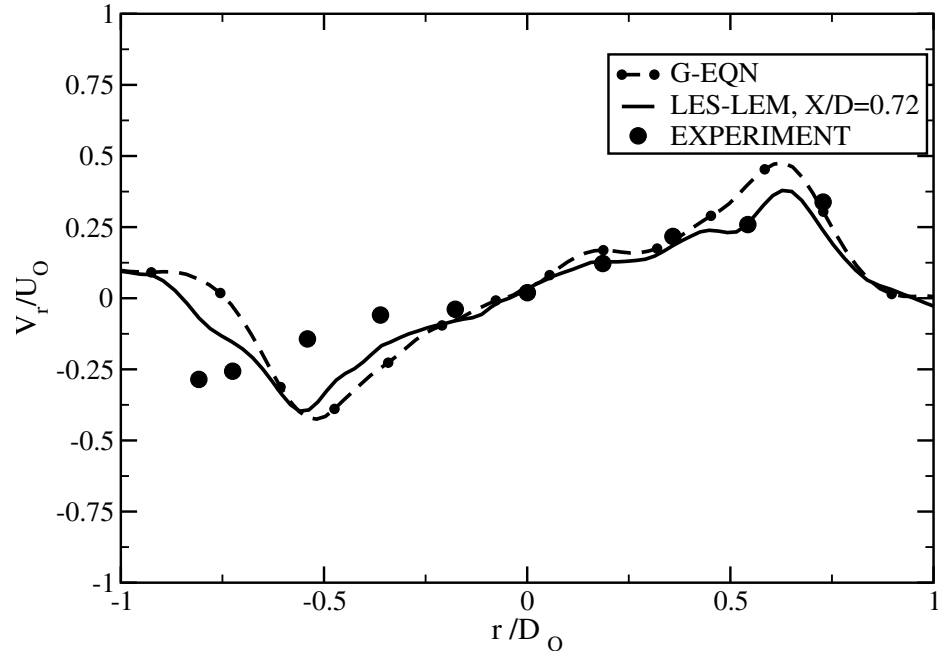
5.1.5 RMS velocity profiles

Figure 5.7(a) shows the RMS profiles of the axial velocity, u' along the center-line of the combustor. Even though both G-LES and LES-LEM under predicts the turbulence, LES-LEM shows a closer agreement than the G-LES. Moreover, LES-LEM predicts the rise in u' at $X/D \approx 1$ as seen in the experiments. This corresponds to the location at which the flame crosses the center-line of the combustor. Flow acceleration through the flame enhances the turbulences hence increases the RMS of the axial velocity fluctuations. Not only does the G-LES under predicts the magnitude, but it also seem to have missed the trend of the experiments. This is because, G-LES only captures the flame wrinkling at the LES scales. Effects of the fine (sub-grid) scale wrinkling are realized only through the flame-speed models which are not valid in the broken-reaction-zone regimes. LES-LEM on the other hand simulates the sub-grid scale wrinkling and hence can capture the flame-turbulence interaction at all the scales. Figure 5.7(b) shows the RMS profiles of the axial velocity, u' at a radial location ($r/D = 0.81$) near the shear layer. Again the magnitude and the trend of LES-LEM predictions are in very good agreement with the experiments.

Figure 5.8(a) shows the radial profiles of RMS axial velocity, u' at a $X/D = 0.18$.

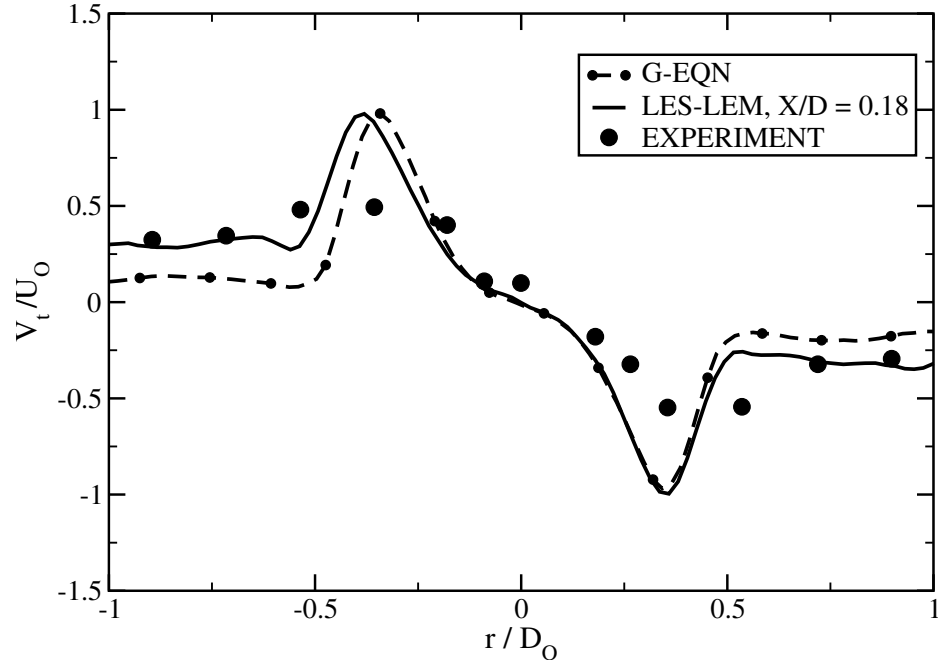


(a) $X/D = 0.18$

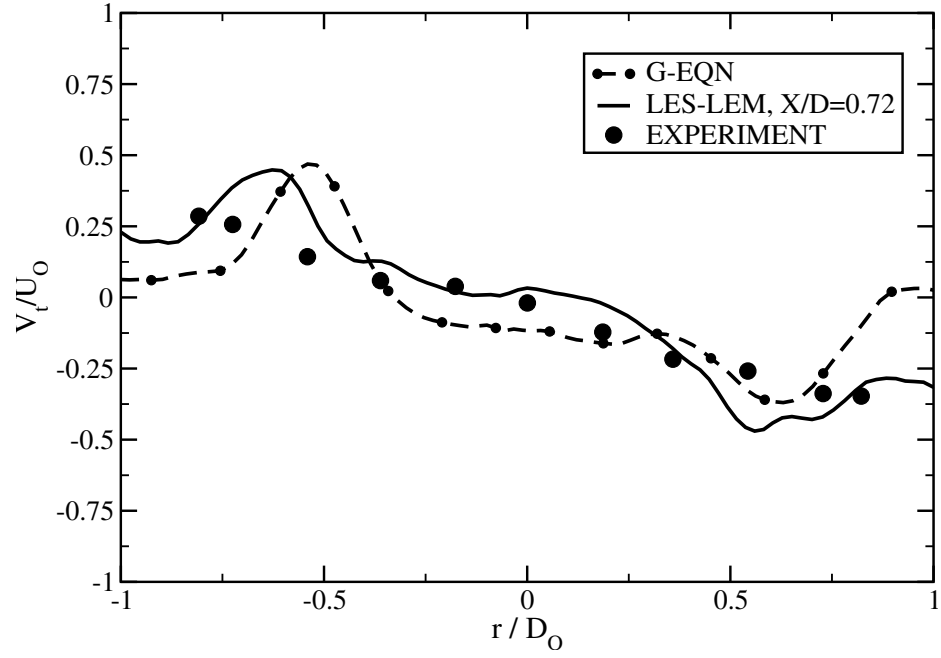


(b) $X/D = 0.72$

Figure 5.5: Radial variation of mean radial velocity profiles.

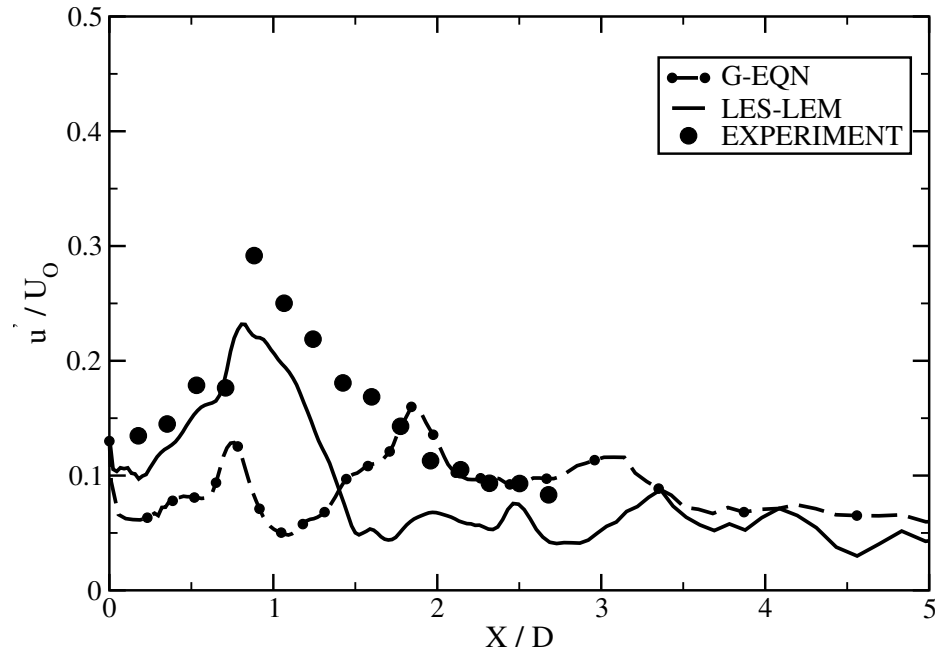


(a) $X/D = 0.18$

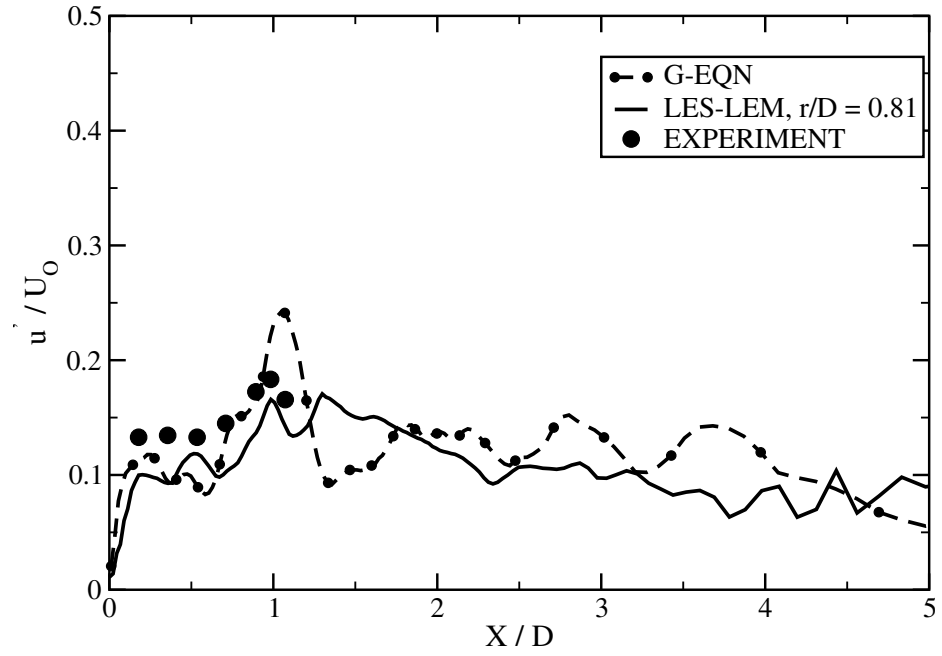


(b) $X/D = 0.72$

Figure 5.6: Radial variation of mean tangential velocity profiles.



(a) $r/D = 0.0$



(b) $r/D = 0.81$

Figure 5.7: Axial variation of axial RMS velocity profiles.

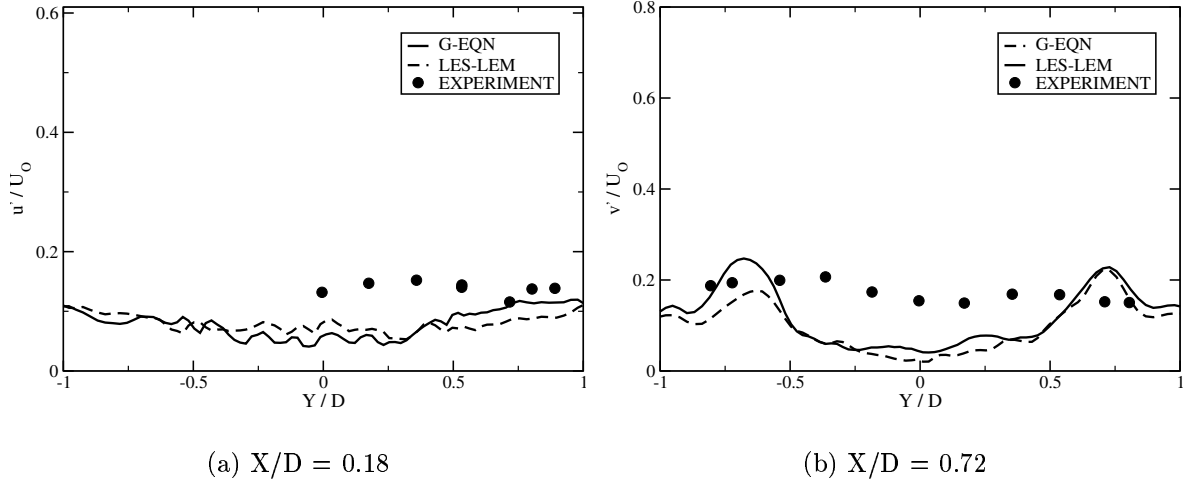


Figure 5.8: Radial variation of axial RMS velocity profiles.

Both G-LES and LES-LEM under-predicts the the fluctuations. However, both G-LES and LES-LEM clearly predict a peak in the RMS velocity at the shear layer consistent with the physics of the flow. It is surprising to note that the experiments do not show any increase in the radial fluctuations (fig. 5.8(b)) near the shear layer or in the flame zone. Since the details of the measurements such as the error margin in the measured fluctuating quantities, are not available, it is difficult to judge the model predictions for the RMS fields.

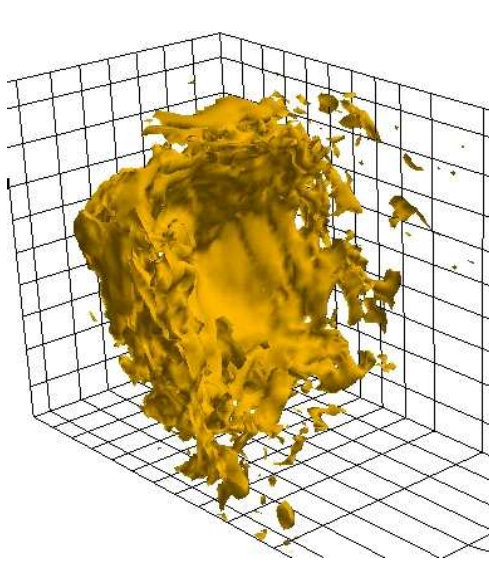
Figure 5.9 shows the instantaneous and mean iso-surface of temperature, obtained using G-LES and LES-LEM approach. The magnitude of the iso-surface shown is 2200 K. Interaction between the turbulence and the chemistry can be clearly seen from the intense wrinkling of the flame surface due to turbulent eddies. The formation of a rib-like structure at the base of the flame (fig. 5.9(b) and (d)) clearly shows the effect of swirl on the flame structure. Swirl imparted to the flame, creates a wide, but short flame that spans approximately 1/3-rd of the length of the combustor. LES-LEM seems to predict a more compact flame compared to G-LES in both the mean and the instantaneous snap-shots. Figure 5.10 shows the temperature contours at the

mid-plane in the span-wise direction.

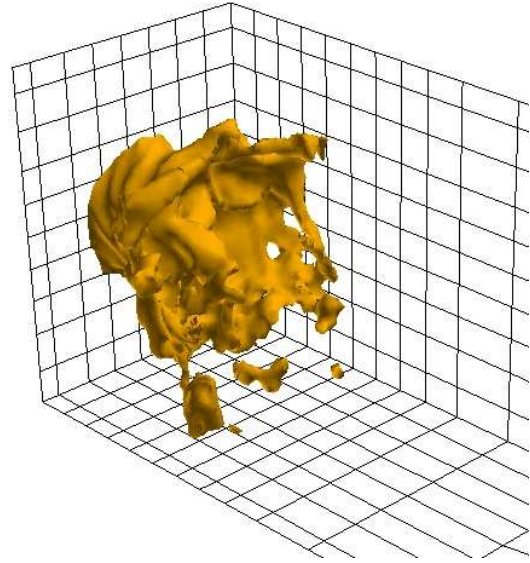
An interesting feature of this figure is that LES-LEM captures the flame in 2 LES grid points and still retains numerical stability. This is possible only because the dynamics of the flow and the flame at the sub-grid scales are resolved completely. The schematic diagram shown in fig. 5.11 is used to explain the numerical stability gained by using LES-LEM approach. Suppose an LES grid as shown in fig. 5.11(a) is used to resolve the flame structure shown as a red curve (temperature profile). There are 2 issues involved here. First, the interaction between the flame and the sub-grid scale eddies cannot be resolved using such a grid and second, the presence of high gradients in the temperature profile would induce numerical oscillations and induce numerical instability. On the other hand, fig. 5.11(b) shows the flame resolved with LES-LEM approach. Even though the flame is captured in 3 LES cells, the fine-scale structure of the flame is well resolved in the sub-grid LEM domain. This aspect of LES-LEM provides a close coupling between the fluid-dynamics and the chemistry and gives a improved numerical stability.

5.2 Conclusions

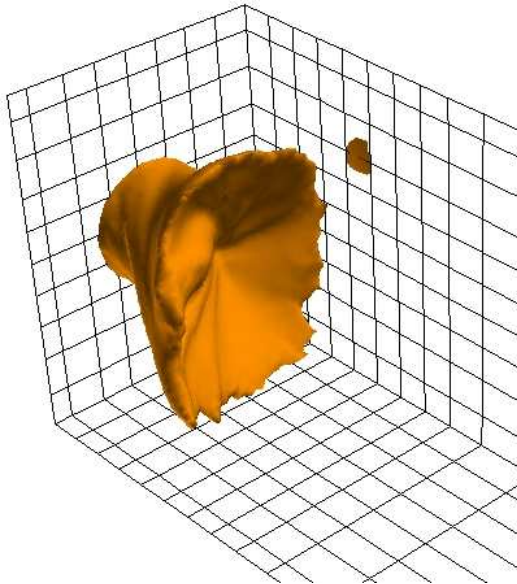
A sub-grid turbulent combustion model, LES-LEM has been developed and used to predict reactive flow-field in a realistic combustor. The results obtained from the LES-LEM approach is compared with LDV measurements and another combustion model, G-LES approach which has been validated in the past (Kim and Menon [2000]). LES-LEM performs reasonably well in predicting the complex (high Re, swirling, compressible, reactive) flow-field encountered in GEPS's operational hardware (LM6000 DLE combustor). The RANS approach has been used in the past (Hura et al. [1998]) to model the LM6000 combustor. However, significant amount of empiricisms and iterations were needed to get similar level of agreement with the measurements. The approach presented here is more robust and physically intuitive,



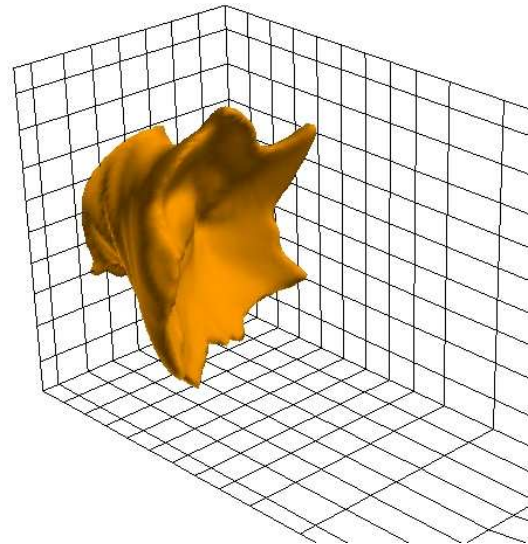
(a) Instantaneous G-LES



(b) Instantaneous LES-LEM

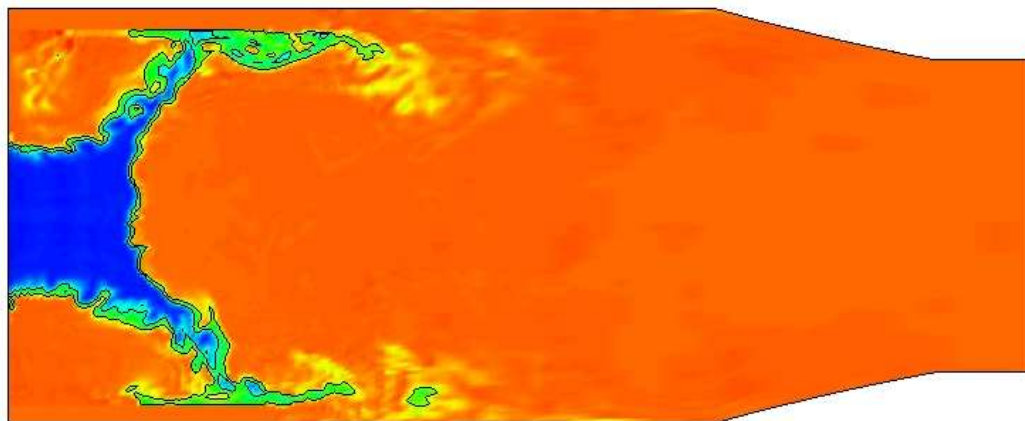


(c) Mean G-LES

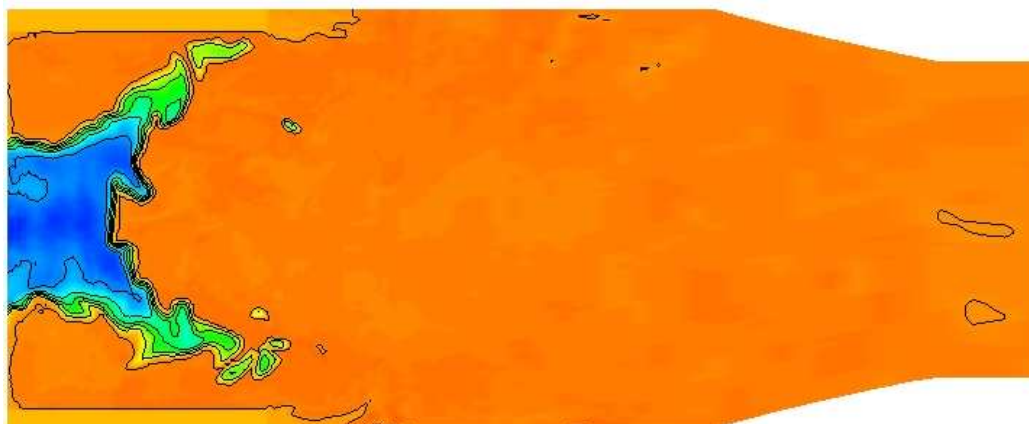


(d) Mean LES-LEM

Figure 5.9: Iso-surface of the instantaneous and mean flame structure.



(a) Instantaneous G-LES



(b) Instantaneous LES-LEM

Figure 5.10: Contours of the instantaneous flame structure.

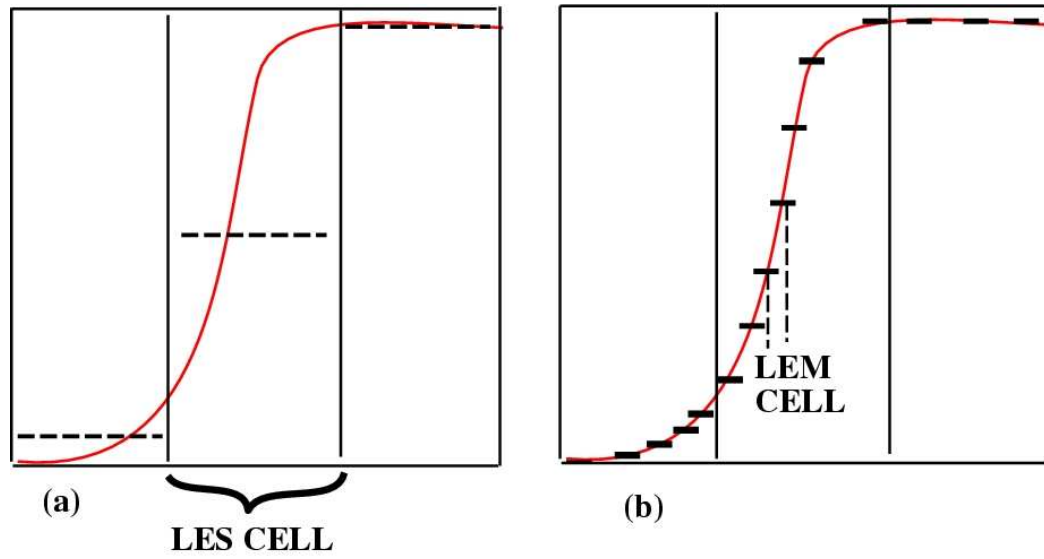


Figure 5.11: Schematic diagram illustrating the LES-LEM numerical stability.

which can capture the flame turbulence interactions at all scales without any need for ad-hoc adjustments that depend on the regime of the combustion

CHAPTER VI

DNS & LES OF TEMPORAL MIXING LAYERS

This chapter describes the validation studies used to demonstrate the accuracy of the droplet tracking algorithms and the spray models for the two-phase flows. For this purpose, studies were conducted using DNS and LES of droplet-laden temporal mixing layers and the results compared with spectral simulations of Ling et al. [1998] using DNS.

Free shear layers have been a subject of intensive research due to their great technological significance in many practical devices such as combustors and scram-jet engines. Free shear layers are those formed by the merging of two streams initially separated. Formation of large scale coherent structures and subsequent pairing are intrinsic feature of these shear layers. Numerous experimental, analytical and numerical studies have been conducted in the past on free shear layers (Brown and Roshko [1974], Winant and Browand [1974], Metcalfe et al. [1987]).

The present study focuses on the DNS & LES of the droplet laden flows applied to a turbulent shear layer configuration. The temporal mixing layer (TML) provides a relatively simple, non-homogeneous flow geometry which removes some of the restrictions imposed by homogeneous turbulence studies. Albeit some differences between spatially evolving shear layers and temporally evolving mixing layers, there are many similarities in the dynamical features of the two shear layers (vortex-roll up and pairing). Hence the results obtained from a TML simulation can be very useful in elucidating many of the fundamental mechanisms of turbulent transport and droplet dispersion in such shear layers. Results presented in this chapter includes the evolution of energy growth in the fundamental and subharmonic modes, evolution of

momentum thickness, Stokes number effect on the dispersion behavior of the particles, and the RMS of the droplet concentration. It should however be noted that these are presented here just to validate the droplet models, particle-tracking algorithms, and the coupling between the continuum and the dispersed phase in compressible flows and not to reveal any new physics.

6.1 Boundary and Initial conditions

The numerical technique uses a second-order-time and fourth-order-space accurate MacCormack explicit scheme as explained in Chapter 3. The computational grid used $64 \times 64 \times 64$ grid points. The grid is uniform in all the directions. Stream wise(X) and span wise(Z) boundary conditions are treated as periodic, whereas the boundaries in the transverse direction are treated as free-slip walls.

The flow configuration studied is that of a 3-D temporally evolving mixing layer, formed by the merging of two co-flowing parallel streams with different momentum. From the linear stability analysis (Michalke [1964]) it is well known that this flow is extremely unstable and responds to any background or imposed perturbation by forming large scale coherent structures, similar to those, that are initiated at the tip of a splitter plate in a spatially evolving mixing layers. In fact, a temporally evolving mixing layer can be related to spatially evolving shear layers by introducing a Galilean transformation which relates the stream wise co-ordinate to the time. In this Galilean transformed domain one may assume periodic boundary conditions in the stream wise direction and the transport variables evolve in time rather than in space.

The initialization of the flow field consists of the mean field and the perturbations on the mean. The initial mean velocity field is given by a tangent hyperbolic profile, $U(y) = (U_o/2)\tanh(y/\delta)$, where δ is the initial vorticity thickness. To trigger the growth of the coherent structures, an initial perturbation that has the form of

the eigenfunction corresponding to the most unstable mode (or the fastest growing mode/disturbance) for the hyperbolic tangent profile has to be applied (Metcalf et al. [1987]). Furthermore, if a vortex pairing event is to be triggered, the initial vorticity is given a subharmonic perturbation, in addition to the fundamental perturbation. The wave length of this subharmonic perturbation is twice that of the basic one, while its phase is such that its amplitude reaches a maximum midway between the two evolving vortices, so that vortex pairing is obtained (Metcalf et al. [1987]). In the present simulation, the size of the computational domain is chosen to be 4π in all three directions and the initial perturbation applied is such that it would allow for a vortex roll-up of the fundamental mode and one pairing event.

6.2 DNS of Mixing Layer

Here, the results obtained from the DNS of a droplet-laden mixing layer for Stokes number ranging from 0 – 100 are presented. Flow conditions are chosen to exactly match the pseudo-spectral simulations of Ling et al. [1998]. The Reynolds number based on the velocity scale $U_o = 10m/s$ and a reference scale $L_o = 1$ is 500 for all the cases simulated. Results are presented for the mixing layers subjected to primary and secondary instabilities. Primary instabilities are usually triggered by two-dimensional perturbations, resulting in the formation of large coherent vortical structures in the span wise planes. On the other hand three-dimensional perturbations trigger secondary instability that leads to the formation of stream wise vortices.

Figure 6.1 shows the contours plots of the vorticity magnitude for $St = 0$ (Unladen) case at different non-dimensional times, where non-dimensional time T , is defined as $T = tU_o/L_o$. This figure clearly shows the unsteady evolution of the flow subjected to perturbations corresponding to the fundamental mode and its first subharmonic mode. Due to the forcing at the fundamental mode, the mixing layer forms large coherent structures at later times. This is evident from the vortex roll-up at

$T = 12$, shown in Fig 6.1. The perturbations corresponding to the first subharmonic mode then causes the neighboring vortices to merge leading to the familiar vortex-pairing event observed in earlier numerical (Metcalf et al. [1987]) and experimental (Brown and Roshko [1974]) studies. This phenomenon of pairing leads to the growth and widening of the mixing layer.

Figure 6.2 shows the growth rate of the energy in fundamental and subharmonic modes, simulated using the second- and fourth-order numerical scheme. Energy in the fundamental mode $E(1, 0)$ corresponding to the stream-wise wave number $k_x = 1$ and subharmonic mode $E(0.5, 0)$ corresponding to $k_x = 0.5$ are computed as

$$E_{k_x} = \int_{-L/2}^{+L/2} u_i(k_x) \hat{u}_i^*(k_x) dy \quad (6.1)$$

where $\hat{u}_i^*$ is the complex conjugate of u_i . Even though both second- and fourth-order schemes show good agreement with the DNS of Ling et al. [1998], fourth-order schemes show excellent agreement for longer times. The energy in the fundamental mode attains its maximum at $T = 10.8$. This is defined as roll-up time (Ling et al. [1998]) corresponding to the unstable fundamental mode. Similarly, another time scale called the pairing/coalescence time can be defined as the time at which the subharmonic mode attains its maximum energy. Pairing time occurs at $T = 22$, approximately twice the roll-up time.

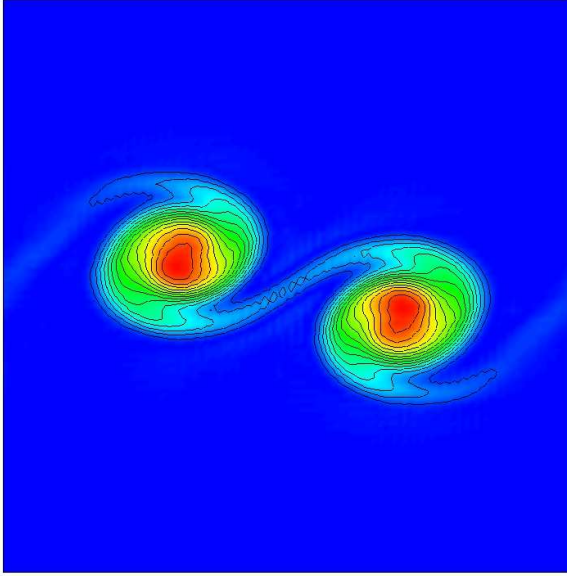
Figure 6.3 shows the evolution of the momentum thickness with time. Momentum thickness is defined as

$$\delta_m = \int (1/4 - \overline{U(y)^2}) dy.$$

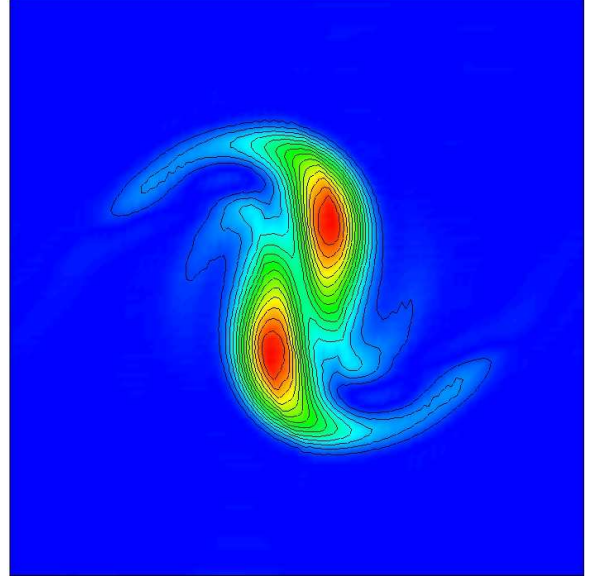
where

$$\overline{U(y)} = \int_{-L_x/2}^{+L_x/2} \int_{-L_x/2}^{+L_x/2} u_1(x, z) dx dz / L_x L_z U_o$$

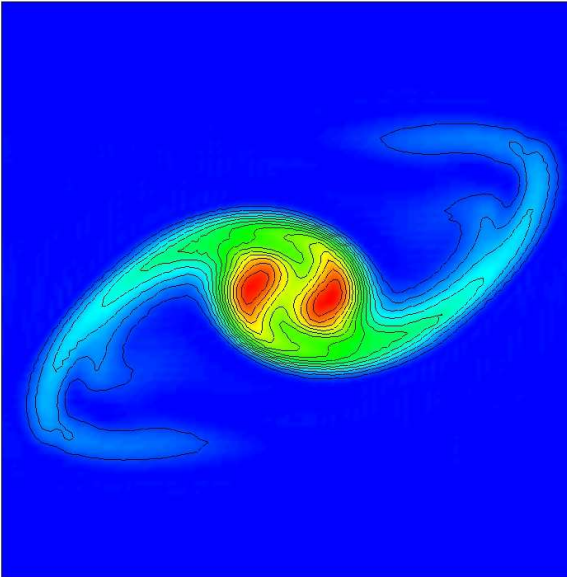
As it can be observed from Fig. 6.3, momentum thickness of the mixing layer grows continuously till the vortex pairing event and then begins to decrease. Since the initial



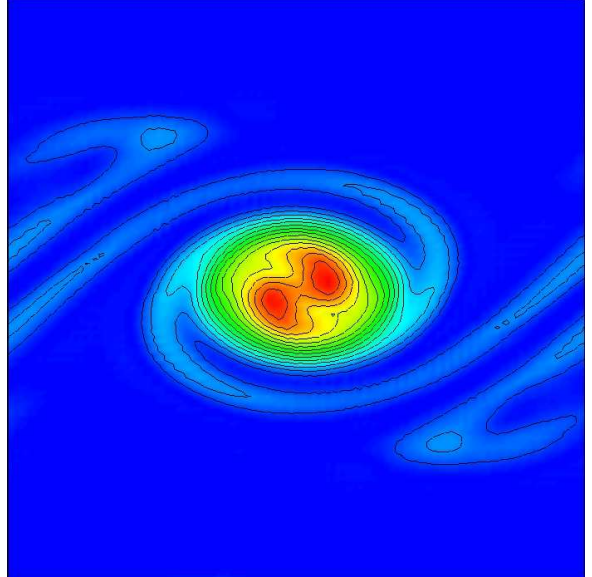
(a) $T = 12$



(b) $T = 24$



(c) $T = 36$



(d) $T = 48$

Figure 6.1: Evolution of span-wise vorticity in a temporal mixing layer.

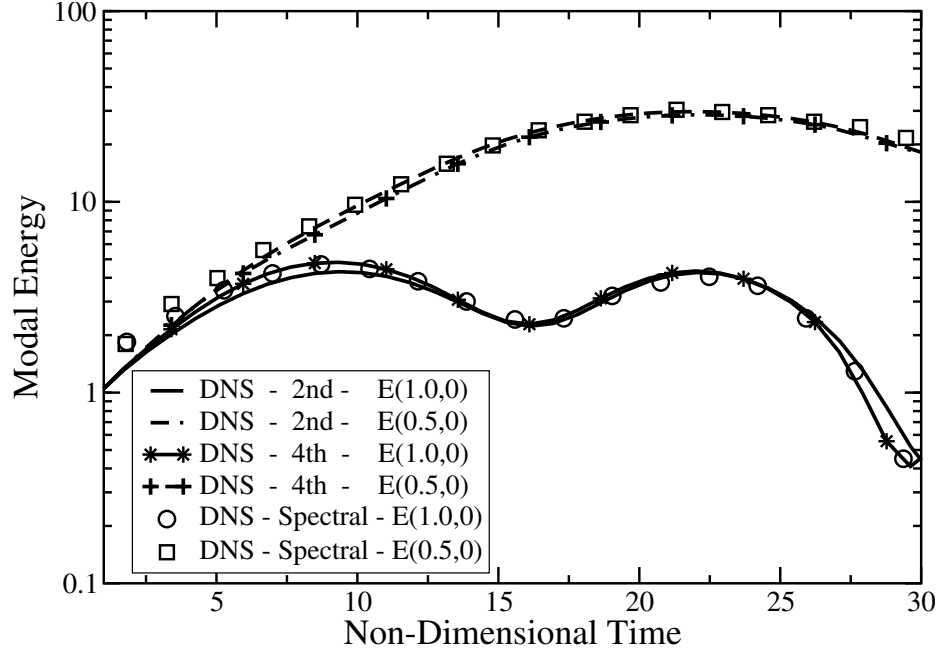


Figure 6.2: Evolution of the modal energy.

perturbation included only one subharmonic, the pairing process marks the end of the mixing layer growth. The pairing process, and hence the mixing layer growth would continue if additional subharmonics are included in the initial perturbation and if the computational domain is large enough to cover the wavelength of the imposed subharmonic modes. In the present case, the momentum thickness reaches its maximum at $T = 21$ corresponding to the pairing time of the unstable subharmonic mode.

6.3 Results on Particle Dispersion

Here, the results are reported for the DNS of a temporal mixing layer subjected to primary instability in the presence of droplets. The Stokes numbers of the droplets studied are 0.1, 1, 4 and 100, corresponding to the cases simulated by Ling et al. [1998]

Initial conditions for the droplets are such that they are uniformly distributed through out the domain and are in dynamic equilibrium with the gas phase. One particle per cell (262144 groups) was placed at the start of the simulation.

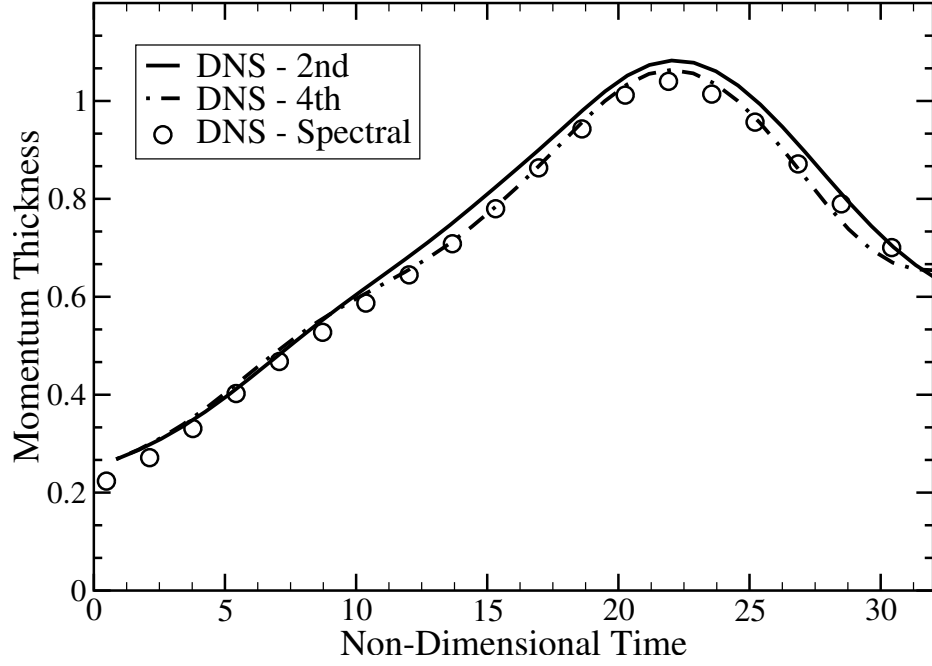


Figure 6.3: Evolution of momentum thickness.

Stokes number effect

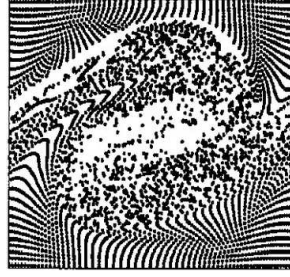
Figure 6.4 shows the distribution of droplets at a non-dimensional time of $T = 28$. Particles with small Stokes number follow the fluid closely and hence more droplets are present even in the core region for $St = 0.1$, compared to droplets of higher Stokes numbers. At such a small Stokes number, viscous effects dominate the inertial effects and hence the droplets are dragged along with the fluid and hence the droplets can even be convected into the core of the span wise vortical structures. As the droplet Stokes number approaches unity, notable difference show up in the behavior of the droplets in following the fluid motion. Now, droplets accumulate near the circumference of large structures. This leads to increased lateral droplet dispersion with droplets marking the lateral boundaries of the rolled-up vortical structures. With further increase in Stokes number, droplet response time further increases and hence the influence of flow on the droplets decreases. Some droplets are observed to pass through the core of these structures and accumulate in the braid regions between the

large vortical structures. This trend is also observed in the two dimensional inviscid simulations of Martin and Meiberg [1993]. The excellent qualitative agreement between the present calculations and that of the simulations of Ling et al. [1998] is also apparent in fig.6.4.

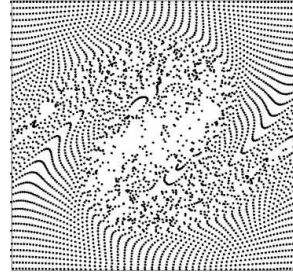
Dispersion of particles with unity Stokes number

Dispersion behavior of droplets of Stokes number of the order of unity is shown in fig.6.5. Particle concentration is shown at four non-dimensional times along with the results from the pseudo-spectral simulations of Ling et al. [1998]. Particles tend to accumulate around the circumference of the large vortical structures that are formed by roll-up at $T = 12$. Numerical and experimental (Ling et al. [1998], Lazaro and Lasheras [1992], Eaton and Fessler [1994]) studies carried out in the past indicate that the particles tend to accumulate in regions of low vorticity and high strain rate. This tendency of the droplets to preferentially accumulate in regions away from high vorticity (like the vortex cores) has been observed even in the isotropic turbulent flow simulations of Squires and Eaton [1990]. Such preferential concentration of particles can have serious implications in liquid-fueled combustors, where, the preferential accumulation may lead to large spatial variation in mixture equivalence ratio, which may lead to incomplete combustion and harmful pollutant emissions. The growth of the mixing layer due to vortex pairing leads to the increased later dispersion of the droplets as is seen at $T = 20$ and $T = 28$. The present study shows excellent qualitative agreement with the spectral DNS of Ling et al. [1998] in predicting the droplet-laden flow-field.

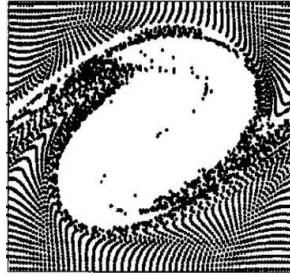
Figure 6.6 shows the vorticity and the droplet distribution for the droplets $St = 4$ at various non-dimensional times. As it can be seen the primary span-wise vortical structure is essentially devoid of droplets. This happens because, the strong vorticity present at the core of the span-wise vorticity centrifuges the droplets away from the



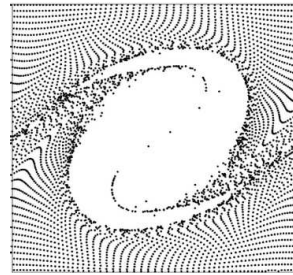
(a) $St=0.1$



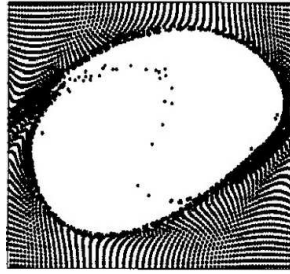
(b) $St=0.1$



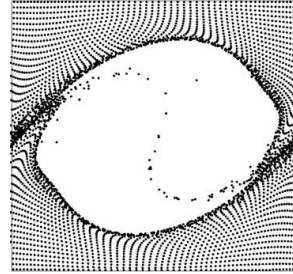
(c) $St=1.0$



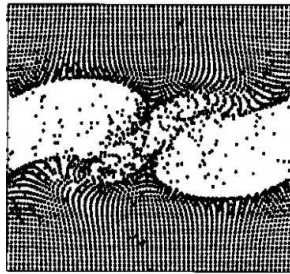
(d) $St=1.0$



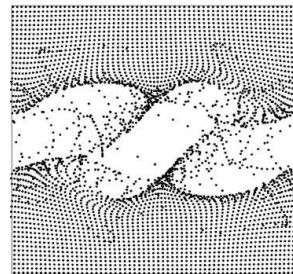
(e) $St=4.0$



(f) $St=4.0$



(g) $St=100$



(h) $St=100$

Figure 6.4: Stokes number effect (spectral DNS (a, c, e, g) and current DNS (b, d, f, h)).

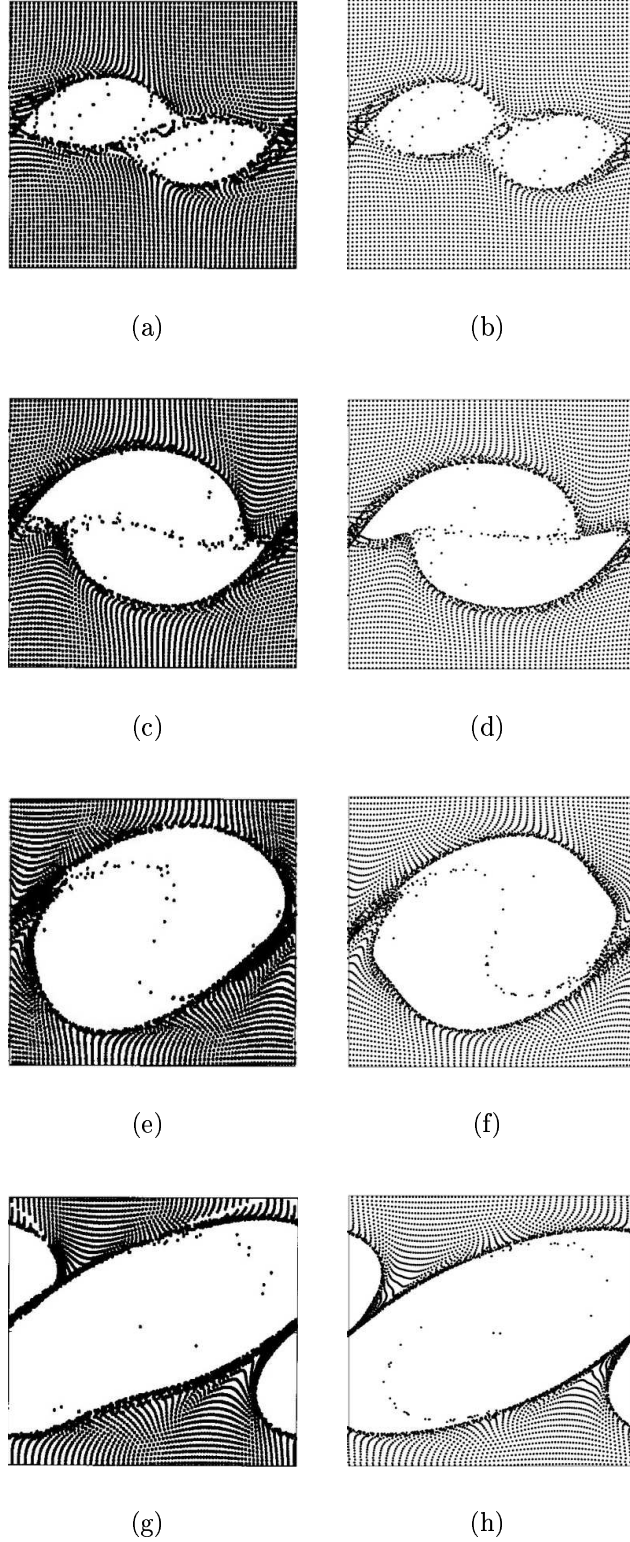


Figure 6.5: Comparison of particle dispersion for $St = 4$. spectral DNS (a, c, e, g) and current DNS (b, d, f, h).

center and augmenting dispersion. These droplets accumulate near the convergence or high strain regions of the flow such as the braid region.

Figure 6.7 shows the RMS of particle number per cell over the whole field. This is used to characterize the overall accumulation tendency of the droplets in the flow-field. The N_{RMS} is defined as (Ling et al. [1998])

$$N_{RMS} = \sqrt{\sum_{i=1}^{N_c} \frac{N_i^2}{N_c}} \quad (6.2)$$

where N_c is total number of computational cells and N_i is the number of particles in the “i-th” cell. From the figure it is clear that the N_{RMS} increases with time indicating the correlation between the growth of the mixing layer and the dispersion of the particles. An interesting feature to be noted is that the particles with Stokes number of the order of unity ($St = 1-4$) have higher N_{RMS} , compared to lighter ($St \ll 1$) and heavier particles ($St \gg 1$). This happens because, lighter particles have a strong tendency to truly follow the carrier(gas) phase and hence they are more uniform dispersed. Heavier particles, on other hand, follow their own inertia and are not dispersed much by the flow. Therefore, their distribution remains more or less like the initial conditions (i.e. uniformly distributed). However, particles with Stokes number of the order of unity ($St = 1-4$) are preferentially dispersed by the flow and hence their distribution becomes non-homogeneous even though their initial distribution is homogeneous. The agreement with the spectral DNS of Ling et al. [1998] is very good over the whole range of Stokes number.

Another way to quantify the dispersion characteristics of particles with different Stokes number is to evaluate the dispersion function in the Y-direction for the particles initially distributed in the mid plane in the transverse (Y) direction. The dispersion function is defined as (Ling et al. [1998])

$$D_y(t) = \sqrt{\left(\frac{1}{N_p} \sum_{i=1}^{N_p} (Y_i(t) - Y_m(t))^2 \right)} \quad (6.3)$$

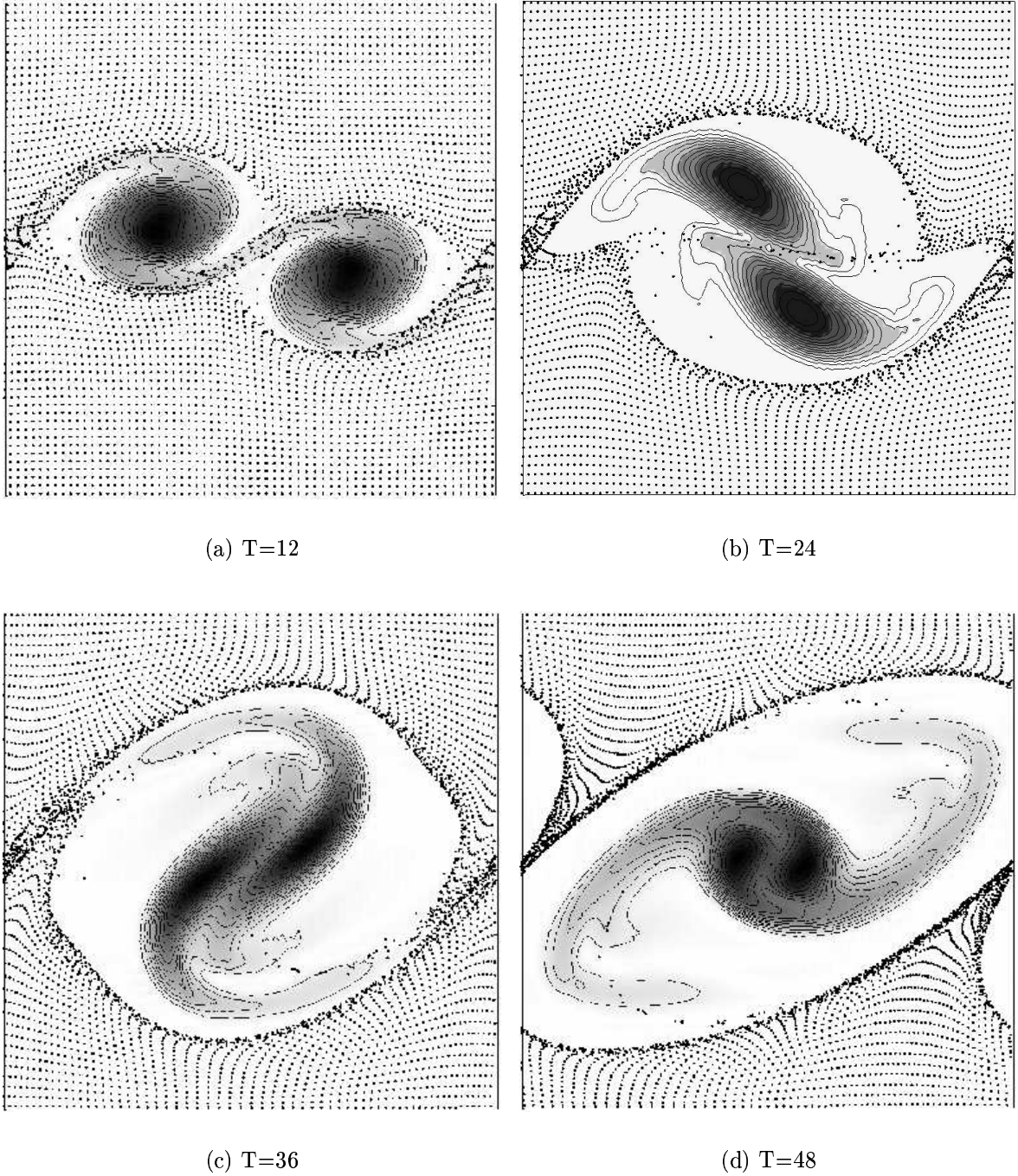


Figure 6.6: Contours of span wise vorticity and particle distribution - $St = 4$.

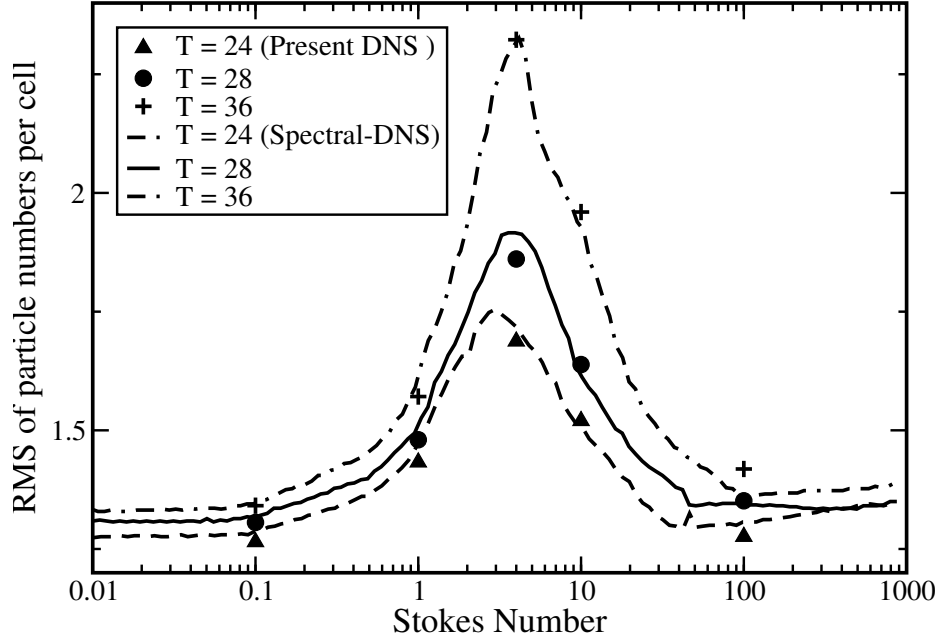


Figure 6.7: RMS of droplet concentration.

where N_p is the total number of the particles, $Y_m(t)$ is the mean value of the particle displacement in the vertical direction at time “t”, and $Y_i(t)$ is the displacement of a particle “i” in the vertical direction at time “t”. Figure 6.8 shows the Y-dispersion for the particles with Stokes number ranging from 0.1 to 100. At small times particles with $St = 0.1$ show more dispersion. However, at later times, particles with $St = 1$, show more dispersion due to the formation of K-H rollers and their subsequent merging, which causes the preferential dispersion of the particles with $St \approx 1$. Particles with larger Stokes number ($St = 100$) have lower dispersion compared to the lighter particles.

This concludes the validation of the DNS studies on the particle laden mixing layers. Even though, this validates the particle-tracking algorithms and the various spray models, the real test is to evaluate their performance within an LES formulation. The following section briefly provides some of the validation results for the LES of particle-laden mixing layers.

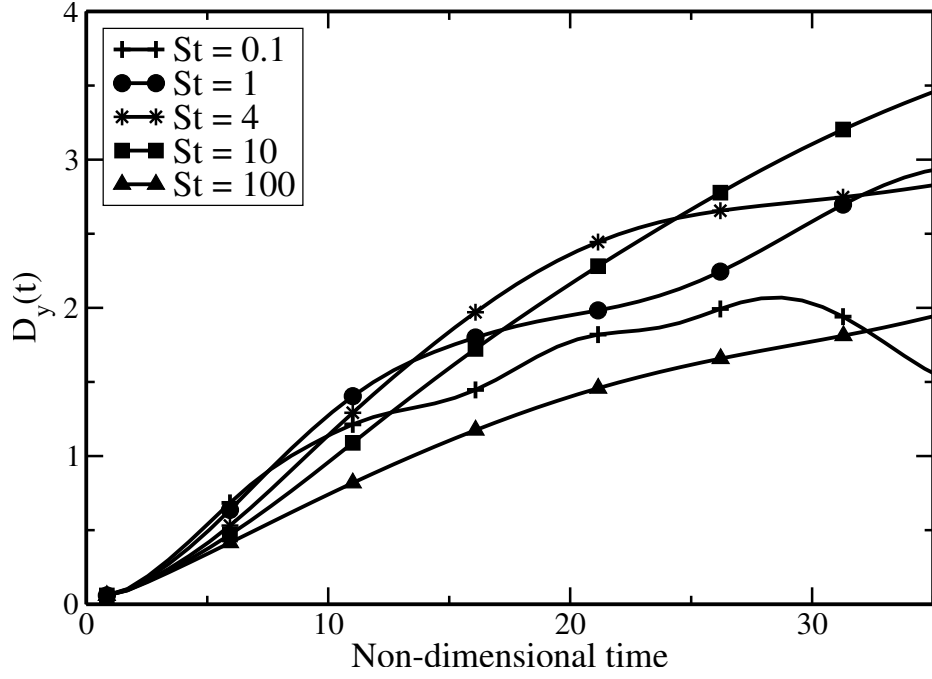


Figure 6.8: Stokes number effect on dispersion behavior.

6.4 LES of Mixing Layers

Results are reported for the LES of particle-laden mixing layer and compared with the DNS to evaluate the spray models and the particle-tracking algorithm used within the LES formulation. All the LES studies were performed with a $32 \times 32 \times 32$ grid. The numerical scheme employed is second-order-accurate in time and fourth-order-accurate in space. The LDKM approach described in chapter 2 is used for all the LES study. In fact only LDKM approach yields good agreement with the DNS. The initial and the boundary conditions are exactly the same as that in the DNS study. The particle mass loading is kept constant between the DNS and the LES study.

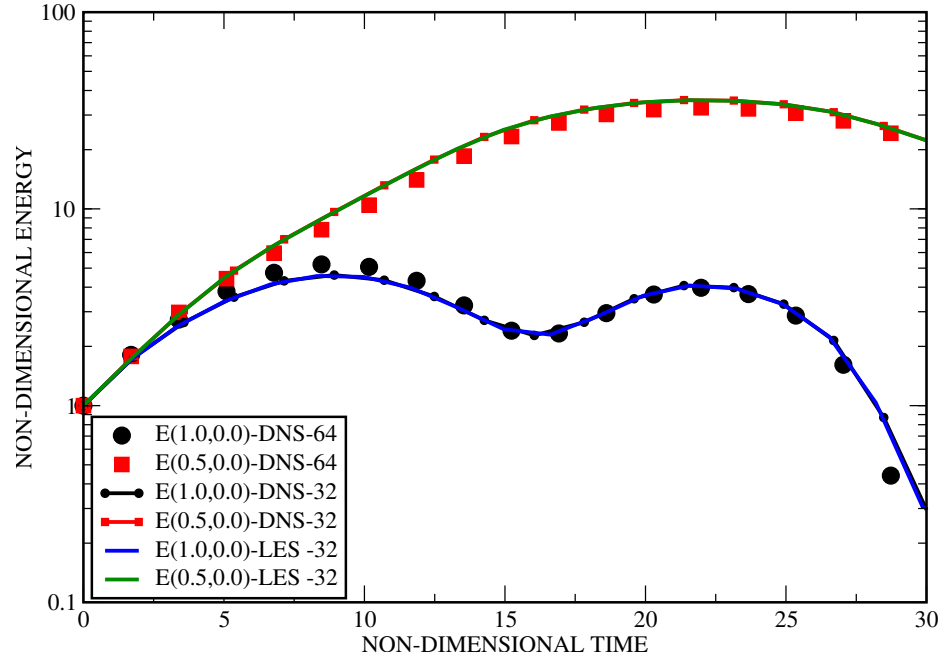
For the purpose of validation, only the modal energy growth rate is compared. First, the results are shown for the single-phase ($St = 0$) simulation. Figure 6.9 shows the energy growth in the fundamental and subharmonic mode obtained from 3 different simulations. A fully-resolved DNS with a 64^3 grid, an under-resolved DNS with a 32^3 grid and a dynamic-LES with 32^3 were performed to compare the solution.

As it can be seen, the present dynamic LES shows an excellent agreement with both the DNS results.

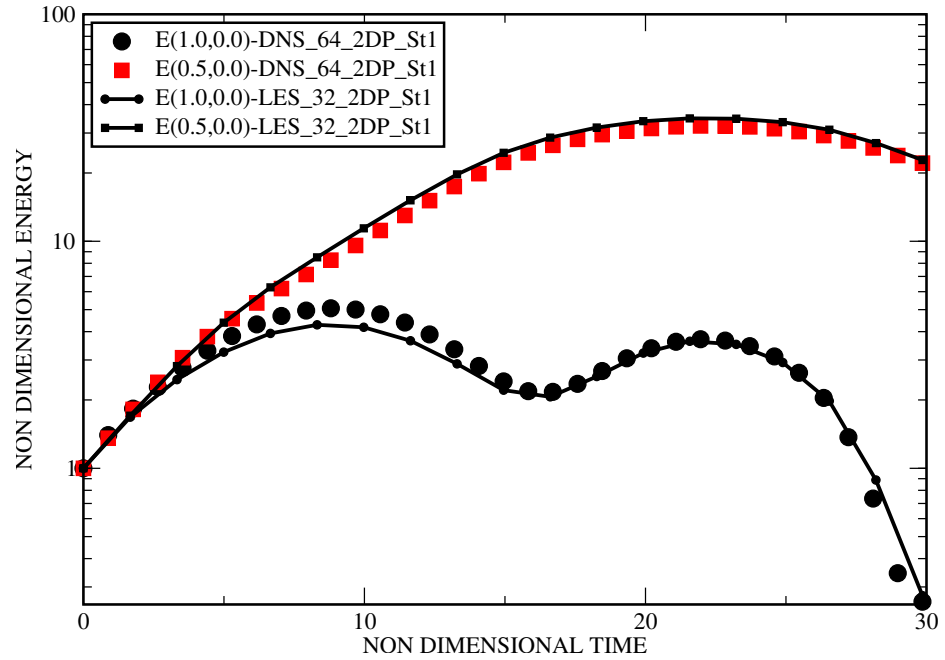
LES was carried out for three particle Stokes number ranging from 1 to 100. Figures 6.9(b), 6.9(a) and 6.10(b) shows the energy growth rate in the fundamental and the subharmonic modes for $St = 1, 10$ and 100. Again, the present two-phase LES shows excellent agreement with the DNS results. LES predicts the vortex roll-up time and the vortex pairing time to be 11 and 21 respectively. This is in very good agreement with the present DNS and the spectral DNS of Ling et al. [1998]. This implies that the dynamics of the interaction between the particle field and the mixing layer are captured very well by the present dynamic LES.

6.5 Conclusions

DNS and LES studies were carried out to simulate particle-laden temporal mixing layers. Results were compared against the pseudo-spectral DNS studies of Ling et al. [1998]. Both qualitative and quantitative comparison shows excellent agreement with the spectral-DNS results. Results obtained from the present dynamic LES approach is in very good agreement with the current DNS and the spectral DNS. Results obtained from the particle dispersion studies shows strong Stokes number dependence on the particle dispersion. Lighter particles ($St \ll 1$) are more uniformly dispersed by the flow, compared to particles whose Stokes number of the order unity. Particles with Stokes number of order unity are preferentially dispersed in regions of low vorticity (away from the vortex core). Heavier particles ($St \gg 1$) follow their own inertia and do not necessarily follow the fluid motion.

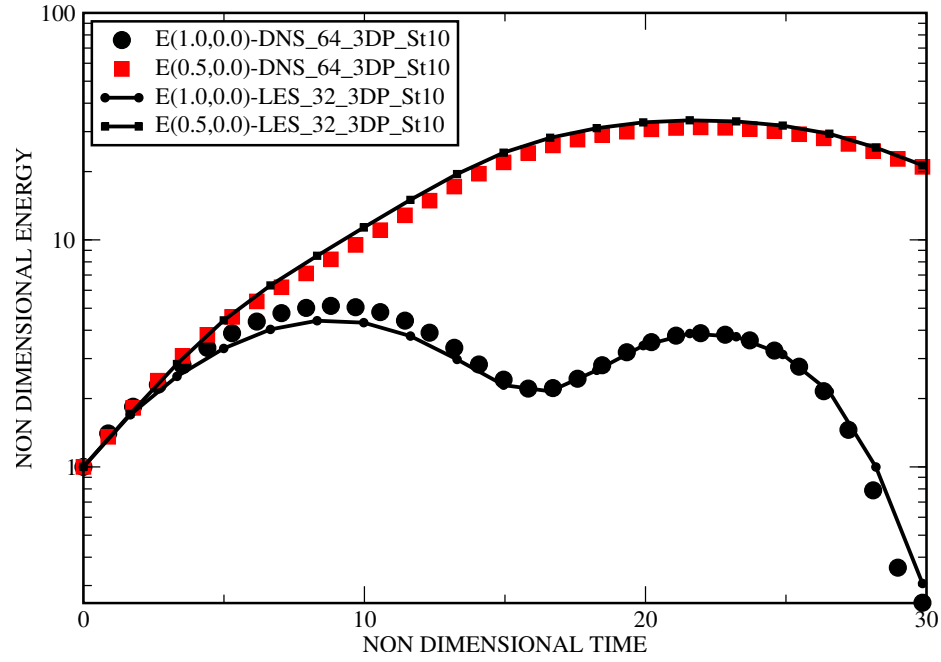


(a) $St=0$

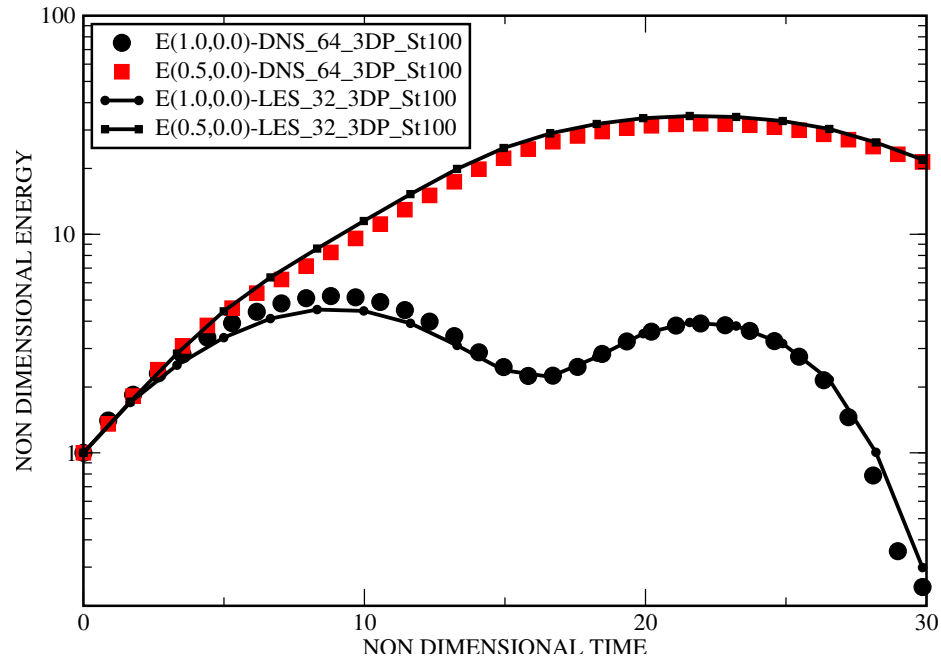


(b) $St=1$

Figure 6.9: DNS-LES comparison: Evolution of energy growth in fundamental and subharmonic mode.



(a) $St=10$



(b) $St=100$

Figure 6.10: DNS-LES comparison: Evolution of energy growth in fundamental and subharmonic mode.

CHAPTER VII

LES OF SPRAY COMBUSTION IN SWIRLING FLOWS

"Teamwork is essential - it allows you to blame someone else"

-Unknown

The spray models within the LES formulation were validated in a temporal mixing layer configuration in the previous chapter. This chapter is used to evaluate the performance of spray models in a spatial shear layer configuration such as those found in a gas-turbine configuration. The results reported here includes the dynamical features of spray-vortex-flame interactions, statistically averaged mean and RMS velocity profiles and the effect of droplets, swirl and heat release on the fine scale structure of the turbulence, by examining the vorticity and strain-rate fields.

The configuration consists of an inlet section with a central injector cone and a cylindrical dump combustor (Fig. 7.1). The geometry and conditions are for an operational General Electric Dual-Annular Counter-Rotating Swirling (GE-DACRS) liquid-fueled gas turbine combustor. At the inlet, stream-wise and tangential velocity profiles with a 7% turbulence level are prescribed (Kim et al. [1999]). The mean inlet mass flow rate is 0.435 kg/sec at 816 K and 1.38 MPa. The bulk velocity at the inlet is 30 m/s and the Reynolds number based on bulk velocity at the dump plane and inlet diameter is $\approx 260,000$. A single-step global chemistry using infinite rate-kinetics for n-heptane is employed in this study. Thus, strain induced extinction is not addressed in this study.

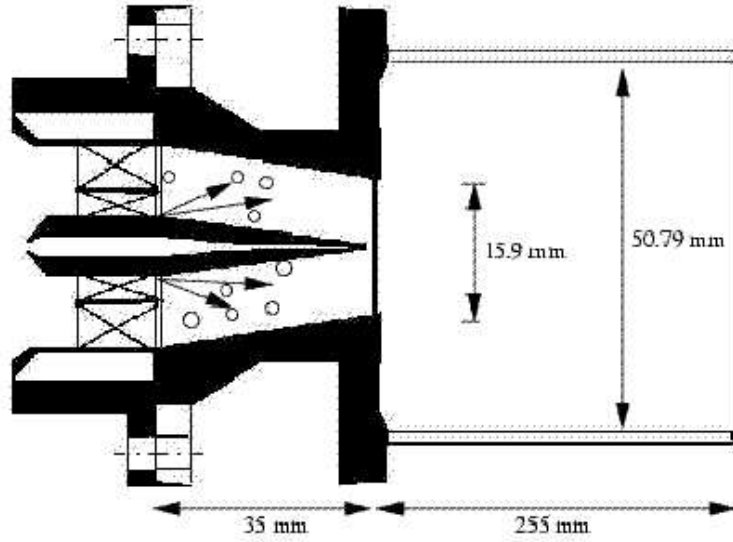


Figure 7.1: GE-DACRS Gas turbine combustor configuration.

The computational grid used 141 x 75 x 81 grid points in the axial, radial and azimuthal directions, respectively with clustering in high shear regions. In terms of wall units, $\Delta y^+ = 3.17$ and $\Delta x^+ = 4.21$, in the shear layer at the dump plane. Earlier studies Kim et al. [1999] in a similar gas turbine configuration have shown that this kind of resolution is sufficient to resolve the large scale features that dominate this flow field. No-slip, adiabatic and non-catalytic wall boundary conditions and non-reflecting characteristic conditions are employed at the inlet and the outflow (Poinso and Lele [1992]). The rapid decay of swirl and the presence of a converging exit section, results in a mean flow that is predominantly axial near the exit. These conditions justify the usage of the non-reflecting outflow boundary conditions in swirling flows.

A large number of particles, typically over 100,000 “computational parcels” (for the non-vaporizing case) are tracked. Each of these parcels contains on an average 150 individual droplets. Thus, on an average 15 million droplets are present in the computation. A mono-disperse distribution with spherical droplets of size $40 \mu m$ (which is approximately the Sauter Mean Diameter of the droplets in DACRS) is

introduced in the inlet through the central shaft at an half-angle of 10 degrees to the flow (see Fig. 7.1). The droplets are assumed to vaporize completely as they reach a cut-off size of $10\mu m$. Droplet-wall collisions are assumed to be perfectly elastic (Sommerfeld and Qiu [1993]). In the reacting spray simulations very few droplets reach the wall before they completely vaporize, and hence elastic wall conditions should not introduce significant errors.

Currently no experimental data is available for GE-DACRS for direct comparison. However, all the models used in this study have been validated in the past (Kim et al. [1999], Pannala and Menon [1998]) and in the present thesis for various flows. For instance, Chapter 4 validated the present LES with a k_{sgs} closure to predict swirling, high Re, premixed combustion in a GE-LM 6000 can combustor. Chapter 6 validated the spray models using DNS and LES studies. In addition, Pannala and Menon [1998] used the above spray models in two-phase temporal mixing layers, decaying and stationary isotropic turbulence and demonstrated that the current spray model correctly captures the physics consistent with previous DNS studies.

In the present study, typically 80 processors have been employed for the LES. Simulations on the IBM SP P3 (375 MHz) typically require 5000 single processor CPU hours for a single flow-through time to track approximately 100,000 droplets (in the non-vaporizing case) and about 2700 single processor CPU hours for the reacting case. The memory requirements is 13.3 Gigabytes.

7.1 Flow Visualization

Four simulations were performed to study the effect of swirl, heat release and particle dispersion on the flow. Two swirling flows defined by the Swirl numbers of 0.75 and 1.5 (at the inlet) are simulated without combustion and with combustion. Swirl number is defined as mentioned in Chapter 4. Note that the actual Swirl number at the dump plane (i.e., at the entrance to the combustor) are approximately 0.5 and

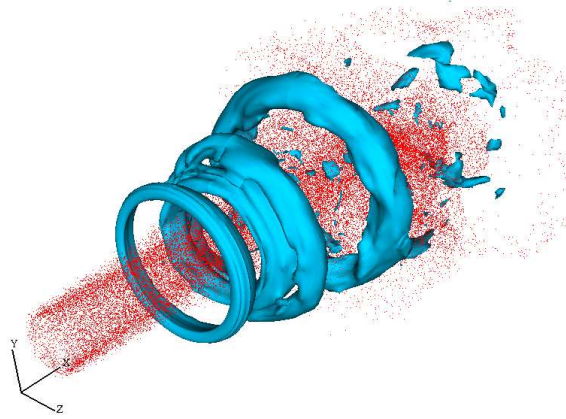
0.8, respectively due to flow relaxation and decay in the inlet. This Swirl number at the dump plane is below the critical value (at which the vortex breakdown occurs) for the low swirl, however, it is well above the critical Swirl number for the high swirl case Kim et al. [1999].

The particle behavior in any flow can be characterized by the Stokes number St , which is defined as the ratio of the flow or aerodynamic time to particle response time. Thus, $St = \tau_f/\tau_p$, where τ_f is the characteristic flow time and τ_p is the representative particle time (for low particle Reynold's number) given by $\tau_p = \rho d_d^2/(18\mu)$ (Sommerfeld and Qiu [1993]). Stokes number characterizes the aerodynamic response time of the particle to the the local flow conditions. Hence, particles with Stokes number much less than unity will behave more like a fluid particle and hence will exhibit more tendency to follow fluid motion. On the other hand, particles with Stokes number much greater than unity will show no tendency to follow fluid motion, and will instead follow its own inertia (Squires and Eaton [1990]). In complex flows with swirl, strong recirculation, large-scale structures and combustion, the local fluid dynamic time scale is likely to vary significantly and therefore, even with an initially mono-disperse spray (as in the case here), a range of St can exist in the flow field. For instance, Stokes number computed using a characteristic flow time based on the step height and the velocity at the dump plane is 0.32, whereas the St computed based on the the length of the combustor from the dump plane to the exit of the combustor is 2.5. This implies that depending on the spatial location, the particles can behave like a lighter or a heavier particle.

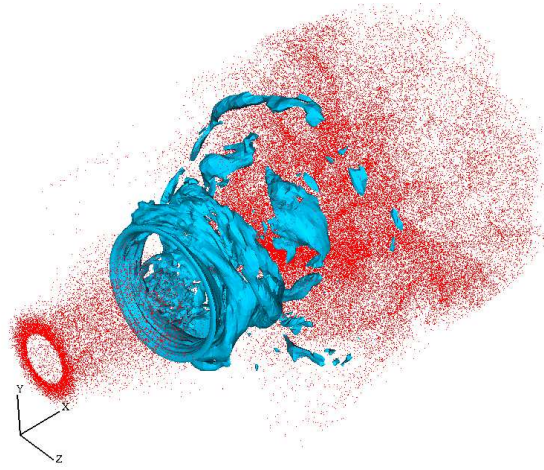
General observation based on the instantaneous images of the vorticity and droplet number distribution (Figs. 7.2 (a) and (b)) shows that low swirl is not very effective in dispersing the particles away from the center-line since the particle inertia exceeds that of the flow. Thus, particles leave the inlet section without any significant lateral dispersion, as shown in Fig. 7.2(a). However, with increase in swirl, a significant

lateral dispersion of particles is clearly seen in Fig. 7.2(b).

Kelvin-Helmholtz type instability leads to the formation and shedding of coherent vortex rings from the dump plane. These vortical structures comprise predominantly of the azimuthal vorticity component. On the other hand, the presence of swirl (and hence, tangential velocity) leads to formation of a strong vortex core (columnar vortex) which is predominantly comprised of the stream-wise vorticity. In addition, the onset of azimuthal instability due to the presence of tangential shear in swirling flows, leads to production of an additional component of stream-wise vorticity. Thus a swirling flow evolves continuously under the action of joint stretch/strain in axial and azimuthal directions. Even though the stream-wise vorticity associated with the columnar vortex decays because of stream-tube divergence and viscous effects, the stream-wise vorticity produced by the azimuthal instability grows continuously due to the growth of the instability itself. The presence of the stream-wise vortical layer is evident in Figs. 7.3(a)-(b) which shows the iso-surface of azimuthal (blue surface, -15000sec^{-1}) and stream-wise (green surface, 6000sec^{-1}) vorticity. Note that the azimuthal vorticity is produced by Kelvin-Helmholtz instability, whereas the stream-wise vorticity is produced by the azimuthal instability. The complex interaction between these vorticity results in the formation of tube-like azimuthal vortical structures of opposite sign (yellow surface, 6000sec^{-1}) as shown in Fig. 7.3(c) and (d). Thus, under the combination of axial and azimuthal stretch effects, the large-scale azimuthal and stream-wise structures rapidly breakdown into smaller structures. The fact that the same phenomenon is observed in both the low and high swirl cases indicate that shear in both axial and azimuthal directions accelerate the transition to small scale vortical structures in swirling flows. This effect also plays a major role in enhancing mixing in swirling flows, compared to swirl-free flows. The presence (and the absence) of these vortices plays a major role in the particle dispersion process as well. When the vortical structures are coherent (as in the low swirl case) the spray



(a) Non-reacting low-swirl



(b) Non-reacting high-swirl

Figure 7.2: Instantaneous azimuthal vorticity and droplet distribution.

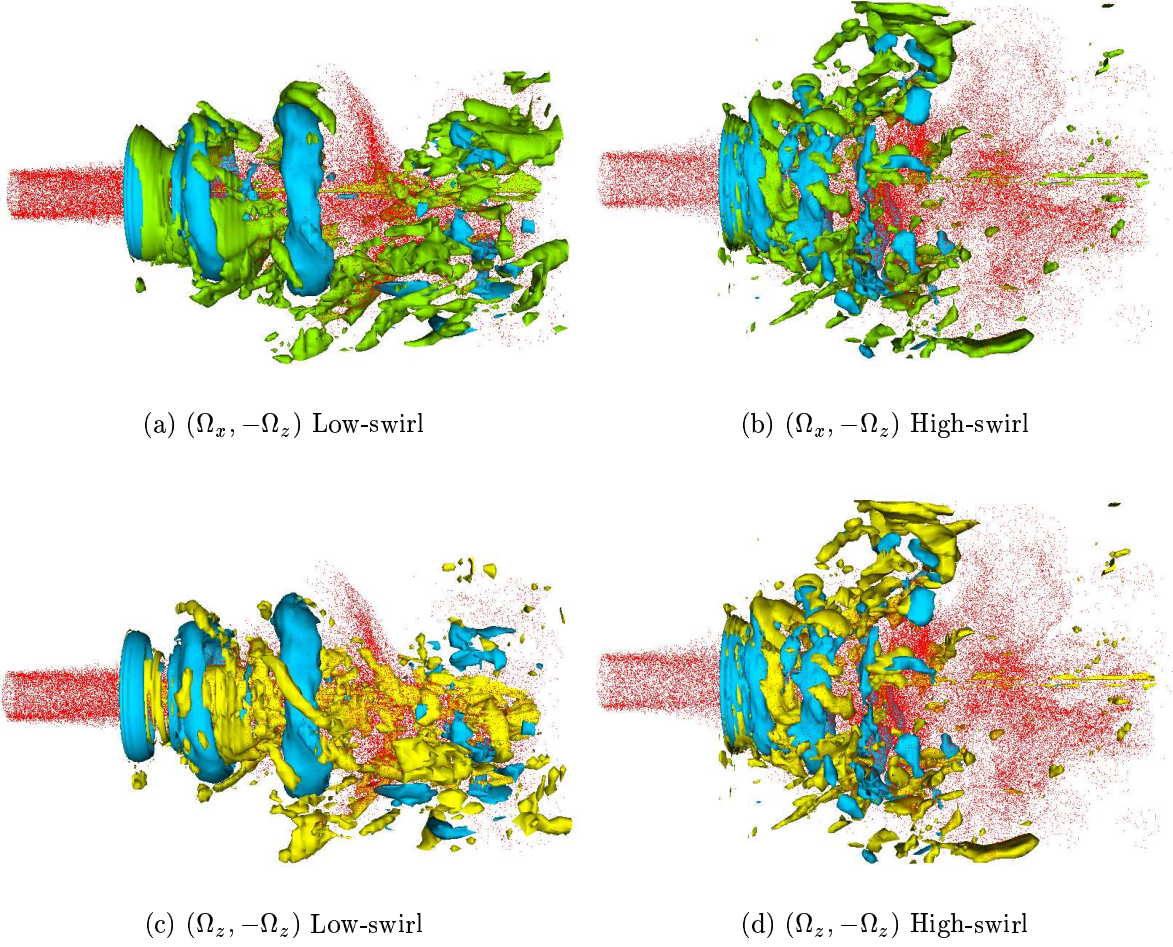


Figure 7.3: Instantaneous vorticity and droplet distribution, green - axial vorticity, yellow - positive azimuthal vorticity and Blue - negative azimuthal vorticity.

dispersion is highly modulated by the motion of these structures, whereas, when these structures are broken up due to increase in swirl, the particle dispersion is uninhibited leading to a much more dispersed pattern.

Figures 7.4 (a) and (b) show the instantaneous flame-vorticity distribution for the low and high swirl reacting cases, respectively. In these images, the blue surface corresponds to the iso-surface of azimuthal vorticity, while the orange surface corresponds to the iso-surface of temperature. The iso-scalar values for the vorticity and temperature are 15000 sec^{-1} and 2200 K , respectively. Here, the flame is identified

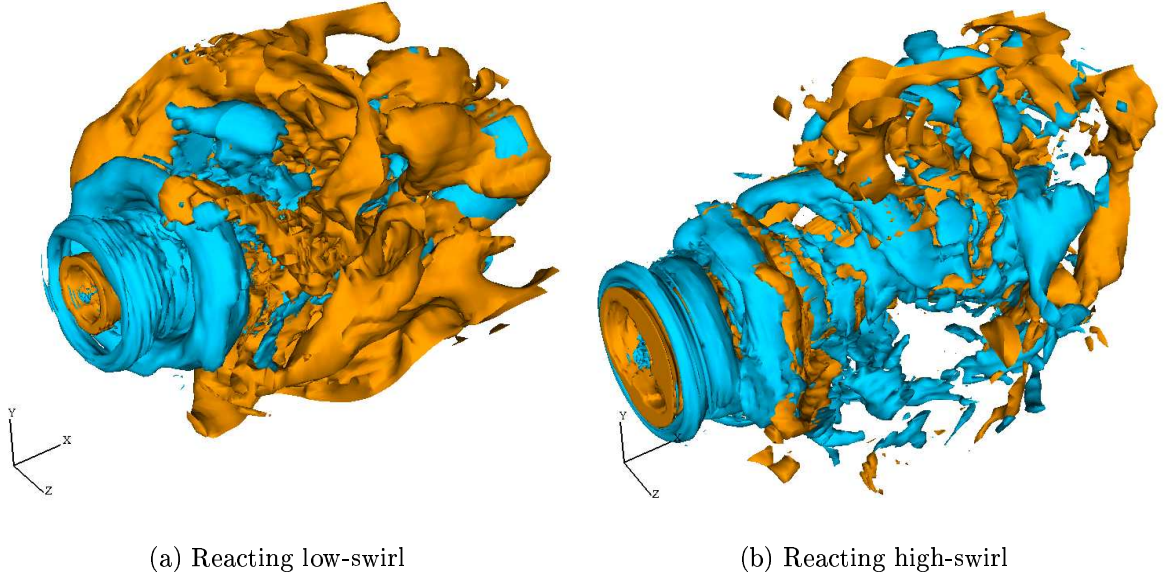


Figure 7.4: Instantaneous azimuthal vorticity and temperature.

as the iso-surface of temperature value corresponding to 2200 K . A longer flame is seen to be present in the low swirl case, whereas a relatively shorter flame is seen in the high swirl case. The flame surface seems to wrap around the large scale structures near the dump plane. On the other hand, farther downstream, the small scale vortical structures formed due to the breakdown of these large scale structures, induces fine scale wrinkling on the flame surface.

7.2 Center Line Axial Velocity Variation

7.2.1 Non-reacting case

In all the figures that follow, the axial distance is non-dimensionalized by the diameter of the premixer inlet (D) at the dump plane (unless otherwise stated). Figure 7.5(a) shows the variation of center-line axial velocity for the low and high swirl case without combustion. It can be observed that for the low swirl case, immediately downstream of the dump-plane ($X/D = 0$) there is a region where the flow accelerates and reaches a maximum, and then begins to decay as in non-swirling flows. Analysis of the velocity

and the pressure gradient variation near the center line indicates that this is due to the combined effect of the geometrical features of the inlet convergence and the sudden expansion, which results in a non-trivial interaction between azimuthal and axial velocities. The presence of convergent section increases the angular velocity of the flow, due to the conservation of angular momentum. This results in the decrease of pressure near the axis which induces a favorable axial pressure gradient ($\partial p/\partial x < 0$) that tends to accelerate the flow near the center line. But the absence of such acceleration in the case of high swirl indicates that this is not a linear phenomena, and that more than one physical processes are involved in determining the flow structure near the center line. This is discussed further below.

Presence of swirl in the flow allows for the natural development of a strong columnar vortex core (stream-wise vorticity) in the inlet pipe with a reduced pressure along its axis. This induces a favorable pressure gradient which tends to accelerate the flow. This is evident in Fig. 7.6 which shows the variation of $\partial p/\partial x$ along the center line. The pressure is normalized by a reference pressure, $p_o = 1atm$. A region of favorable pressure gradient extending up to $0.33D$ from the dump plane (for low swirl) can be seen in Fig. 7.6. However, as the flow expands suddenly into the combustor, the radius of the columnar vortex core increases, leading to a decrease of the angular velocity (by conservation of angular momentum) with the subsequent increase of pressure along its axis. This imposes an adverse pressure gradient ($\partial p/\partial x > 0$) that tends to decelerate the flow near the center line.

Thus, it is the balance between the favorable and the adverse pressure gradient, that determines the flow structure near the center line of the combustor. As it can be observed in Fig. 7.6, the favorable pressure gradient near the dump plane, imposed by the convergence/swirl accelerates the flow in the low swirl case. As the flow expands into the dump combustor the stream-wise vorticity decays (vorticity of the columnar vortex core), and the pressure along the axis increases, thereby decelerating the flow.

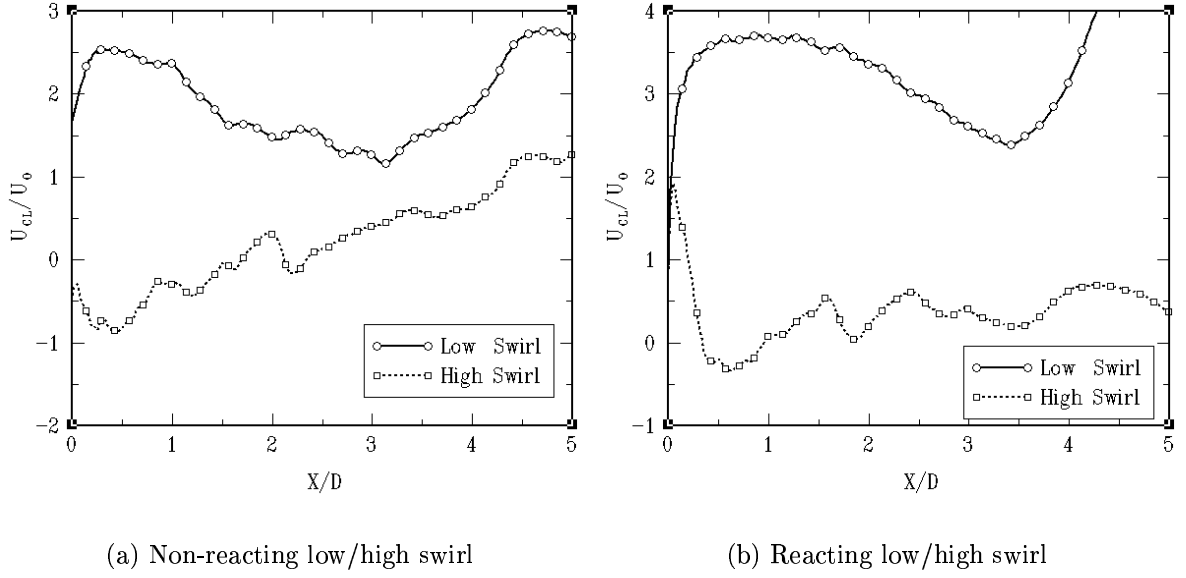


Figure 7.5: Center-line axial velocity profiles.

On the other hand, for the high swirl case, the rapid divergence of the columnar vortex core, imposes a very strong adverse pressure gradient, that causes flow reversal at the dump plane, leading to the formation of the central toroidal recirculation zone (CTRZ) near the center line. For the high swirl case, the size of the CTRZ extends upto $1.83D$ downstream of the dump plane. This is approximately the same distance over which the adverse pressure gradient exists near the center line. It can also be observed that the high swirl case appears to show a secondary CTRZ extending to $0.2D$, slightly downstream of the larger primary CTRZ. Figure 7.6 also shows a corresponding small region of adverse pressure gradient at this location.

7.2.2 Reacting case

Figure 7.5(b) shows the variation of center-line axial velocity for the low and high swirl case with combustion. Center-line axial velocity variation is significantly altered by combustion and heat release. Observation shows that the rate of decay of the axial velocity for the low swirl reacting case is less than its corresponding non-reacting case.

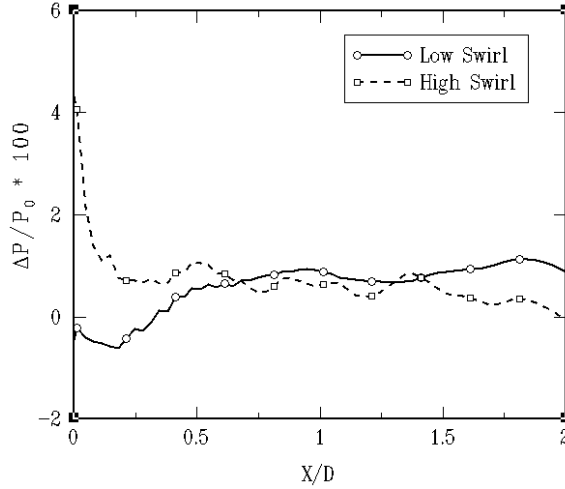


Figure 7.6: Center-line variation of the pressure gradient for non-reacting low/high swirl.

Heat release and the associated gas expansion leads to a large increase of the axial velocity in the low swirl case, whereas, the axial extent of the CTRZ is reduced to $0.6D$. The increased axial velocity in the case of low swirl might lead to difficulty in burning and possibly leading to instability. On the other hand, high swirl provides a region of hot recirculating zone near the center line which helps to anchor the flame.

7.3 Mean Stream wise Velocity Profiles

Figures 7.7(a) and (b) show respectively, the radial profiles of the mean stream-wise velocity for the low and high swirl case without combustion. The velocity is non-dimensionalized by the bulk velocity at the premixer inlet, $U_o = 30 \text{ m/sec}$.

While vortex break-down is absent in the low swirl case, its presence is clearly seen in the high swirl case. The dashed lines mark the extent of central toroidal recirculation zone (CTRZ) and the corner recirculation zone (CRZ) in each of the cases considered. The CTRZ extends upto $1.83D$ downstream from the dump plane for the high swirl case (CTRZ is absent in the low swirl case). The peak reverse velocity in the CTRZ for the high swirl case occurs at $X/D = 0.5$, where the axial

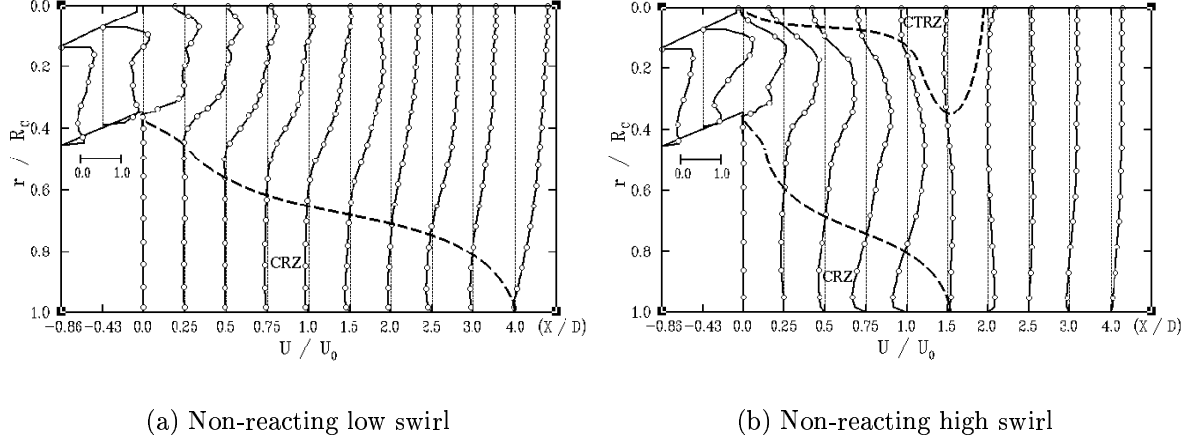


Figure 7.7: Radial variation of the stream-wise mean velocity profiles in the combustor. The radial dimension is normalized by the radius of the combustor, R_C .

velocity is $-9.9m/s$. It is believed that the rapid divergence of the streamline toward the end of the CRZ, results in the broad region of the CTRZ around $X/D = 1.5$. For both swirl cases, a large CRZ is observed. The CRZ extends upto $4D$ for the low swirl and $1.5D$ downstream for the high swirl case. This is because, the peak tangential velocity ($28.7m/s$) occurs radially outwards away from the centerline. This peak in the tangential velocity profile toward the corner results in a strong primary CRZ in conjunction with a weak CTRZ. For the low swirl case, the peak reverse velocity in the CRZ occurs at $X/D = 2.5$ where the axial velocity reaches $-4m/s$, whereas for the high swirl case, the region of high reverse velocity occurs at $X/D = 0.75$, where the axial velocity is $-9.8m/s$. This indicates that a stronger CRZ is present in the case of high swirl compared to the low swirl, even though the extent of the CRZ itself is smaller. It can also be observed from the axial velocity profiles for the high swirl case, that the flow expands rapidly soon after entering the combustor, when compared to the low swirl case. This divergence of streamlines in high swirl case leads to the reduction in size of the CRZ as compared to the low swirl case.

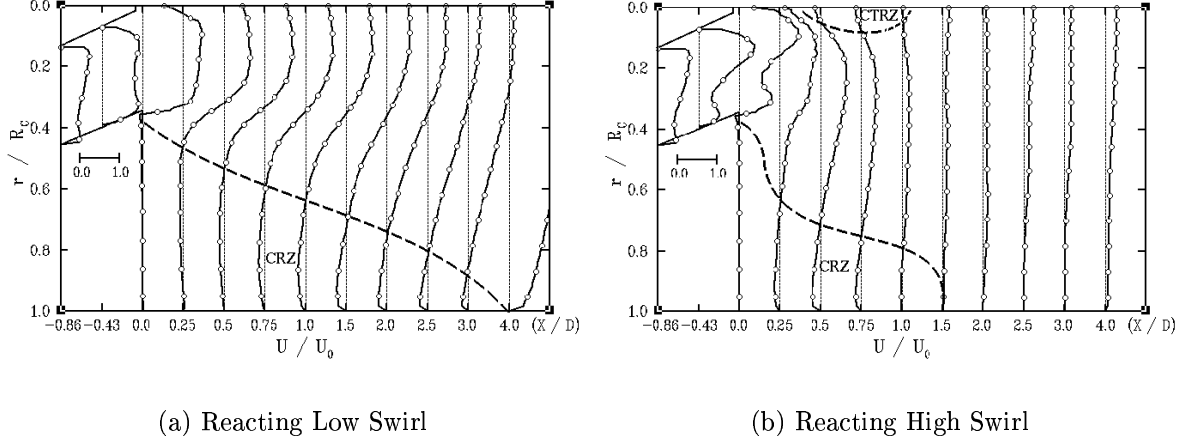


Figure 7.8: Stream wise mean velocity profiles.

Examination of stream-wise mean velocities profiles (not shown) slightly downstream the dump plane ($X/D \approx 0.2$) revealed that the presence of particles reduces the mean velocity. This is primarily due to the drag exerted by the particulate phase on the carrier phase. Thus, particle drag reduces the large radial variation in the velocity profiles and smoothen out radial gradients.

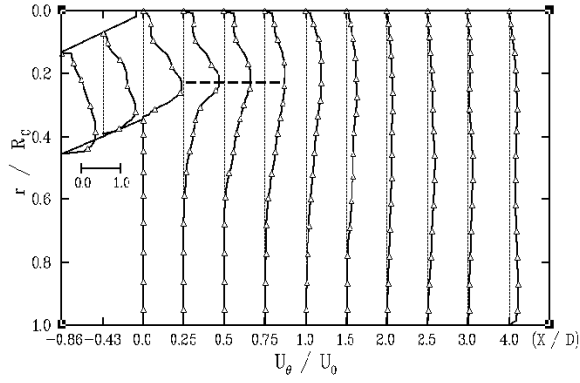
Figures 7.8 (a) and (b) show respectively, the radial profiles of the mean stream-wise velocity for the low and high swirl case with combustion. Again, the presence of the CTRZ is evident for the high swirl case, whereas it is clearly absent in the low swirl case. However, the size of the CTRZ is significantly reduced. This is because, the axial flow acceleration due to combustion imposes a favorable pressure gradient in the combustor, which counter-balances the adverse pressure gradient of the sudden expansion. Examination of the pressure gradient plots (not shown) indicates that the the region of favorable pressure gradient imposed by the combustion dominates the adverse pressure gradient imposed by the sudden expansion. Thus, heat release effect seems to counter-balance the mechanism that leads to the formation of the CTRZ.

7.4 Mean Tangential Velocity Profiles

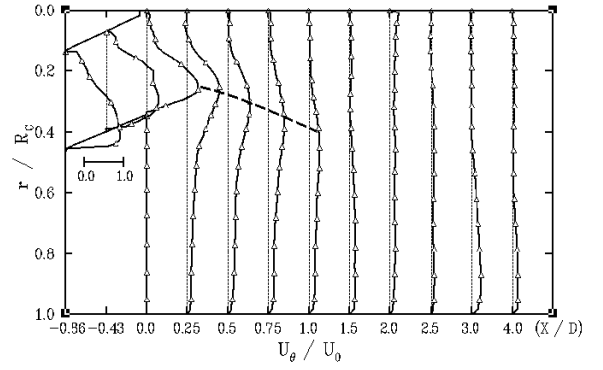
Figures 7.9(a) - (d) show respectively, the radial profiles of the mean tangential velocity for the low and high swirl case, with and without combustion. The bold dashed line marks the radial location where the tangential velocity peaks. The location of the peak tangential velocity in the low swirl case (with and without combustion) varies a little, and remains unchanged for large axial distance. On the other hand, in the case of high swirl (both with and without combustion), the location of peak tangential velocity moves away from the center line. The rate of this movement is more pronounced for the non-reacting case, as compared to the reacting case. If the columnar vortex core (stream-wise vorticity) radius is defined as the radial location where the peak tangential velocity occurs then the movement of this peak location away from the center line indicates the growth of the radius of the vortex core. As the radius of the vortex core increases, it creates an adverse pressure gradient near the center line of the combustor, which eventually leads to the vortex break down, thus creating the CTRZ. The fact that the peak location moves rapidly for the non-reacting (high swirl) case than the reacting case, explains why the size of the CTRZ is larger in the former case. Also, for the low swirl case, the small variation in the location of peak tangential velocity with axial distance, provides a consistent explanation for the absence of the vortex break-down and CTRZ.

7.5 RMS Velocity Profiles

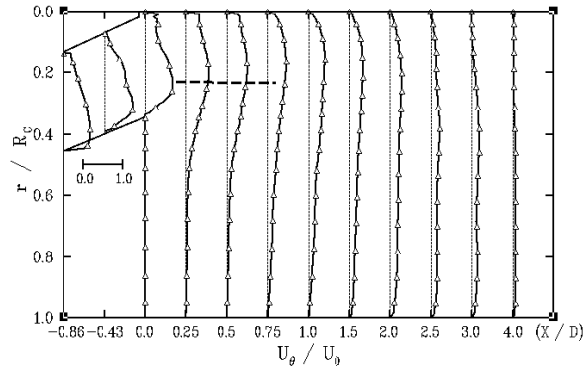
Figures 7.10(a) - (b) shows the profiles of stream-wise RMS velocity reveals that the turbulent fluctuations peak in regions of shear layer and also near the center line. Augmented turbulence levels are observed near center line in the CTRZ for both reacting/non-reacting high swirl case. Higher values of stream-wise RMS velocities are observed near the dump plane. Since the shear layer from the inlet cone tip is not strong enough to induce such large oscillations, it is believed that it could not



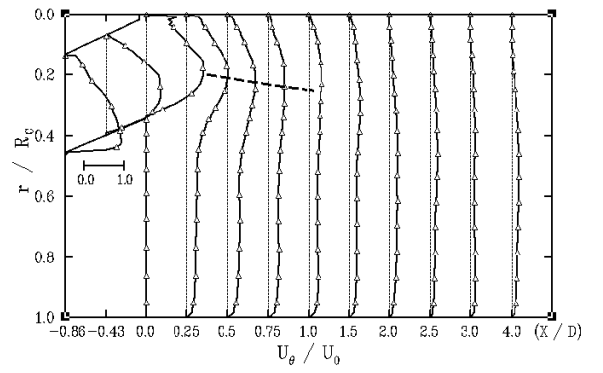
(a) Non-reacting low swirl



(b) Non-reacting high swirl

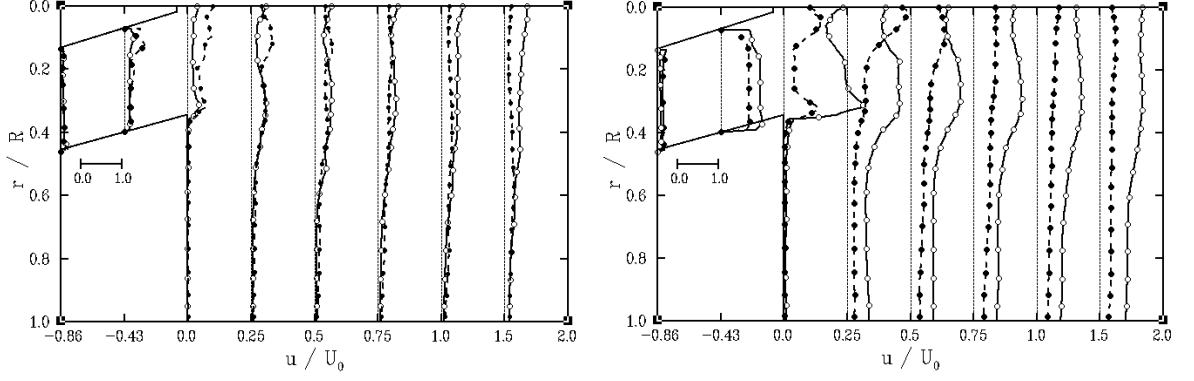


(c) Reacting low swirl



(d) Reacting high swirl

Figure 7.9: Tangential mean velocity profiles.



(a) Non-Reacting Low/High Swirl

(b) Reacting Low/High Swirl

Figure 7.10: Stream wise RMS velocity profiles.

be entirely due to the boundary layer separation from the cone tip. An alternate, rather possible explanation could be that the CTRZ is exhibiting a precessing motion near the forward stagnation point of the CTRZ (closer to the inlet) which results in the pronounced velocity fluctuations. However, the variation in stream-wise RMS velocity near the rear stagnation point is lower, indicating a lesser bubble oscillation. Increased levels of turbulent activity were observed in the presence of reaction and heat release induced flow acceleration.

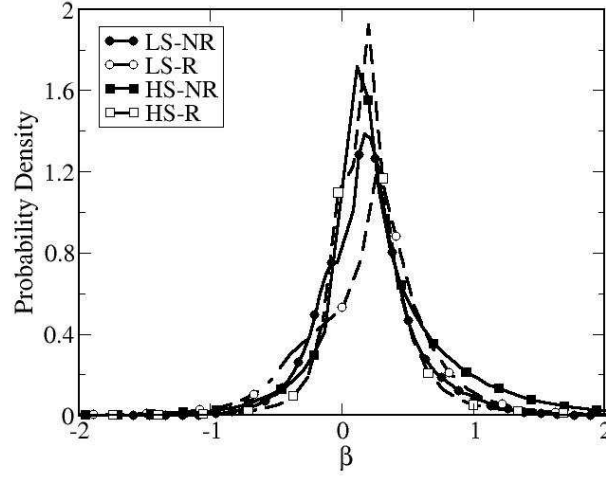
Examination of tangential RMS velocities (not shown), slightly downstream the dump plane ($X/D \approx 0.2$) also revealed that turbulent fluctuations have been attenuated in the presence of the particles. The presence of droplets decreases the turbulence level by introducing additional dissipation. In particular, velocity fluctuations have been attenuated significantly in regions where turbulent intensities are high in unladen flows (i.e., in regions of high shear). This is because in these regions the local Stokes number based on the turbulent time scales are high leading to increased attenuation of the turbulence closer to the shear layer where the turbulence production is very high. Turbulence levels in the recirculating zones are not affected significantly, due to the presence of fewer particles there.

7.6 Large and Small-Scale Flow Structures

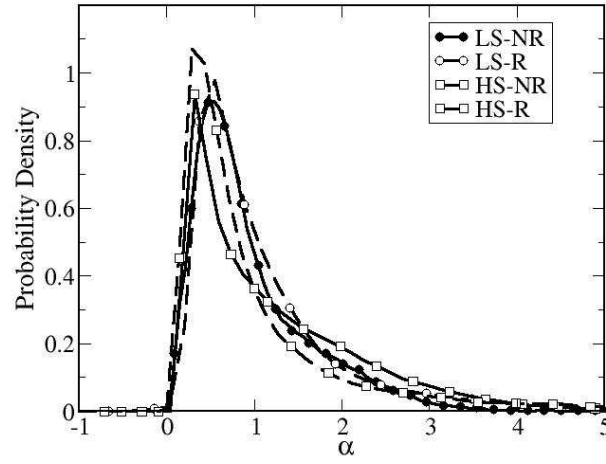
PDFs of the eigenvalues of the $\tilde{S}_{ij}$ in isotropic turbulence have shown that the most likely turbulent state is axis-symmetric extension corresponding to one of the two positive (α and β eigenvalues) strain rates (Betchov [1956], Kerr [1987]). This result suggests the formation of sheet-like structures in the flow. On the other hand, based on the sign (i.e., $\alpha, \beta > 0, \gamma < 0$) and the relative magnitudes of these eigenvalues, it has been suggested that the dominant structures are often tubular, especially in regions of high dissipation (Kerr [1985], Ashurst et al. [1987]).

Since large coherent structures with significant spatial anisotropy exist in the present case (and absent in these earlier studies), these features are revisited here. Statistics are obtained by sampling points only inside the combustor (the inlet and the converging exit section are excluded). Ten instantaneous (and statistically independent) fields, yielding a total sample set of ≈ 5 million points, are used for analysis. Figures 7.11(a) and (b) show respectively, the PDF of the β and α eigenvalues (γ PDF is not shown, for brevity). The shapes are similar for all cases considered, suggesting an universal strain structure (Ashurst et al. [1987]). In all simulations, the intermediate strain-rate (β) is predominantly positive. On an average, this corresponds to a topology of two extensional (positive) and one compressive (negative) strain, consistent with earlier results (Ashurst et al. [1987], Kerr [1985]). Heat release increases the volumetric dilatation, resulting in the overall increase in positive strain rates (Fig. 7.11 (b)).

Both sheet and tubular structures are observed in the present study. The presence of mean axial and azimuthal shear (due to swirl) leads to the formation of quasi-two dimensional sheet-like structures near the dump plane. Figure 7.3 shows that the sheet-like structures associated with Ω_z form tubular rings, while structures associated with Ω_x form tubular (braid-like) connection between the rings. Further



(a) Intermediate (β) eigenvalue PDF



(b) Extensive (α) eigenvalue PDF

Figure 7.11: PDF of the eigenvalues of $\tilde{S}_{ij}$ normalized by the RMS values $\sqrt{2\tilde{S}_{ij}}$. Here, LS: Low Swirl, HS: High Swirl, NR: Non-Reacting and R: Reacting case.

downstream, the large scale structures break down into smaller more randomly oriented tubular structures. Thus, it appears that in the presence of mean shear both sheet-like and tube-like structures are likely. This result is consistent with earlier observations in homogeneous sheared turbulent flows (Nomura and Elgobashi [1992]) where high vorticity magnitude regions were tube-like while moderate vorticity regions were sheet-like. It is believed that the magnitude of the strain field controls the breakdown of the large scale structures into smaller tube-like eddies. The significance of these geometrical structures on scalar mixing and combustion is discussed in the next section.

A strain state parameter, $s^* = -3\sqrt{6}\alpha\beta\gamma/(\alpha^2 + \beta^2 + \gamma^2)^{3/2}$, has been used to characterize the deformation of the fluid element in an incompressible flow (Lund and Rogers [1994]) with its magnitude interpreted as a measure of the shape, such that axi-symmetric extension corresponds to $s^* = 1$, plane shear occurs at $s^* = 0$ and axi-symmetric compression occurs when $s^* = -1$. For compressible flows, all three eigenvalues are independent (only two of them are independent in an incompressible flow), and therefore, an additional non-dimensional parameter is required to characterize the state of the strain-rate. This parameter is the dilatational part of the strain-rate tensor and is obtained by decomposing the tensor into isotropic and anisotropic parts. The isotropic part is given by $D^* = (\alpha + \beta + \gamma)/\sqrt{3(\alpha^2 + \beta^2 + \gamma^2)}$ Lund and Rogers [1994]. This quantity is bounded by ± 1 corresponding to pure isotropic expansion and compression. The anisotropic strain state, s^{**} is computed using the eigenvalues of the anisotropic part of the strain rate tensor, $\tilde{S}_{ij}^* = \tilde{S}_{ij} - \tilde{S}_{kk}\delta_{ij}/3$. The complete state of the strain is then characterized by D^* and s^{**} .

Figure 7.12 shows the PDF of s^{**} and D^* for the reacting cases. Consistent with earlier observations (Lund and Rogers [1994]), axi-symmetric extension is most probable flow structure ($s^{**} = 1$). It is also clear that, increase in swirl increases the likelihood of $s^{**} = 1$ state. By comparing Fig. 7.12 with non-reacting results (not

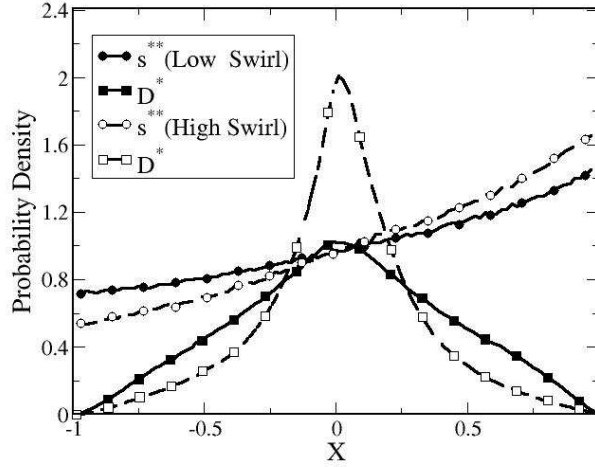


Figure 7.12: PDF of the strain state parameter s^{**} and D^* for low and high swirl reacting cases.

shown) it is found that the variance of D^* increases significantly in the presence of heat release indicating that the volumetric dilatation significantly alters the strain rate field.

7.7 Vorticity-Strain Alignment

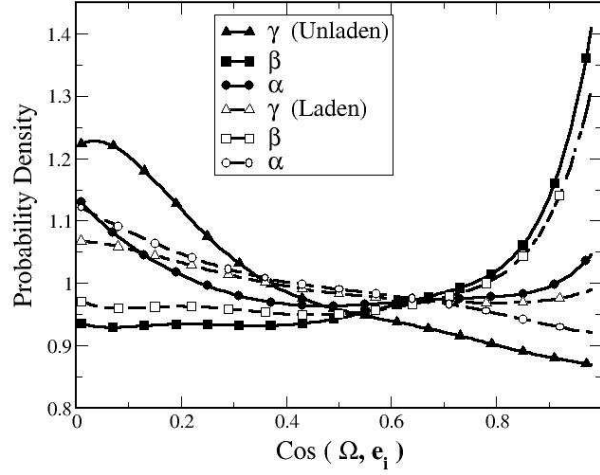
The interaction of the vorticity, $\vec{\Omega}$ and strain-rate, $\tilde{S}_{ij}$ through vortex stretching is considered a primary mechanism for the generation of small scales. The orientation of $\vec{\Omega}$ relative to the principal strain-rate directions (e_α , e_β and e_γ) is determined by evaluating the cosine of the angle, $(\vec{\Omega}, \vec{e}_i)$. Past numerical (Ashurst et al. [1987]) and experimental (Tsinober et al. [1992]) studies have established that in incompressible flow, $\vec{\Omega}$ aligns with e_β and this preferential alignment appears to be a universal feature of small scale structures (Nomura and Elgobashi [1992]). Here, we investigate the effect of particle, swirl magnitude and heat release on this alignment behavior.

Figures 7.13(a) and (b) show that in general, $\vec{\Omega}$ aligns with e_β , and normal to the e_α and e_γ . Presence of particles reduces the magnitude of the alignment in

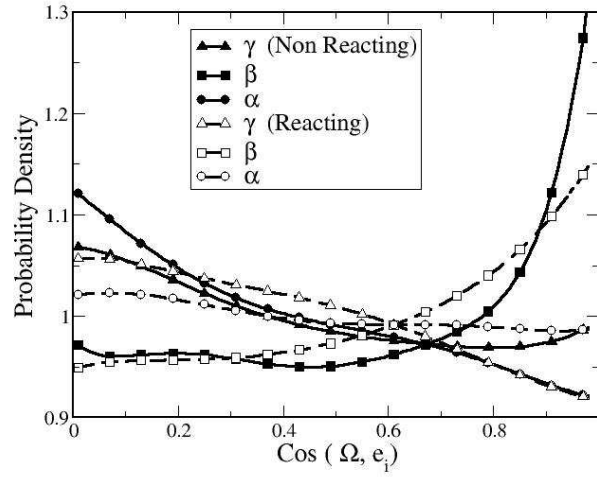
all three directions (Fig. 7.13(a)). Earlier, Ahmed and Elgobashi. [2000] showed that an increase in vorticity stretching increases this alignment, whereas an increase in a rotational effect decreases this alignment. The effect of particles appears to contribute as a rotational effect in their formulation (Ahmed and Elgobashi. [2000]). The observed reduction in $\vec{\Omega} - e_\beta$ alignment in the present (and the earlier case) suggests that the rotational effect is more dominant in particle laden flows.

Figure 7.13(b) shows that heat release decreases $\vec{\Omega} - e_\beta$ alignment as well. This is primarily due to volumetric expansion effect which dominates near the reaction zone. This agrees qualitatively with results obtained in reacting gas-phase homogeneous turbulence (Jaberi et al. [2000]). Increase in swirl (not shown) also reduces this alignment. It has been shown in a previous study (Nomura and Elgobashi [1992]) that this alignment increases for flows with imposed gradient/mean shear compared to isotropic case. It will be shown in the next section that the strain-rate field for high swirl case behaves more like those in isotropic flows, due to the rapid breakdown of the large scale structures, compared with the low swirl case. Hence, it is not surprising that high swirl with an isotropic strain rate field, shows a lower $\vec{\Omega} - e_\beta$ alignment compared to low swirl case, which behaves more like flows with imposed mean gradient/shear, consistent with the earlier observation (Nomura and Elgobashi [1992]). Smith and Menon [1996c] also observed that the $\vec{\Omega} - e_\beta$ alignment reduces in a spatially evolving flows, as the organized structures in the shear layer break down into randomly oriented structures.

Betchov [1956] suggested that vortex stretching and the associated vorticity production occurs when the intermediate strain rate is positive. Results obtained here seems to corroborate this conclusion. Analysis shows β is predominantly positive and that e_β aligns with $\vec{\Omega}$. Thus, a vortex tube experiences stretching along its axis and compression normal to its axis. As vortex tube is stretched, its radius decreases and therefore, by conservation of angular momentum, its rotational kinetic energy



(a) With and without particles, non-reacting case



(b) Non-reacting and reacting cases

Figure 7.13: Alignment of vorticity vector with principal strain-rate directions for low swirl cases.

increases and so does the vorticity along its axis.

7.8 Scalar Gradient - Strain Alignment

Earlier studies (Ashurst et al. [1987], Nomura and Elgobashi [1992]) have established that the scalar gradient tends to align with the most compressive strain axis in isotropic and homogeneous sheared turbulence. However, in flows with imposed mean shear, the alignment between the scalar gradient and the compressive strain tends to occur at $\theta \approx 30$ ($\cos\theta = 0.86$). Figure 7.14 shows the alignment probability for low and high swirl reacting cases. Both cases show that the scalar gradient tends to align perpendicular ($\cos\theta = 0$) to the intermediate (e_β) strain directions. While the scalar gradient is aligned at $\theta \approx 30$ degrees from the compressive strain (e_γ) for the low swirl case, the most probable alignment between the scalar gradient and the e_γ shifts toward zero ($\cos\theta = 1.0$) for the high swirl case. The apparent discrepancy is due to competing effects of the mean and fluctuating (local) strain rates.

Examination of the instantaneous flow field helps to clarify this issue. In the presence of swirl, the flow is subjected to a mean tangential shear in addition to the mean shear in the axial direction. Presence of mean axial shear leads to the formation and shedding of large scale vortices that comprise predominantly of the azimuthal vorticity component. On the other hand, the presence of tangential shear leads to the formation of stream-wise vorticity due to the onset of the azimuthal instability (see Fig. 7.3(a) and (b)). Thus, a swirling flow evolves continuously under stretch/strain in both axial and azimuthal directions. The complex interaction between these vorticity results in the formation of tube-like azimuthal vortical structures of opposite sign and contribute to the breakdown into smaller structures. This is further explained below.

In the low swirl case, the large-scale structures persist longer and provide a relatively higher mean strain rate field over longer spatial extent. In contrast, in high

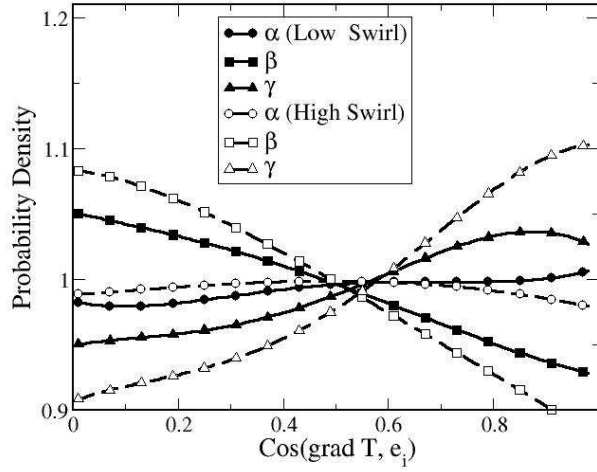


Figure 7.14: Alignment of scalar (temperature) gradient with principal strain rate directions for low and high swirl reacting cases.

swirl these structures break down quickly into smaller tubular structures (Fig. 7.3(b)). This leads to a more isotropic strain-rate field and hence, the alignment between the scalar gradient and the compressive strain is similar as in isotropic flows. However, when large-scale structures persist longer (low swirl), the mean strain-rate dominates the local fluctuating strain-rate and the alignment is similar to those in flows with imposed shear/mean gradient. This result is consistent with earlier observations in premixed combustion in a dump combustor (Smith and Menon [1996c]). Thus it appears that, since the scalar gradient is aligned parallel to the most compressive strain, iso-scalar surfaces are pushed closer together, amplifying the scalar gradient fluctuations at small scales. Molecular diffusion then smears out the gradients and thus, enhances the molecular mixing rate. Furthermore, since vorticity aligns normal to the compressive strain it implies that vorticity is aligned normal to the scalar gradient. This suggests that sheets of iso-scalar surface wraps around tubular vortex rings. While vorticity gradient is intense in these tubes, the scalar gradient is intense in the sheets wrapping them.

7.9 Conclusions

In the present study, the effects of droplets, heat release and swirl on the fine-scale structure of turbulence are studied by examining the vorticity, the strain-rate and the scalar fields obtained in LES of swirling spray combustion in a gas turbine engine. Results show that the vorticity tends to align with the intermediate strain rate, and the presence of particles, heat release and swirl decreases the magnitude of this alignment. The scalar gradient aligns with the most compressive strain rate and this alignment tends to increase the scalar gradient fluctuations at the small scales, thereby increasing molecular diffusion which smears out the gradients. This alignment was also found to increase with swirl. PDF of the strain-rate eigenvalues reveals the presence of both tube-like and sheet-like structures. Tube-like structures of vorticity are found to coincide with regions of intense vorticity gradients. On the other hand, regions of high scalar gradient form sheet-like structures that wrap around the tubular vortical structures. The scalar gradient in these sheets are amplified by the interaction between the strain rate field and vorticity field, thus increasing mixing and reaction in these regions. The fact that different studies (incompressible and compressible, premixed and two-phase, isotropic and shear driven flows) all display strikingly similar features suggests an universality in the manner in which vortical structures and flames interact.

CHAPTER VIII

LES-LEM OF SPRAY COMBUSTION

This chapter discusses the studies carried out to test (qualitatively and quantitatively) the performance of all the models developed in this study, viz. two phase models (Chapter 2) and the sub-grid combustion model, LES-LEM (Chapter 4). For this purpose, a full three-dimensional simulation of a partially premixed flame in a droplet-laden isotropic turbulent flow has been conducted. Before presenting the results obtained from the spray simulations, results for the freely propagating premixed flame in a isotropic field are presented. The motivation behind this study is to gauge and validate the ability of the present LES-LEM in capturing the essential physics observed in similar studies (conducted using DNS) in the past (Ashurst [1990], Rutland and Trouve [1990], Trouve and Poinso [1994], Echekki and Chen [1996]).

The statistical results presented here includes flame surface area evolution, PDF of tangential strain-rate and curvature, alignment between strain-rate eigenvectors and vorticity, and finally the alignment statistics for strain-rate eigenvectors and flame normal direction. These statistics characterize the vorticity and the small scale strain-rate field and hence elucidates the mutual effect of combustion and turbulence on each other.

8.1 Initial Conditions and the Flame Parameters

Flow conditions are chosen to match the DNS of Trouve and Poinso [1994]. The LES is initialized in a three-dimensional domain with a plane laminar premixed flame separating reactants (on the left) from the products (on to the right). A snapshot of the initial conditions is shown in Fig. 8.1. The parameters for this simulation are

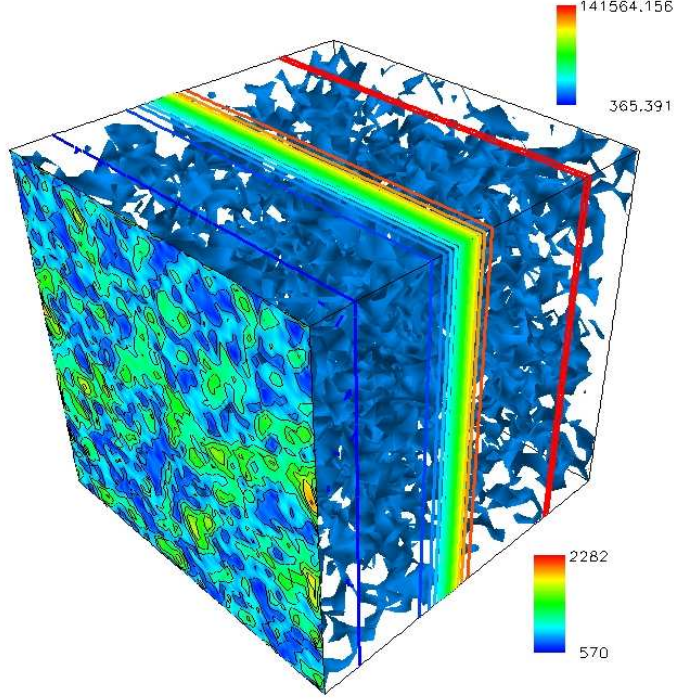


Figure 8.1: Iso-surface of vorticity magnitude and contours of temperature. Also shown are the contours of vorticity magnitude at the inlet plane.

Table 8.1: Turbulent Flame Properties

ϕ	u'/S_L	L/δ_F	η/δ_F	Re_L	Re_λ	T_{Ref}	L_{Box}	L_{Box}/L
0.8	10.0	5.2	0.27	70.0	50.0	570 K	1.5 cm	31

given in Table 8.1. It is worthwhile to note that the Kolmogorov scale is smaller than the flame thickness for this case, which implies that the smaller turbulent scales can penetrate into the flame and influence the species and energy transport. However, the flame is thin compared to the integral scales ($L/\delta_F = 5.2$) and the turbulence is intense relative to the laminar flame speed ($u'/S_L = 10$). According to Trounev and Poinot [1994] the simulated flames are in the border between the flamelet regime and the distributed regime, which is typical of most practical turbulent flames.

A few contrasting features between the two simulations (DNS and LES-LEM) are

reported here.

- DNS studies of Trouve and Poinso [1994] used 129^3 mesh points to resolve the chemical and the turbulent scales. The present LES-LEM study, on the other hand, utilizes only 65^3 grid for the LES and 12 linear eddy cells per LES cell. Since, the chemical scales are resolved by the sub-grid LEM field, LES can be used to resolve only the large scale part of the turbulent spectrum. This implies that the LES-LEM can be used to simulate very high Reynold's number, on a much coarser grid, while still resolving the flame structure.
- The present study employs a single-step, five species ($CH_4, O_2, CO_2, H_2O, N_2$), methane-air mechanism (Westbrook and Dryer [1981]) with finite-rate effects. On the other hand the DNS study used a global single-step mechanism (Williams [1988]) such as



with equal molecular weights for reactants and the products. The chemical reaction rate is computed using the Arrhenius Kinetics. Thus, the DNS study did not include the effect of variable molecular weight and real species and employed a single Lewis number for all the species. The flame structure was controlled by altering the pre-exponential factor, reduced activation energy and the heat release factor.

- In the present study all the transport properties are temperature dependent similar to the DNS study of Trouve and Poinso [1994]. Each species was given a constant (not time varying) but different Lewis number based on its mass diffusivity at the product temperature.

8.2 Results and Discussion

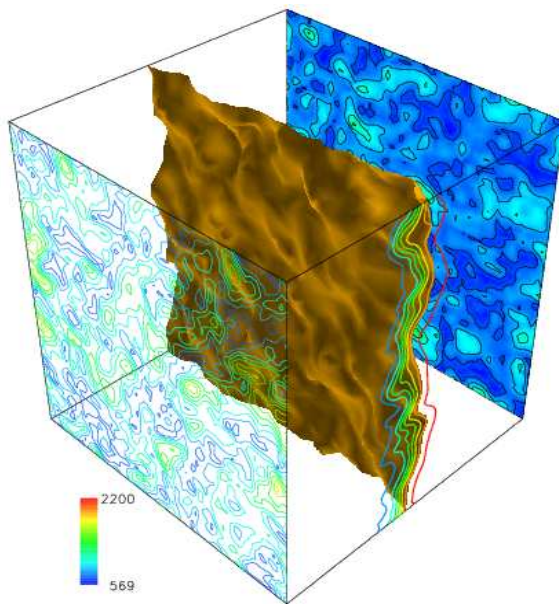
To define the quantities based on the flame, the location of the flame has to be identified. It is a common practice to use a constant progress-variable surface to define the flame front location, where the progress variable c is defined as follows:

$$c = 1 - Y_{CH_4}/Y_{CH_4,u} \quad (8.2)$$

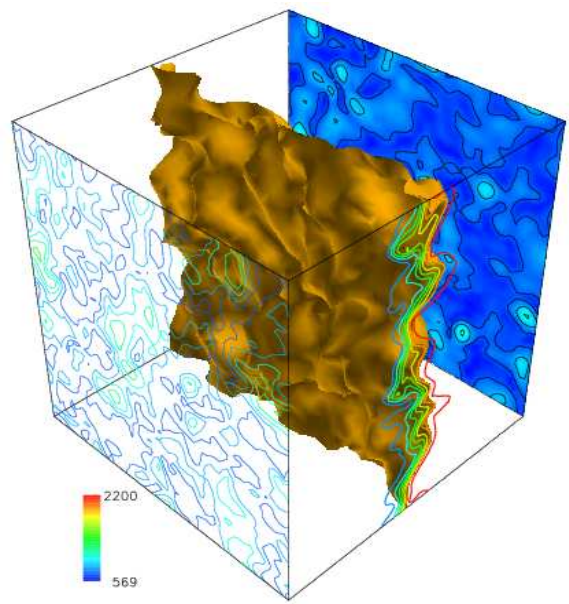
where Y_{CH_4} and $Y_{CH_4,u}$ are respectively, the local methane mass fraction and the mass fraction of methane in the unburnt mixture. The progress variable c varies monotonously through the flame from 0 in the reactants to 1 in the products. $c = 0.8$ surface is used as the flame surface in the present study similar to the study by Trounev and Poinso [1994]. At any location, on this surface the local gradient of c defines the normal direction to the flame front: $\mathbf{n} = -\nabla c/|\nabla c|$ where $\mathbf{n}$ points into the reactants

Figure 8.2 shows the evolution of flame surface at four different instances (non-dimensional) in time. Also shown are the vorticity magnitude at a plane before (inflow plane) and a plane behind (outflow plane) the flame. The interaction of the vorticity field with the flame front causes intense wrinkling. This wrinkling increases the flame surface area of the initially flat laminar flame. The suppression of the post flame velocity fluctuations due to the increased viscosity is evident in these figures. The broadening of the preheat zones due to the penetration of the small scales is also apparent in these images. It is well known (Trounev and Poinso [1994]) that the large scale wrinkling of the flame causes a dramatic increase in flame surface area. To quantify and confirm this notion, the flame surface area is computed as a function of time and is shown in Fig. 8.3.

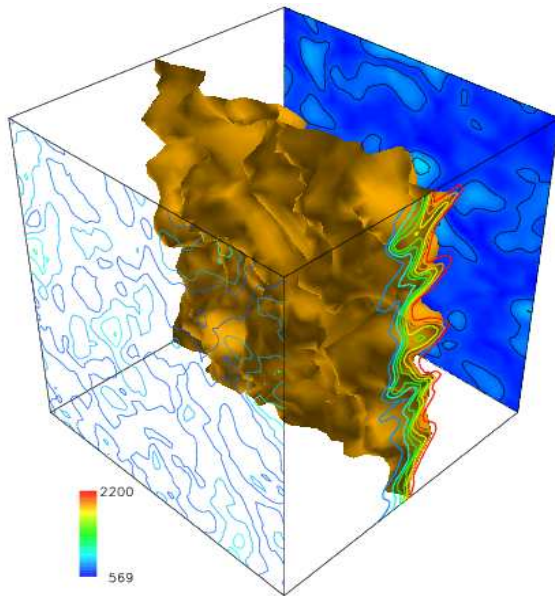
As it can be observed, the flame surface area increases monotonously in time, even though the slope decreases eventually. The increase occurs as the turbulence wrinkles the initially plane laminar flame. As the turbulence decays, the rate of increase of the flame surface area decreases and hence the slope of the plot decreases. Also



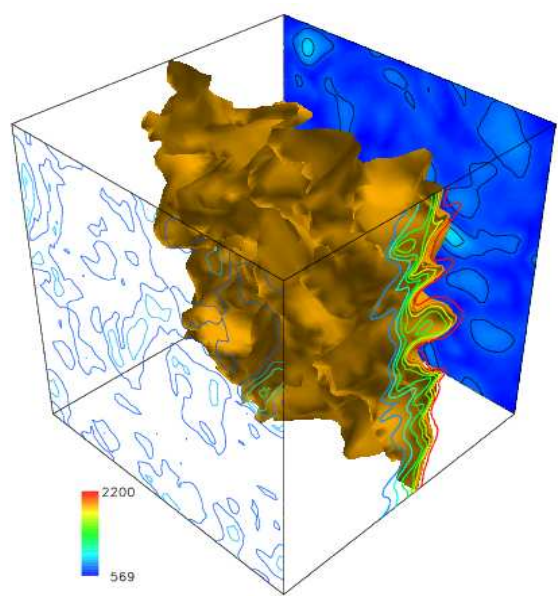
(a) $T = 1$



(b) $T = 6$



(c) $T = 20$



(d) $T = 40$

Figure 8.2: Iso-surface of instantaneous flame structure and contours of vorticity magnitude.

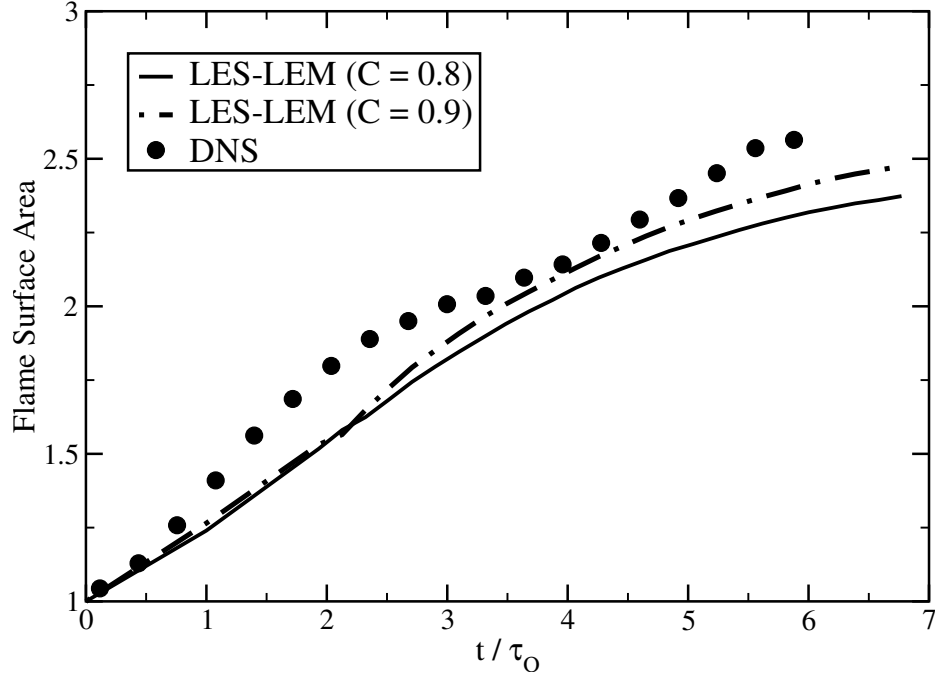


Figure 8.3: Evolution of the flame surface area.

shown in Fig. 8.3 is the result from the DNS study of Trounev and Poinso [1994]. The discrepancy between the current LES-LEM results and DNS can be explained as follows.

- In LES, only the large scales are resolved and not the smaller, sub-grid scales. In LES-LEM, the effect of three-dimensional fine (sub-grid) scales is modeled using stochastic one-dimensional maps. Thus, even though LES-LEM includes the fine-scale wrinkling in the LES averaged reaction rates, there is no direct way to compute the increase in the flame surface area at the sub-grid scales. Therefore, the flame surface area calculated on the LES grid does not include the area contribution from the fine scale wrinkling of the sub-grid scales. Consequently, the flame surface area calculated from LES will be smaller than that of the DNS.
- DNS study included only two species, i.e., reactants and the products, with equal molecular weights. Hence, the effect of nitrogen dilution in the reactant

mixture is not included in the DNS study.

- Finally, the present study includes differential diffusion effects which was not considered in the DNS study.

8.2.1 Curvature statistics

Flame curvature influences the local burning rate and hence, the propagation speed for flames with non-unity Lewis numbers ($Le = \frac{\text{thermal diffusivity}}{\text{mass diffusivity}}$). For the case $Le < 1$, the reactant diffuses more rapidly than thermal energy. Therefore, in convex (to the reactants) regions of the flame, the reactants have a focusing effect. This in turn increases the local reaction rate. The opposite situation occurs if the flame is concave to the reactants. Similarly, for the case $Le > 1$, thermal energy is defocused in convex regions of the flame, thus reducing the flame propagation speeds. On the other hand, the thermal energy is focused in concave regions of the flame. This coupling of differential diffusion with the curvature has been studied extensively in strained laminar flames and two dimensional (Echekki and Chen [1996]) and three dimensional DNS (Rutland and Trouve [1990], Trouve and Poinso [1994]). Results from the current LES-LEM simulations on curvature effects are reported here and discussed in context.

To investigate the effect of curvature on the flame structure, instantaneous reaction rates of methane are computed. Figure 8.4(a) shows the instantaneous reaction rate contours along a two-dimensional plane. The flow is from left to right, i.e., the reactants are on the left and the products are on the right side of the flame. The variation in the colors mark the intensity of the reaction rate and red regions are high reaction rate regions compared to green and blue colors. For $Le > 1$, elements of flame surface that are concave toward the reactants are expected to burn faster, while the elements that are concave toward the products are expected to have a lower burning rates compared to a planar flame. This is also evident in Fig. 8.4(a). The

flame elements that are concave to the reactants have a higher reaction rate compared to the convex flame elements. This is also confirmed in Fig. 8.4(b) which shows the line plot of the reaction rate along two flame elements that are concave and convex to the reactants, respectively.

Typical PDF of flame curvature normalized by the laminar flame thickness is shown in Fig. 8.5. Curvature of the flame surface is given by

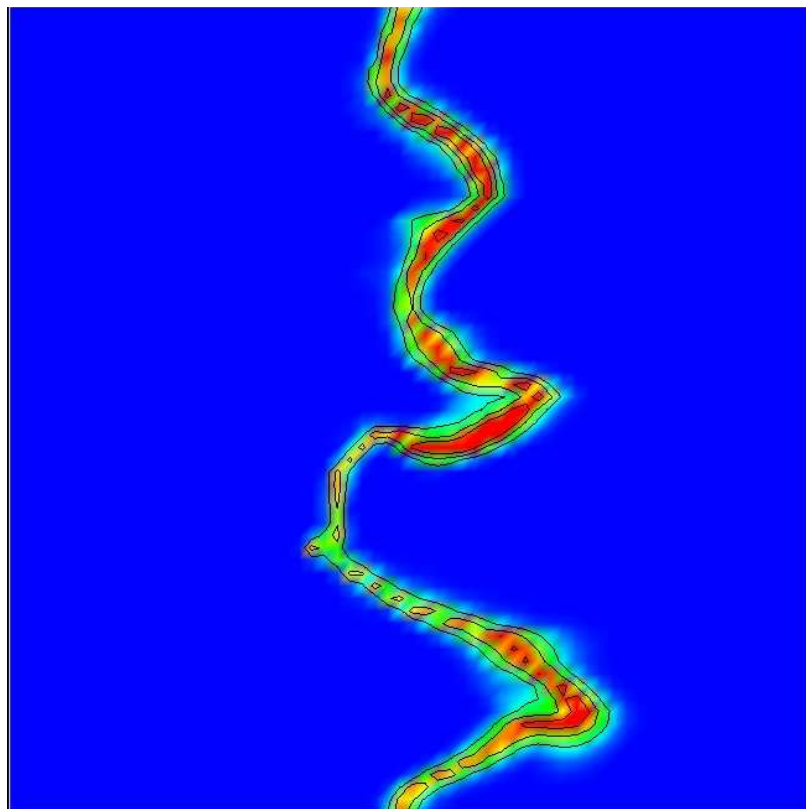
$$1/R = \nabla \cdot \mathbf{n}_{\mathbf{c}=\mathbf{c}_0} \quad (8.3)$$

Note the divergence, $\nabla \cdot \mathbf{n}(x; c)$ is not same as the above definition (Ashurst [1990]) and hence the curvature is usually computed from the following curvature tensor.

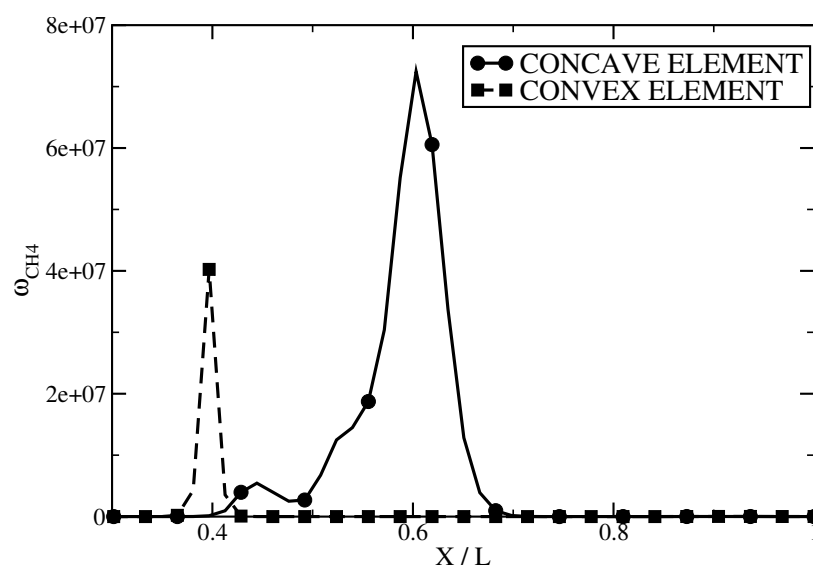
$$h_{ij} = -\frac{\partial^2 c}{\partial x_i \partial x_j} \frac{1}{g} + \frac{1}{g^3} \sum_{k=1}^3 \frac{\partial c}{\partial x_i} \frac{\partial c}{\partial x_k} \frac{\partial^2 c}{\partial x_j \partial x_k} \quad (8.4)$$

where $g = |\nabla c|$. This tensor has two real eigenvalues h_1, h_2 and one zero eigenvalue. Using this tensor and its eigenvalues, the two principal curvatures of a surface are defined as $h_1 = 1/R_1$ and $h_2 = 1/R_2$. The surface curvature of a flame is then defined as the half-sum of the two local principal radii of curvature. Negative curvature represents a concave flame with respect to the reactants. The mean curvature normalized by flame thickness, plotted in Fig. 8.5(a) shows that the mean is zero and the distribution is slightly skewed toward positive curvature. This is consistent with the results obtained in the past (Rutland and Trounev [1990], Echekki and Chen [1996]).

A convenient way to describe the local flame shape is a ratio of the principal radii of curvatures, called curvature shape factor. The curvature shape factor is a quantitative measure of the shape of the surface element. It is defined as $S_h = h_s/h_l$, where h_s and h_l are the smaller and larger magnitude of the eigenvalues of the curvature tensor h_{ij} . By definition, the shape factor, is confined to the range $[-1, +1]$ and its value indicates the shape of the surface element: $S = +1$ means that the two principal curvatures are equal in magnitude and sign, implying a spherical surface element; $S = -1$ implies a pseudo-spherical (saddle-shaped) surface element; and $S = 0$ means



(a) $\dot{\omega}_{CH_4}$



(b) $\dot{\omega}_{CH_4}$

Figure 8.4: Reaction rate contours.

that the element is cylindrical in shape. Intermediate values of the shape factor represent combinations of the above shapes. The shape factor of a flat surface is indeterminate. Figure 8.5(b) shows the PDF of curvature shape factor and it is evident that the cylindrical curvature predominates in the flow. It is also clear that the spherical curvature simply does not occur and the spherical saddle points occur with a small but finite probability. These results are in excellent agreement with the two dimensional (Echekki and Chen [1996]) and three dimensional (Rutland and Trouve [1990]) DNS studies conducted in the past.

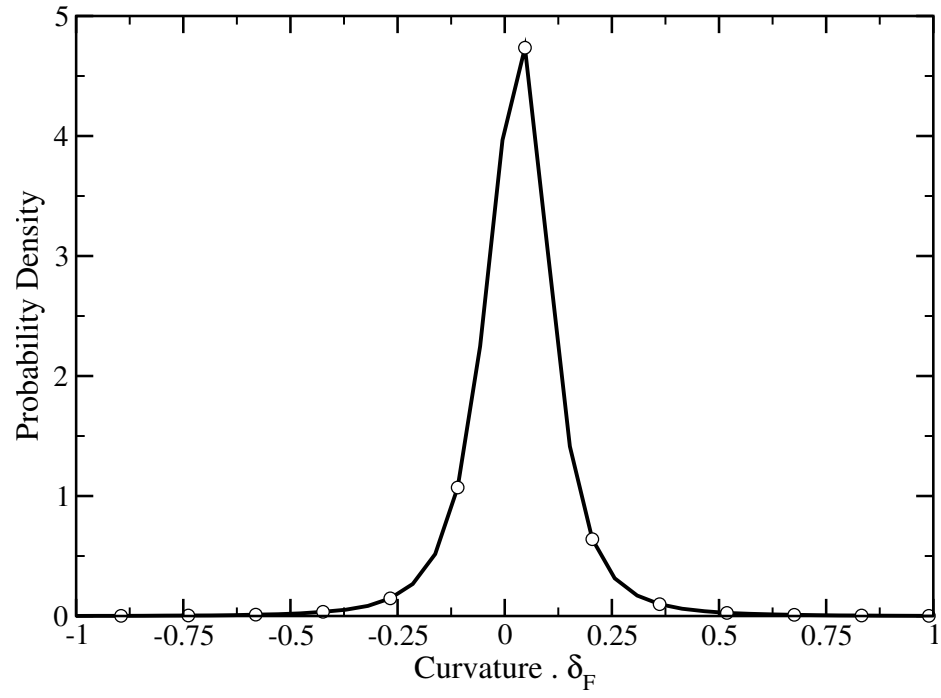
8.2.2 Strain-rate statistics

Strain-rate modifies the convective transport along normal and the tangential directions to the flame, thus modifying the flame structure. Strain-rate also influences the evolution of the flame surface area and hence alters the flame propagation speed. Figure 8.6 shows the PDF of the tangential strain-rate on the flame, normalized by the Kolmogorov time scale. The tangential strain-rate is computed as follows.

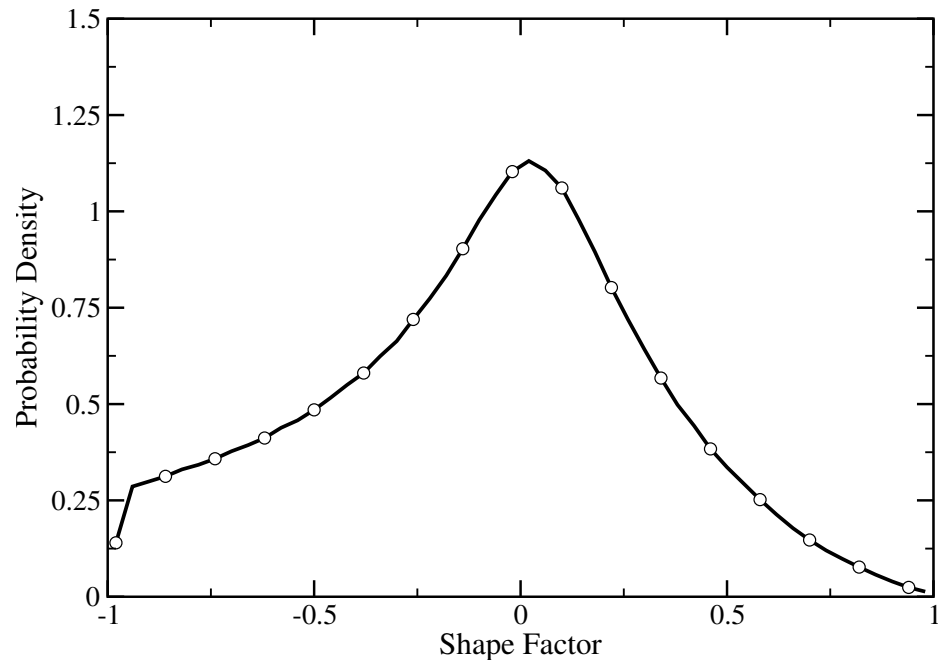
$$a_T = \frac{\partial u_k}{\partial x_k} - n_i S_{ij} n_i \quad (8.5)$$

The PDF reveals that the mean tangential strain-rate is positive and thus, the extensive strain rates are dominant in the flame. This dominance of the positive extensive strain-rate is also seen in the three dimensional simulations of Rutland and Trouve [1990]. In the presence of heat release, flow acceleration normal to the flame is high and hence, the extensive (positive) strain-rate is high across the flame. Since the probability of positive strain-rate is high and since the positive strain-rate is high across the flame, this indicates that the flame aligns preferentially with extensive (positive) strain-rates.

To quantify and confirm this, it is decided to compute the orientation of the flame normal with the strain-rate direction. The principal values and the principal direction of the strain-rate tensor is usually characterized by the eigenvalues and the



(a) Flame Curvature



(b) Shape Factor

Figure 8.5: PDF of flame curvature and shape-factor.

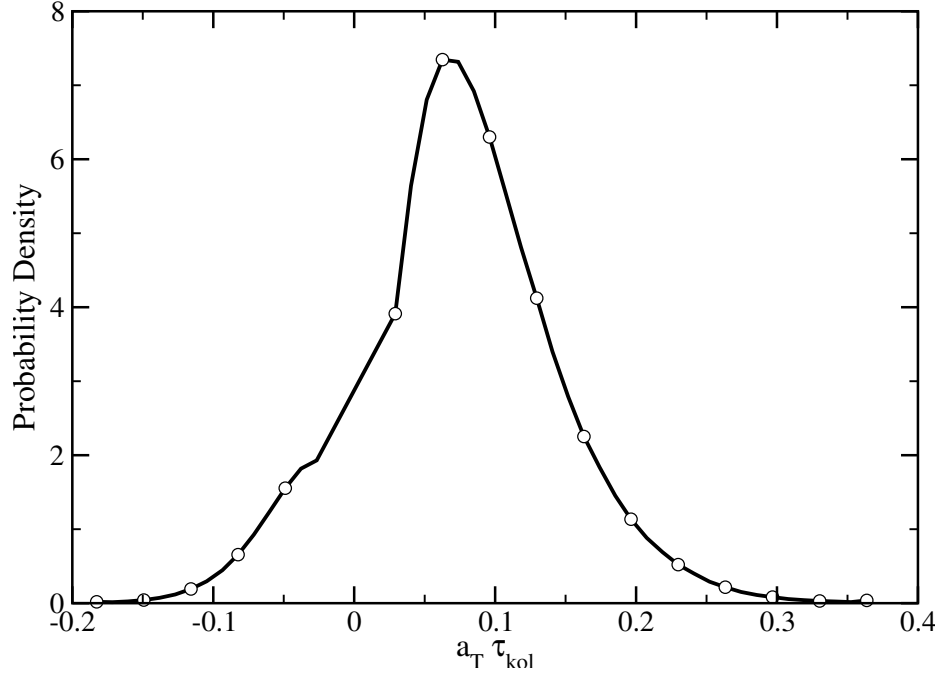


Figure 8.6: PDF of the tangential strain rate.

eigenvectors of the strain-rate tensor, respectively. For a real symmetric matrix like the strain-rate tensor, there are three real eigenvalues α, β, γ and three corresponding eigen directions e_α, e_β and e_γ . Past studies (Betchov [1956], Kerr [1987], Ashurst et al. [1987]) on the PDF of the eigenvalues of the strain-rate tensor in isotropic turbulence have shown that the principal strain-rate structure is such that $\alpha > \beta > \gamma$ with $\alpha > 0$ and $\gamma < 0$ and β taking positive and negative values. Based on the sign, α is associated with the extensive strain-rate and $\gamma < 0$ is associated with the compressive strain-rates. With this information and using the cosine of the angle between the eigen directions and the flame normal direction, the orientation of the flame surface and the strain-rate field can be determined.

Figure 8.7 shows the alignment statistics between the flame normal and the strain-rate eigenvectors. As noted earlier, the PDF is maximum for the cosine of the angle between the flame normal and e_α , the eigenvector corresponding to the extensive strain-rate. This implies that the flame normal is most likely to be aligned with the

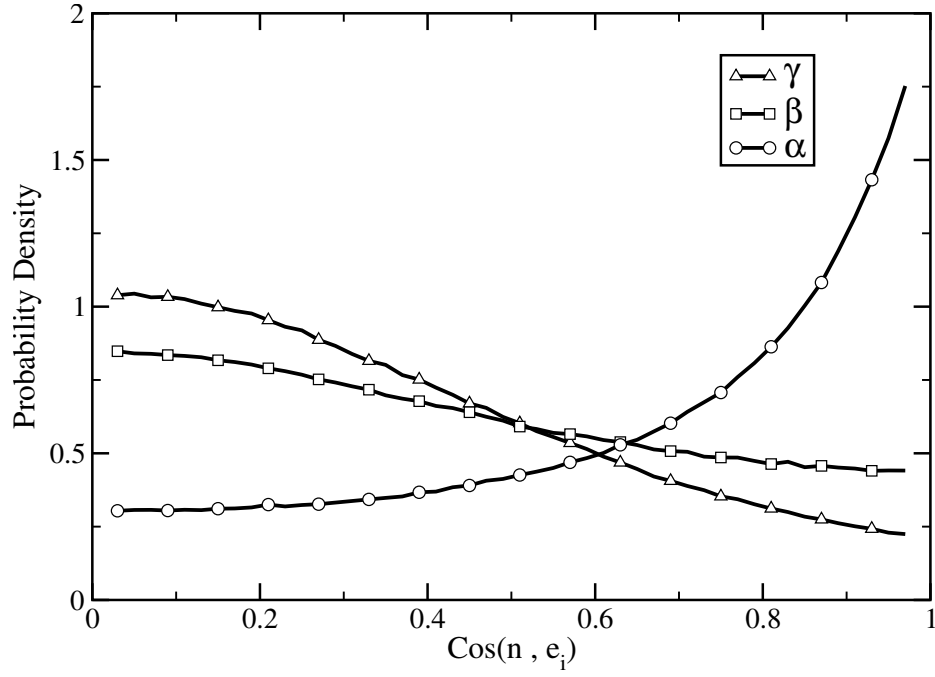


Figure 8.7: Alignment of strain-rate eigenvectors with flame-normal direction.

positive strain-rates as was indicated by the PDF of the tangential strain-rate previously. This result is in good agreement with the earlier observation by Chakravarthy and Menon [2000] for exothermic stagnation point flames.

All these results also indicate an interesting fact that the curvature and the strain-rate are not independent. It is proved earlier that the flame surface is predominantly cylindrical (zero mean curvature) and that the probability of the positive strain-rate is high on the flame surface. This would imply that any other shaped surface such as a spherically curved surface (with a finite non-zero probability) is strained into a more cylindrical shape by the predominant extensive strain-rate. In other words, the regions of high (non-zero) curvature are possible only if the local strain-rate is low and vice versa. This notion is also strongly purported by Rutland and Trouve [1990] from their three-dimensional DNS, where they observed the maximum strain is found in regions of zero curvature while the maximum positive and negative curvature occurs where the strain is nearly zero.

Finally, results are reported for the PDF of the eigenvalue structure of the strain-rate field and for the orientation of strain rate eigen directions relative to the vorticity field. Fig. 8.8(a) shows the PDF of the eigenvalues of the strain-rate tensor. As seen in the past (Betchov [1956], Kerr [1987], Ashurst et al. [1987]), the intermediate strain rate eigenvalue is predominantly positive. In addition, it is clear from fig. 8.8(b) that the vorticity field is aligned preferentially with the intermediate eigenvalue of the strain-rate tensor. This study suggests that the LES-LEM approach can capture the physics of flame-turbulence interaction correctly.

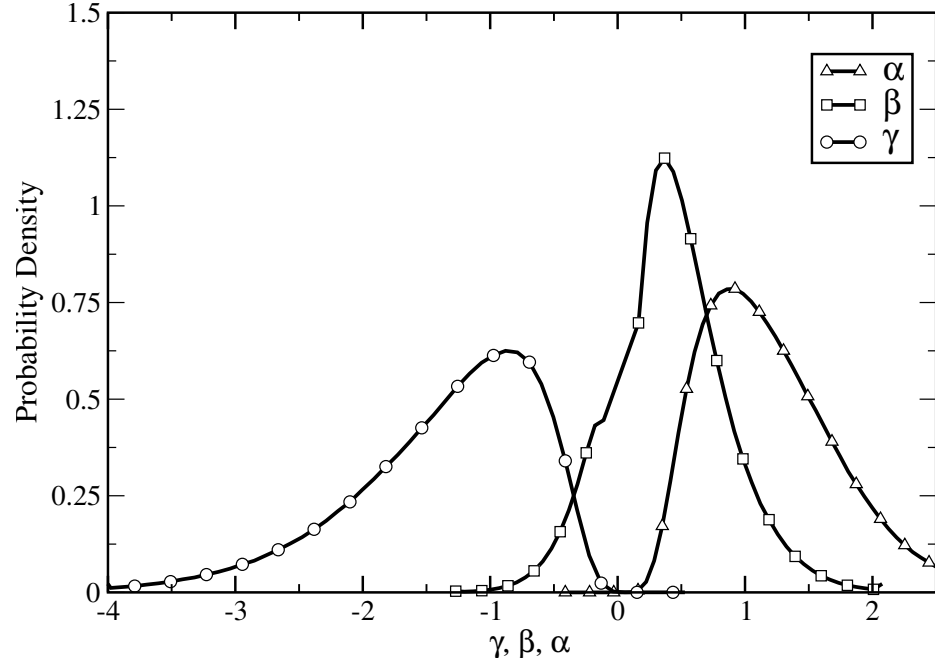
8.3 Partially Premixed Combustion

In many practical circumstances, a certain degree of premixing of reactants can be expected to occur even when the reactants are initially non-premixed. This can lead to partially premixed combustion. Partially premixed combustion occurs in a wide variety of technical applications, for example in direct injection gasoline engines, diesel engines with high pressure injection systems or gas turbines.

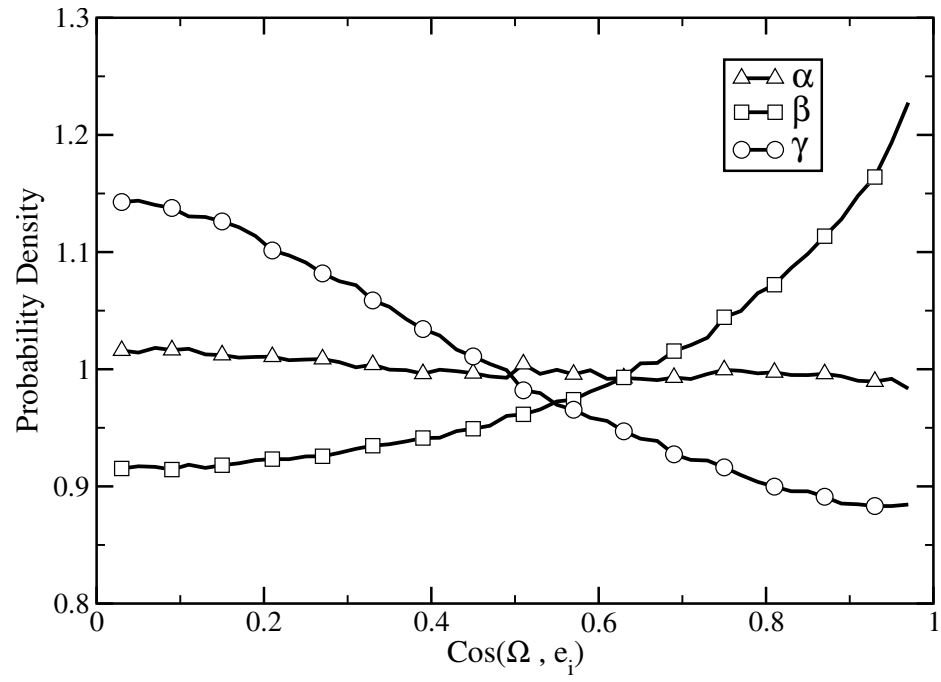
To test the capability of the present LES-LEM approach to deal with this problem, and the spray models to capture the physics of partially premixed combustion, studies were conducted on the freely propagating flame in a partially premixed reactant, laden with droplets of fuel (methane droplets). Results on the flame-structure are reported here.

Initial conditions

- A plane laminar flame (corresponding to an equivalence ratio of $\phi = 0.8$) is initialized in an isotropic flow field, near the middle of the domain.
- The equivalence ratio of the gas mixture is gradually reduced from $\Phi = 0.8$, before the flame to $\Phi = 0.35$ at the inlet. This gives a smooth spatial variation to the equivalence ratio ahead of the flame front.



(a) α, β, γ



(b) $\Omega \cdot e_i$

Figure 8.8: (a) PDF of the eigenvalues of the strain-rate tensor (b) Alignment of strain-rate eigenvectors with vorticity.

- To make up for the reduction of methane mass fraction in the gas mixture, the incoming mixture is seeded with droplets at a rate that would give an over-all equivalence ratio of 0.8.
- The initial diameter of the injected droplets are, $d_d = 20\mu m$ and the droplets are assumed to complete vaporize when they reach a cut-off radius of $r_d = 5\mu m$.
- The droplets are injected randomly (with an uniform distribution) at the inlet plane
- The initial temperature of the injected droplets are $107K$ and are injected at a velocity of $50m/sec$.
- The incoming gas mixture is at a temperature of $570K$ and the system is at a constant pressure of $1atm$.
- Characteristic conditions of Poinot and Lele [1992] are used at the inflow and outflow and periodic conditions are specified in the transverse (Y) and span-wise (Z) directions.

An uniform grid of $64 \times 64 \times 64$ is used for the present study. The initial conditions for the turbulent field are given in Table 8.1.

Figure 8.9 shows the iso-surface of the temperature and the droplet distribution. The magnitude of the temperature iso-surface is 2200 K. The droplets are colored based on their radius. As it can be observed, the size of the droplets are continuously decreasing, due to vaporization and combustion. The reduction is drastic as the droplets move closer to the flame. The evaporated droplets along with the oxygen rich environment surrounding them, provide a continuous supply of premixed reactants to the flame zone. Even though most of the droplets vaporize near the flame, some droplets seem to traverse through the flame and vaporize in the post flame zone. The

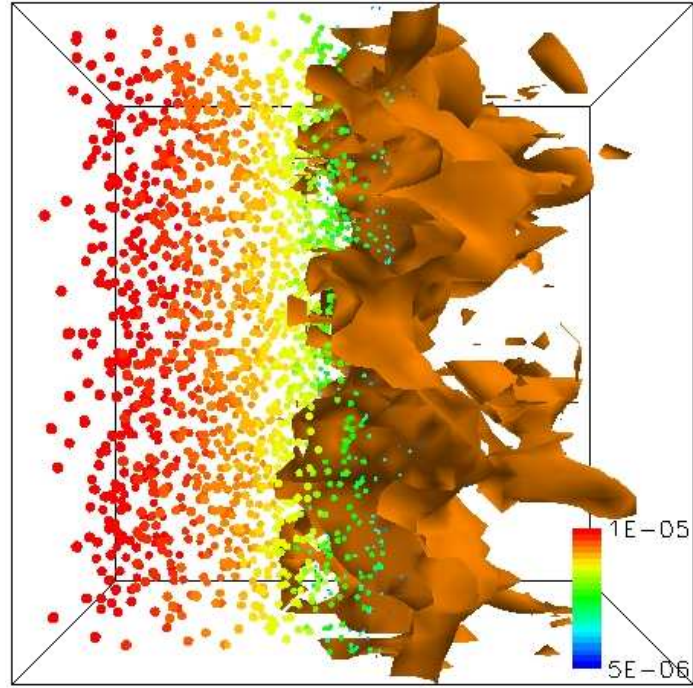


Figure 8.9: Iso-surface of temperature and droplet distribution.

droplets that vaporize behind the flame, burn individually or as groups beyond the flame, as shown in Fig. 8.10.

Figure 8.10 shows the contours of methane mass fraction (in color) along with the contours of methane reaction rate ($\dot{w}_{CH_4}$) (in gray scale). As it can be observed from this figure, vaporization of more droplets near the flame zone creates an large patches of methane mass fraction ahead of the flame (green patches). Some of the yellow and red patches ($Y_{CH_4} > 0.06$) seen at the flame and slightly behind the flame are due to the sudden evaporation of droplets that managed to reach the flame and even traverse through the flame zone.

Figure 8.11 shows the contours of vorticity magnitude along with the droplet distribution. The intense vorticity field before the flame zone dissipates quickly in the post flame region, due to the increased viscosity in the hot products. Figure 8.12(a) shows the contours of temperature. In general, a broad flame is predicted due to the strong variation in equivalence ratio. This is also confirmed by the contours of

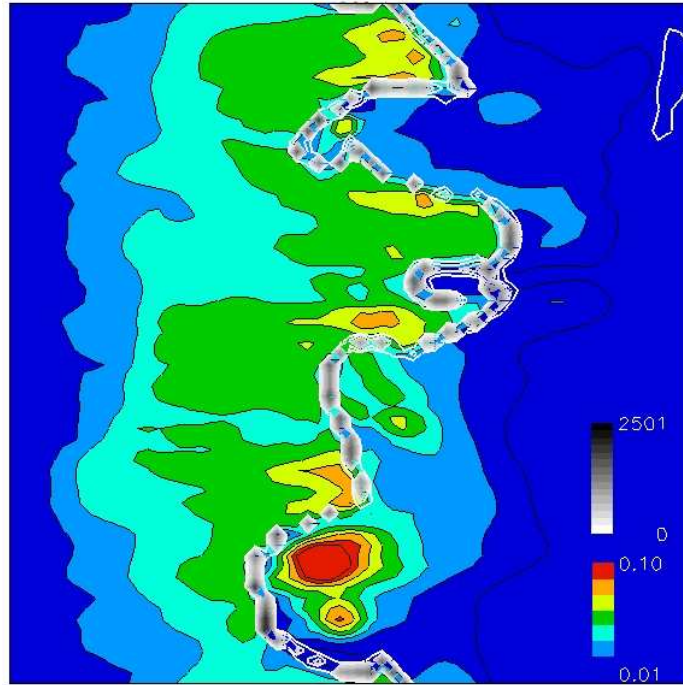


Figure 8.10: Contours of methane reaction-rate and mass fraction.

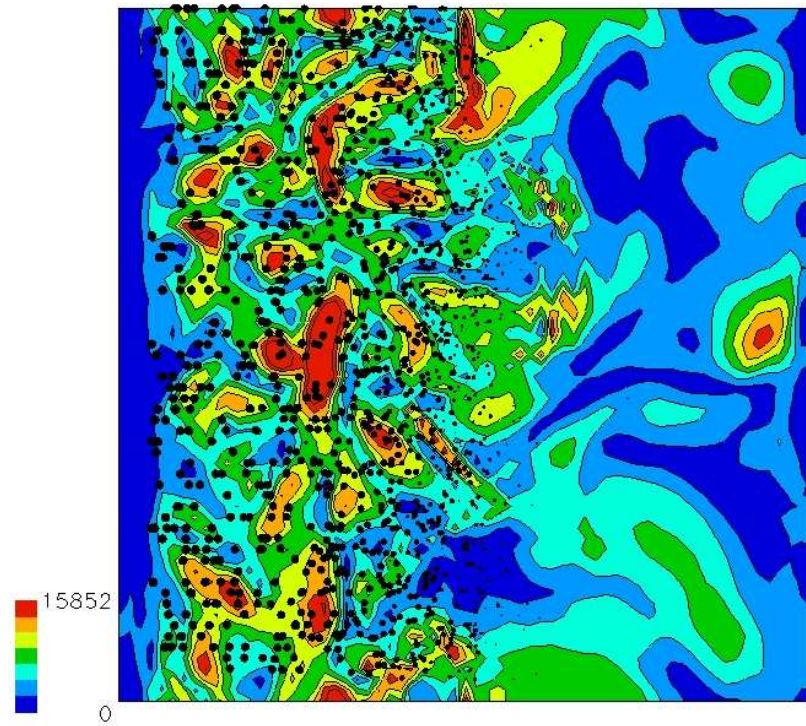


Figure 8.11: Contours of vorticity and droplet distribution.

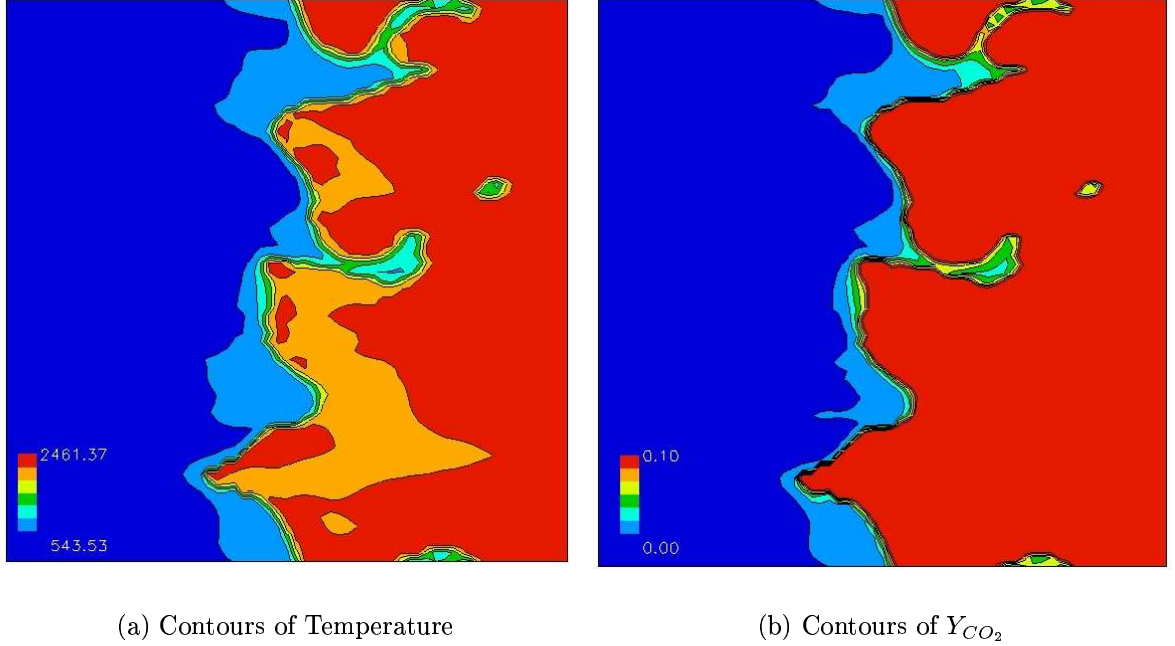
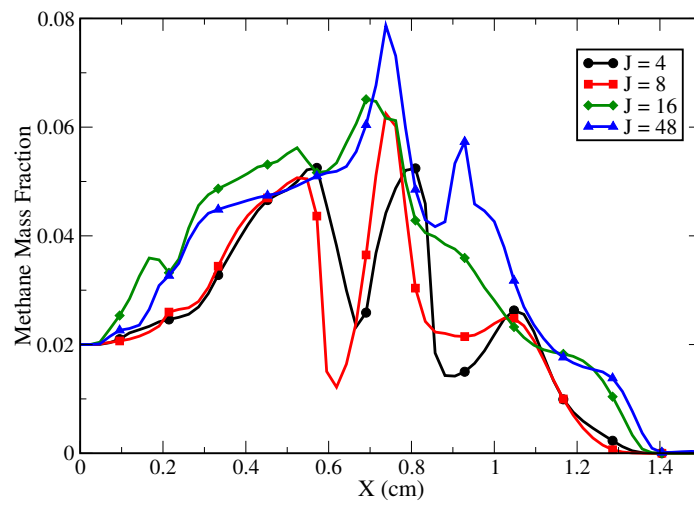
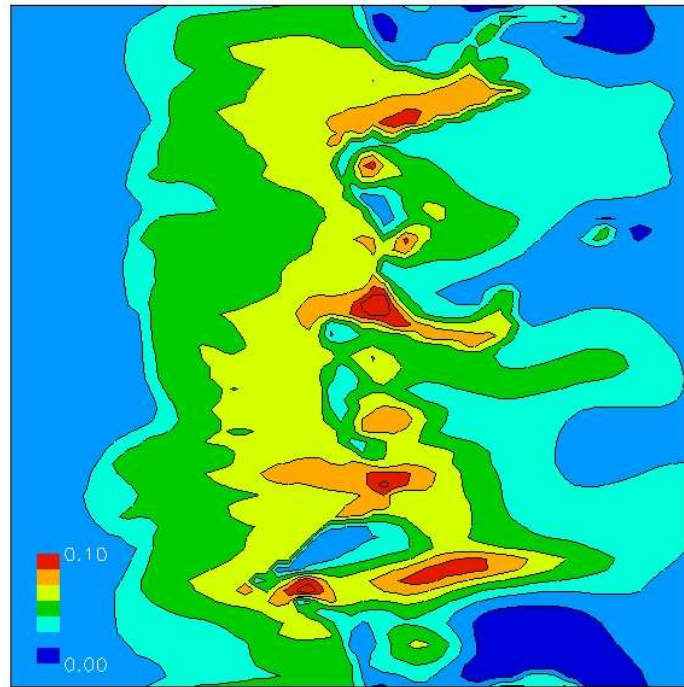


Figure 8.12: Instantaneous contours of temperature and Y_{CO_2} .

methane mass fraction shown in Figure 8.13. As a result of increased evaporation of droplets near the flame, a strong variation in the methane mass fraction can be observed. In addition, droplets that traverse through the flame vaporize behind the flame and burn in clusters seen by the red and orange patches behind the flame. Some of the methane vapor leaves unburned due to the depletion of oxygen at the flame zone.

Figure 8.13(b) shows the variation of methane mass fraction along the stream-wise direction, at 4 transverse (J) planes. Large amounts of methane concentration (Note: $Y_{stoic,CH_4} = 0.055$) are found even beyond the flame zone (for instance at $j = 48$) and eventually gets consumed before leaving the domain.

Figure 8.14(a) shows the evolution of the average droplet diameter across the flame at different times. Average droplet diameter is computed as an arithmetic mean of all the droplets present in each (Y-Z) plane. Also, shown in the figure is the line plot of the mean gas temperature across the flame. As it can be observed, the average



(b) Mass fraction Contours of CO_2

Figure 8.13: Instantaneous contours of methane mass fraction.

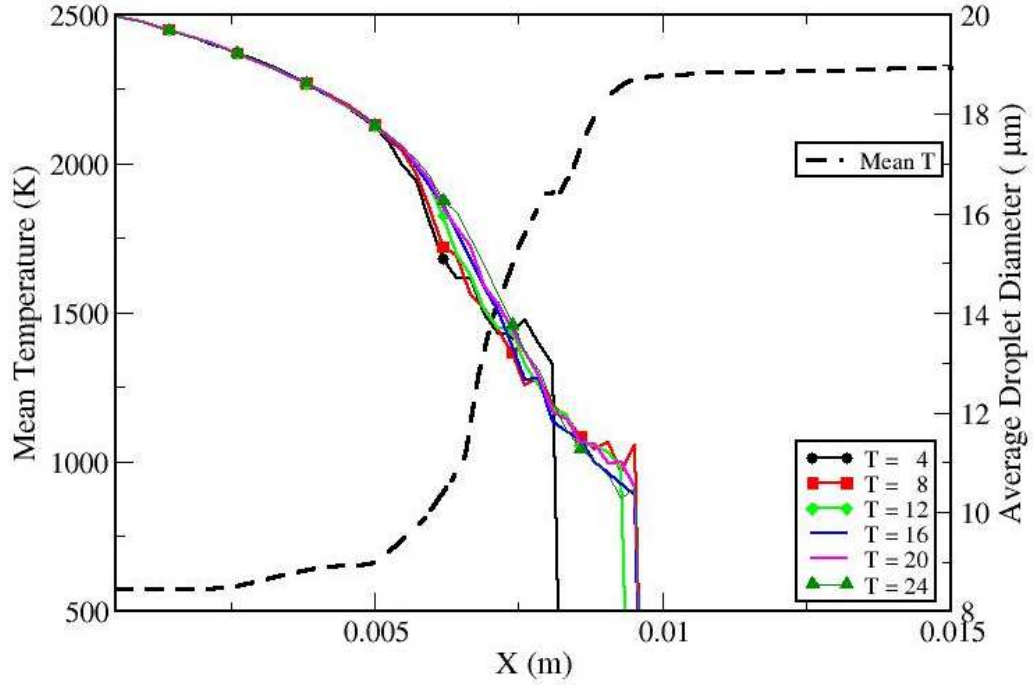


Figure 8.14: Evolution of droplet diameter.

droplet diameter decreases gradually at first almost linearly. This is the period of unsteady heat-up when most of the energy from the gas-phase goes to increase the droplet temperature. As a result, the evaporation rate is less and the rate of decrease of the droplet diameter is also low. As the droplet begins to move closer to the reaction zone, heat transfer rate to the drop increases and hence, begin to vaporize quickly with an abrupt reduction in the droplet diameter. An interesting feature to note is that, the average droplet diameter behind the flame ($T \approx 1800K$) is a finite non-zero number. This again indicates the presence of droplets at and beyond the flame zone. The abrupt drop in the droplet diameter at $X = 0.095\text{ m}$ is because of the absence of droplets beyond that axial location.

Figure 8.15 shows the evolution of the droplet temperature along the stream-wise direction at different times. Lefebvre [1989] points out the following about the evaporation rate of a droplet placed in a hot environment,

Initially, almost all of the heat supplied to the drop serves to raise the temperature. As

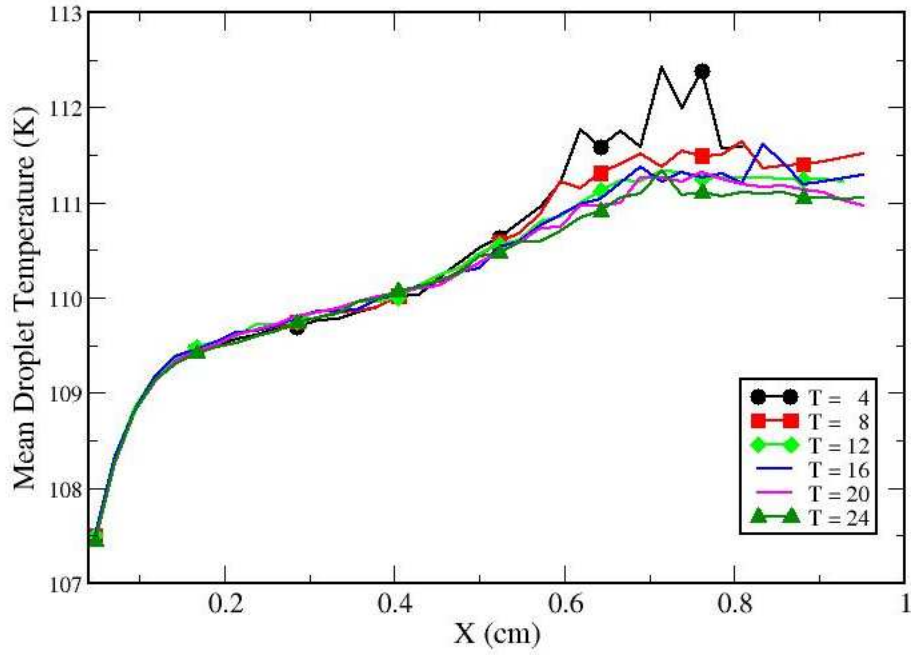


Figure 8.15: Evolution of droplet temperature.

the temperature rises, (1) part of the heat transferred to the drop is needed to furnish the heat of vaporization of the liquid (2) the outward flow of fuel vapor impedes the rate of heat transfer to the drop. This tends to diminish the rate of increase of drop surface temperature. Eventually a stage is reached where all of the heat transferred to the drop is used as heat of vaporization and the drop stabilizes at its wet-bulb temperature. Note, in reality, it would take the droplet lifetime for the droplet interior to reach an approximately uniform temperature at the wet-bulb value.

Even though, this analysis holds strictly for a single droplet, this can be extended to the present situation. The droplets at each stream-wise plane ($X = \text{constant}$, Y-Z plane), faces nearly similar conditions, because of the following reasons

1. the turbulence is homogeneous and isotropic
2. the initial conditions for the droplets (diameter, temperature, velocity, etc.) are identical for each droplet.

3. the gas phase temperature field is nearly homogeneous in every stream-wise plane ($X = \text{constant}$, Y-Z plane)

As a result, the average droplet property (such as the droplet temperature) should evolve similar to that of a single-droplet. As it can be seen from Figure 8.15, the evolution of the mean droplet temperature as it passes through the flame (along the stream-wise direction) exhibits all the character of the single-droplet. Initially for a short distance ($X < 0.2\text{cm}$) the temperature of the droplet rises steeply and then the rate of increase decreases and finally stabilizes around its wet bulb-temperature (For Methane, $T_{WB} = 111K$).

8.4 Triple Flame Structure

Triple flames arise in a number of practical configurations where the fuel and oxidizer are partially premixed. If a range of mixture fractions $Z_{min} < Z < Z_{max}$ exists in the partially premixed field, and the stoichiometric mixture fraction lies between these limits, flame propagation generates a flame structure called a triple flame. From the fuel-rich to fuel-lean sides of the mixture, rich and lean premixed flame branches emerge. Behind these premixed flames, at the locus of their intersection, along a surface in the vicinity of the stoichiometric surface, a diffusion flame is stabilized, and is fed by the excess fuel and oxidizer from the fuel rich and fuel lean (oxidizer rich) sides. The diffusion flame is anchored to the premixed branches by both preheating and diffusion of the radicals from the premixed flame zones. Past studies (Echekki and Chen [1998], Veynante et al. [1994]) have shown that flame propagation in a partially premixed gas occurs along the surface of stoichiometric mixture by triple flames.

Triple flame structure certainly plays an important role in turbulent combustion systems, due to their increased stability limits. For instance, Domingo and Vervisch [1996], Favier and Vervisch [1998] using vortex-triple flame interaction studies, showed

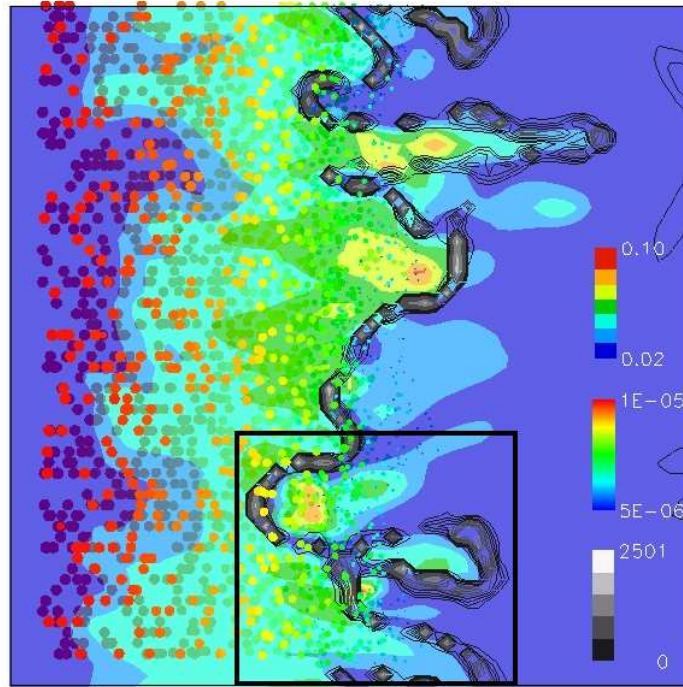
that the triple flames dynamically adjust their structure to unsteady environment and for complete extinction, it is required to extinguish both the premixed wings and the diffusion flame. Moreover, it has been observed (Veynante et al. [1994]) that even if one of the wings is quenched due to intense vorticity, the other reactive zones are able to sustain combustion.

Due to the presence of vaporizing droplets in a partially premixed reactants, regions of fuel rich and fuel lean mixture strengths are expected to be present in the flow field. Figure 8.16(a) shows the contours of reaction rate and methane mass fraction along with the droplet distribution at an instant (in X-Y plane). The area under the black rectangle is magnified and shown in fig. 8.16(b). Triple flame structure is evident in these figures showing the presence of a lean-premixed wing and a rich premixed wing along with a diffusion flame in between them. The colored contours mark the fuel rich (red and yellow patch near the rich premixed wing) and fuel lean (purple patch near the lean premixed wing) regions.

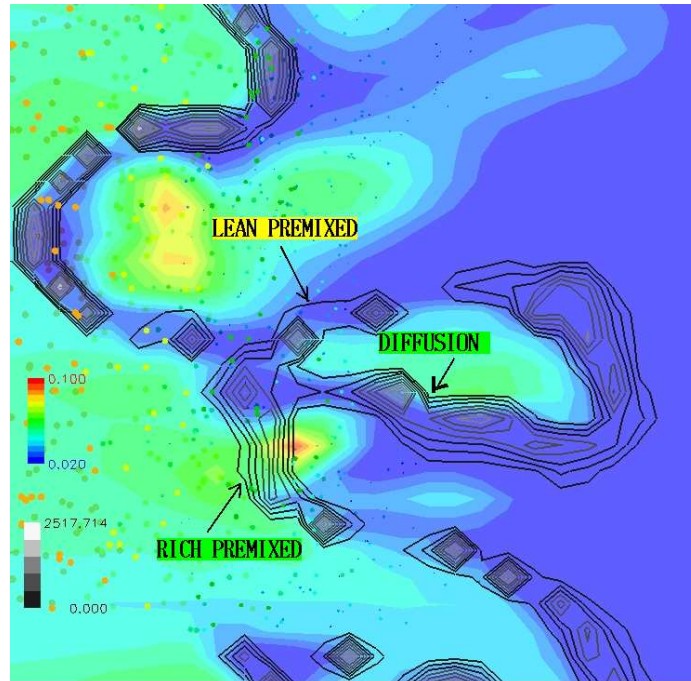
Figure 8.17 shows the presence of triple flame structure in another plane (Y-Z plane). Contours of methane reaction rate (black contours) and mixture fraction (colored contours) are used to identify the triple flame structure. Triple flame structures are shown in the marked rectangular boxes. The orange lines indicate the value of mixture fraction in the vicinity of the stoichiometric mixture fraction ($Z_{i_{Stoic}} = 0.055$). It can be clearly seen, in each of the marked boxes the orange lines that run in-between the reaction-rate contours mark the diffusion flame.

8.5 Conclusions

Results from the LES of partially premixed flames propagating in a droplet laden decaying isotropic flow-fields are reported. A sub-grid combustion model, using a global one step methane-air chemistry is used in this study. A fully coupled (momentum and thermal coupling, two-way) spray models are used to track the evaporating droplets

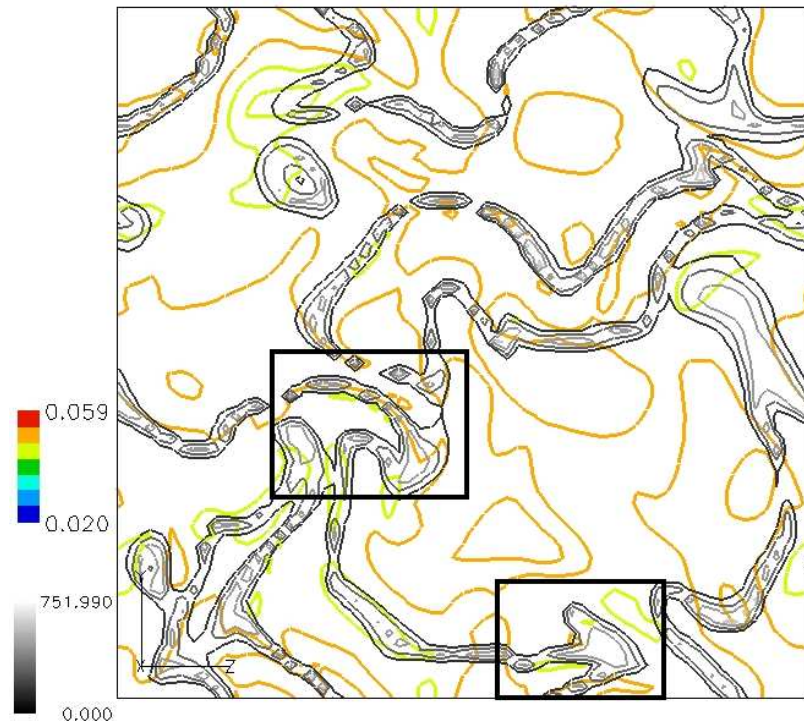


(a) Contours of $\dot{\omega}_{CH_4}$, Y_{CH_2} and droplet distribution

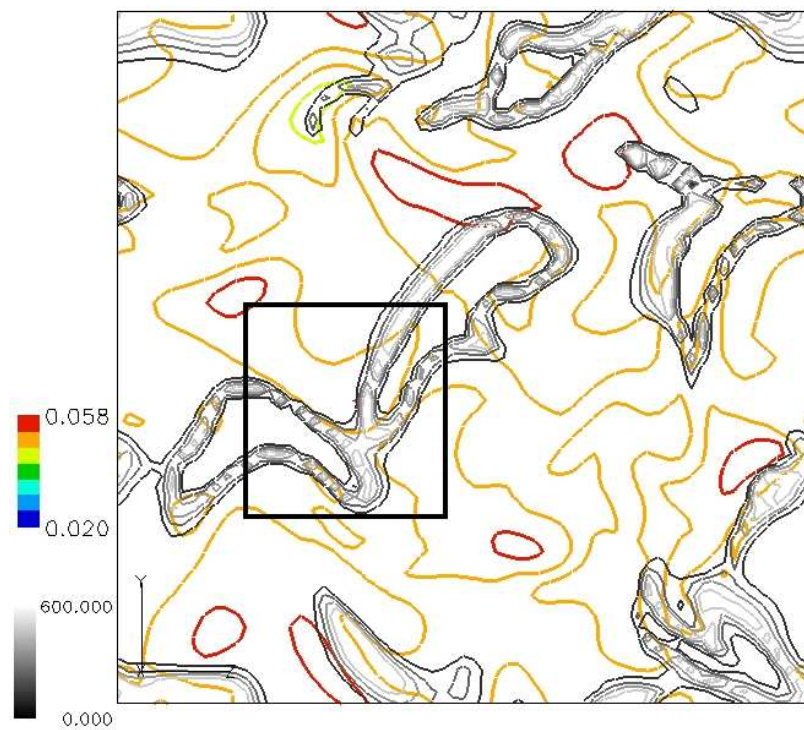


(b) Magnified image of triple flame shown in (a)

Figure 8.16: Contours of $\dot{\omega}_{CH_4}$ showing triple flame structure.



(a) Triple flame structure in Y-Z plane



(b) Reaction rate Contours of CH_4

Figure 8.17: Instantaneous contours showing triple flame structure.

in a partially premixed reactant mixture. The flame structure and other statistics such as the flame area evolution, curvature and strain-rate statistics computed using the LES-LEM are in good agreement with the DNS studies. LES-LEM along with the spray models correctly captures the flame structure in the partially premixed flames. Most of the fuel droplets completely vaporize before reaching the flame, and hence provides a continuous supply of reactants, which results in an intense reaction zone similar to a premixed flame. Some of the droplets that did not evaporate completely, traverse through the flame and vaporize suddenly in the post flame zone. Due to the strong spatial variation of equivalence ratio a broad flame similar to a premixed flame is realized. Triple flame structures are observed in the flow-field due to the equivalence ratio fluctuations.

CHAPTER IX

CONCLUSIONS

The primary focus of this thesis is to develop a generic combustion model for Large Eddy Simulation in compressible, reactive flows. The word generic is used here in the sense that the combustion model should be able to predict premixed, partially premixed and non-premixed flames without any *ad-hoc* modifications. Simultaneous presence of multiple modes of combustion is a common feature and characteristic in most of the modern power generation devices. As a result, a single predictive combustion model than can handle a wide range of operating conditions in complex geometries and complex flow fields (compressible, swirling, high Reynolds number) are needed. A sub-grid combustion model LES-LEM is developed and validated in this study and is employed to predict premixed, partially premixed and multi-phase combustion in high Reynolds number compressible flows.

LES-LEM is used to predict reactive flow-field in a realistic LM6000 combustor. The results obtained from the LES-LEM approach are compared with LDV measurements and another combustion model, the G-LES approach which has been validated in the past (Kim and Menon [2000]). LES-LEM performs reasonably well in predicting the complex (high Re, swirling, compressible, reactive) flow-field encountered in GEPS's operational hardware (LM6000 DLE combustor). The RANS approach has been used in the past (Hura et al. [1998]) to model the LM6000 combustor. However, significant amount of empiricism and iteration were needed to get similar level of agreement with the measurements. This application demonstrates that LES-LEM is a robust sub-grid combustion model which can predict the flame-structure under wide range of operating conditions such as those encountered in gas-turbine combustors,

without any empirical interventions. Besides, this application of LES-LEM tests and meets eight of the nine objectives identified in the Research Objectives section of (Chapter 1) of this thesis.

Following the demonstration of LES-LEM for premixed combustion, DNS and LES studies of particle laden temporal mixing layers were carried out to validate the droplet-models and the particle-tracking algorithms used in this study. Results are compared against the pseudo-spectral DNS studies of Ling et al. [1998]. Results obtained from the present dynamic LES approach are in very good agreement with the current DNS and the spectral DNS. Results obtained from the particle dispersion studies show strong Stokes number dependence on the particle dispersion. Both qualitative and quantitative comparison shows excellent agreement with the spectral-DNS results thus validating the two-phase LES approach.

LES studies of spray combustion in a gas turbine combustor are carried out using global chemistry with infinite-rate kinetics to study the effects of droplets, heat release and swirl on the fine-scale structure of turbulence. Results show that the vorticity tends to align with the intermediate strain rate, and the presence of particles, heat release and swirl decreases the magnitude of this alignment. The scalar gradient aligns with the most compressive strain rate and this alignment tends to increase the scalar gradient fluctuations at the small scales, thereby increasing molecular diffusion which smears out the gradients. This alignment is also found to increase with swirl. PDF of the strain-rate eigenvalues reveals the presence of both tube-like and sheet-like structures. Tube-like structures of vorticity are found to coincide with regions of intense vorticity gradients. On the other hand, regions of high scalar gradient form sheet-like structures that wrap around the tubular vortical structures. The scalar gradient in these sheets are amplified by the interaction between the strain rate field and vorticity field, thus increasing mixing and reaction in these regions.

LES of partially premixed flames propagating in a droplet laden decaying isotropic

flow-fields is carried using using a global one step methane-air chemistry. The flame structure and other statistics such as the flame area evolution, curvature and strain-rate statistics computed using the LES-LEM are in good agreement with the DNS studies. LES-LEM along with the spray models correctly captures the flame structure in the partially premixed flames. Most of the fuel droplets completely vaporize before reaching the flame, and hence, provides a continuous supply of reactants, which results in an intense reaction zone similar to a premixed flame. Some of the droplets that did not evaporate completely, traverse through the flame and vaporize suddenly in the post flame zone. Due to the strong spatial variation of equivalence ratio a broad flame similar to a premixed flame is realized. Triple flame structures are observed in the flow-field due to the equivalence ratio fluctuations. This application of LES-LEM completes the last requirement of the proposed objectives in the Research Objectives section of this thesis.

Thus, a generic, sub-grid combustion model for the LES of turbulent reacting flows has been developed, tested and validated. It has been shown that this model, namely LES-LEM can capture the flame-structure in the premixed, partially premixed, non-premixed, single phase and in multi-phase reacting flows.

CHAPTER X

FUTURE RECOMMENDATIONS

Supersonic Combustion Besides fundamental interest in turbulent mixing, research in compressible mixing is driven by numerous technological applications. Foremost among them are scramjet engines for hypersonic propulsion systems, where fuel and oxidizer must be mixed rapidly and efficiently. However, the reduced mixing observed in the compressible shear layers found in such systems solicit longer combustor lengths resulting in severe weight and drag penalties. A better understanding of the turbulent mixing and combustion in the presence of compressible effects and shocks is still lacking. Even though LES-LEM captures the correct physics of subsonic combustion, its extension to supersonic combustion is not trivial. The effect of high compressibility and the presence of shock would require relaxing some of the assumptions (such as constant sub-grid pressure) used in the present study.

Modeling Liquid Jet Break-up Due to the small dimensions and the complex geometries of the injectors, it is nearly impossible to characterize the spray using measurement techniques. This can be modeled computationally using a level-set approach to track the interface of the liquid jet and the gas phase. This can provide valuable information on the process of formation of the droplets and their subsequent interaction with the other droplets. This method can be coupled with the Lagrangian tracking used in this study to model the complete spray field evolution. But the computational complexity of a disintegrating liquid stream can rival the the simulation of the combustor flow. There exists a great scope for the development of advanced models in this area.

Dense-Spray Modeling One of the important limitation of the the two-phase models used in the present study is the dilute-spray approximation. This implies that the model in the current state cannot be used for modeling the the near field of the injectors. Dense-spray models can be used along with the level-set approach mentioned earlier to model the droplet coalescence and break-up effects.

General comment Use of better/improved droplet models that include non-uniform droplet temperatures, secondary liquid motion in the interior of the droplet can improve the accuracy of the predictions.

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